A Regime-Switching Continuous Hidden Markov Model as a Reference Synthetic-Data Generator for Equity Returns: Extended Evaluation, Semi-Markov Ablation, and Regime-Conditional Value-at-Risk

Abdulrahman Alswaidan* Cade Jin† Jeffrey D. Varner‡

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Abstract

We develop a family of regime-switching continuous hidden Markov models (CHMMs) as a reference synthetic-data generator for daily equity returns and evaluate it against a broad panel of non-parametric, classical, discrete-state, and deep-generative baselines. The three CHMM variants share a log-space Expectation–Maximization (EM) scaffold with a common quantile-initialized transition matrix and differ only in their per-state emission law: Gaussian (CHMM-N), Student-t with data-selected per-state degrees of freedom (CHMM-t), and Laplace (CHMM-L). At a single moderate state resolution the continuous models reproduce the canonical stylized-fact set of daily equity returns (heavy tails, negligible linear autocorrelation, persistent volatility clustering, leverage effect, and aggregational Gaussianity) without any jump augmentation or return-space discretization.

On ten years of daily SPDR S&P 500 exchange-traded fund (ETF) data with ticker SPY, we benchmark the generator family along three operational axes. First, on distributional fidelity, flat CHMM-t attains the smallest Maximum Mean Discrepancy (MMD) of any row in the panel and the discriminator Area Under the Curve (AUC) closest to 0.5 among parametric and Markov-switching rows, second only to a window-diffusion baseline on AUC and ahead of generative adversarial and Markov-Switching GARCH (MS-GARCH) baselines. Second, on unconditional Value-at-Risk (VaR) calibration, a semi-Markov CHMM ablation and a two-regime MS-GARCH baseline tie for the best-in-panel Kupiec likelihood-ratio statistic at 1% ($\mathrm{LR}_{\mathrm{uc}} = 0.01$), with MS-GARCH alone leading at 5%. Third, on regime-conditional VaR, a Viterbi-decoded state-conditional quantile of flat CHMM-t passes both the Kupiec unconditional-coverage and the Christoffersen independence likelihood-ratio tests cleanly at 1% and 5% on a held-out 2024 to 2026 window, closing the breach-clustering gap that the unconditional panel leaves open. We also report a concrete negative finding against a discrete-state plus Poisson-jump baseline, whose 5% VaR breach rate is overly conservative at 1.7% and fails Kupiec with a likelihood ratio of 16.81. We extend the framework to multi-asset synthesis through a Single Index Model and Gaussian and Student-t copulas on six equities, with a truncated C-vine variant reported as a visual reference on the same universe; the flat Student-t copula remains the most competitive dependence layer in our setup.

*Cornell University, Robert Frederick Smith School of Chemical and Biomolecular Engineering, Ithaca, NY, USA. Email: aa2725@cornell.edu.

†Cornell University, Ithaca, NY, USA. Email: cj383@cornell.edu.

‡Cornell University, Robert Frederick Smith School of Chemical and Biomolecular Engineering, Ithaca, NY, USA. Email: jdv27@cornell.edu.

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1 Introduction

A synthetic generator of financial time series is useful as a *digital twin* only if it reproduces the canonical stylized facts of the underlying observable simultaneously: for equity returns, heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [1, 2]. Any such generator, intended for downstream tasks such as risk measurement, scenario generation, or regime-aware dynamic asset allocation [3], must reproduce all three simultaneously [4, 5], and should be *practical to train*, so that the scaffold can be reused across observables without architectural surgery. GARCH models [6, 7] capture volatility clustering but cannot represent the multi-regime structure of the marginal, while i.i.d. generators match the marginal but destroy temporal persistence. Hidden Markov models (HMMs), popularized in macroeconometrics by Hamilton [8] and extended to stock-market returns by Schaller and Van Norden [9], offer a natural middle path through a latent regime structure.

An influential negative result has shaped two decades of work on this problem. Rydén et al. [10] fitted Gaussian-emission HMMs with $K = 2$–3 states and concluded that, while HMMs reproduce heavy tails and absent return autocorrelation, they *fail* to capture the slow ACF decay of absolute returns: the geometric sojourn-time distribution of a standard Markov chain produces exponential decay that is too fast. Bulla and Bulla [11] restored ACF fidelity with hidden semi-Markov models (HSMMs) at the cost of substantially increased complexity, and Alswaidan and Varner [12] reached the same end within a discrete HMM by adding a Poisson jump-duration mechanism calibrated through two extra hyperparameters.

We show that these complexities are unnecessary. Our work builds on the established continuous-emission HMM-for-finance literature of Hamilton [8] (two-regime Gaussian), Bulla and Bulla [11] (semi-Markov dwell-time fix), and Nystrup et al. [13] (continuous emissions for long-memory fit), but departs from each on four specific axes: (i) per-state Student-t degrees-of-freedom ν_k updated inside an Expectation–Conditional-Maximization (ECM) step with a golden-section bracket, (ii) a closed-form weighted-mean-absolute-deviation M-step for Laplace emissions, (iii) a head-to-head benchmark against the jump-augmented discrete HMM of Alswaidan and Varner [12] under the same evaluation protocol, and (iv) a two-pipeline composition in which a shared continuous emission scaffold (Pipeline A) composes with a copula on the rank-transformed path (Pipeline B) for cross-asset synthesis. A continuous HMM trained by the Baum-Welch algorithm [14, 15] with quantile-based initialization reproduces all three stylized facts at moderate state resolution, without jump augmentation, discretization, or separate emission fitting. We further observe that the emission family is a first-class design choice that the same EM scaffold accommodates at no architectural cost. Alongside the classical Gaussian variant (CHMM-N) we introduce and evaluate two heavy-tailed variants sharing identical forward-backward recursions: CHMM-t, with per-state Student-t emissions whose degrees-of-freedom ν_k are updated inside the M-step by a golden-section search on the ECM Q-function in the tradition of Peel and McLachlan [16] and Liu and Rubin [17]; and CHMM-L, with per-state Laplace emissions whose location is the weighted median and whose scale is the weighted mean absolute deviation, both in closed form. We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$; $K = 18$ sits inside the IC-minimizing band $\{15, 18, 21\}$ under the four information criteria (AIC, BIC, HQC, CAIC) used for HMM state selection in stock-price prediction by Nguyen [18] and, within that band, wins or ties every IS and OoS distributional pass-rate metric. The negative result of Rydén et al. [10]

is a low-K artifact: once the transition matrix has heterogeneous diagonal entries across regimes, its eigenvalue spectrum provides a mixture of geometric sojourn times that approximates slow ACF decay, achieving an HSMM-like effect within the standard HMM framework.

Against eight alternative generators on ten years of daily SPY data (i.i.d. bootstrap, stationary block bootstrap, Gaussian and Laplace i.i.d., the discrete HMM of Alswaidan and Varner [12] in no-jump (NJ) and Poisson-jump (WJ) variants at the authoritative prior-paper hyperparameters $K_{\text{disc}} = 90$, $\epsilon = 5 \times 10^{-5}$, $\lambda = 67$, GARCH(1,1), and a single-layer GRU + Gaussian-head autoregressive density baseline trained by negative log-likelihood), the Gaussian CHMM-N at $K = 18$ attains 93.8% in-sample (IS) and 84.2% out-of-sample (OoS) Kolmogorov–Smirnov pass rates and ACF-MAE of 0.0507 on absolute returns, essentially matching GARCH's temporal fidelity without sacrificing distributional fit. The heavy-tailed CHMM-t and CHMM-L variants retain this ACF and pass-rate fidelity; CHMM-L brings the simulated IS excess kurtosis to 6.75, close to the observed 7.68, while CHMM-t overshoots at 14.57 (see Section 7). At the low-K end of the sweep CHMM-t already beats CHMM-N on IS KS (91.1% at $K = 3$ versus 89.8%) because the per-state heavy tails of Student-t compensate for under-resolved regime structure: a small number of heavy-tailed regimes can cover the return distribution that a larger number of Gaussian regimes is needed to tile. We diagnose the CHMM-t IS kurtosis over-correction (14.57 versus observed 7.68) by reporting the per-state ν_k histogram and a $\nu_{\min}$ bracket-sensitivity study, and quantify its downstream consequence with a 1%/5% Value-at-Risk and Expected-Shortfall calibration against the historical series. Univariate generalization is confirmed on NVDA, JNJ, JPM, AAPL, and QQQ for all three emission families; the OoS gap on NVDA and JPM under the 2024–2026 regime is acknowledged as a stationarity scope point and a quarterly refit diagnostic is reported in Appendix E, but is not a headline methodology of the paper. For multi-asset synthesis, we extend the univariate framework with a Single Index Model [19] and Gaussian / Student-t copulas [20–22], both of which preserve each asset's fitted CHMM marginal. On six SPY-member assets the copulas substantially outperform SIM on per-asset KS fidelity (median 96.0–97.8% vs. 78.5%) and correlation-matrix reproduction (off-diagonal MAE 0.027 Student-t, 0.031 Gaussian, 0.076 SIM), with the Student-t copula's profile MLE selecting $\nu^* = 6$ degrees of freedom. Option pricing, volatility-index generalization, and leverage-effect modelling are explicitly out of scope and are deferred to future work.

Scope. The seven-metric panel in Section 3.5 evaluates distributional, tail, and temporal fidelity of the generated return series. The 1%/5% VaR and Expected-Shortfall back-test in Section 6.1 grounds those fidelity numbers in a concrete downstream risk-measurement consumer. The input data are public daily U.S. equity prices, so no privacy or individual-level considerations apply.

The remainder of the paper is organized as follows. Section 2 positions the work against related literature. Section 3 describes the data, the CHMM, the benchmark models, the cross-asset constructions, and the evaluation protocol; Section 4 develops the EM estimator and state-count selection; formal algorithms and metric definitions are in Appendices A and B. Section 5 presents the empirical study and Section 6 the VaR back-test; Sections 7–8 discuss and conclude.

2 Related Work

Stylized facts and parametric volatility. The canonical set of stylized facts [2], namely heavy tails [1], absence of linear autocorrelation [23], and slow decay of the absolute-return ACF [24], must be reproduced jointly for a generator to be useful downstream. Autoregressive conditional heteroscedasticity (ARCH) and its generalised variant (GARCH) [6, 7] encode volatility persistence through a single decay parameter but impose a unimodal innovation distribution and, as we confirm, fail distributional tests on equity returns. Jump-diffusion models [25, 26] generate heavy tails but lack a clustering mechanism: after a jump the conditional variance returns immediately to baseline. Maximum-likelihood fitting of heavy-tailed innovations in a latent-volatility framework has been studied for stochastic-volatility-in-mean models by Abanto-Valle et al. [27], who compare Normal, Student-t, generalized error distribution (GED), and Slash errors; our CHMM-t and CHMM-L are the HMM analog of that emission-family choice.

Regime-switching and the Rydén et al. limitation. Hamilton [8] introduced Markov-switching models to economics, and Schaller and Van Norden [9] provided an early demonstration that monthly equity returns from the Center for Research in Security Prices (CRSP) are well-described by a two-state Markov-switching specification with strongly different mean and variance between regimes; Ang and Bekaert [28] subsequently showed that short-rate dynamics in several developed markets also admit a regime-switching representation, establishing regime-switching as a cross-asset-class phenomenon rather than one confined to macro series or equities. The seminal evaluation of HMMs against the Cont stylized facts is due to Rydén et al. [10], who fitted Gaussian-emission HMMs with $K = 2$–3 states to daily equity-index return series and showed that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay. Bulla and Bulla [11] restored fidelity with hidden semi-Markov models (HSMMs) that replace the geometric sojourn distribution with flexible alternatives. Alswaidan and Varner [12] instead kept the discrete HMM framework and introduced a Poisson jump-duration process to force dwell time in tail states.

Within the same family, Nguyen [18] applies a four-state continuous HMM to monthly S&P 500 prices for OoS trading signal generation, selecting the state count through the AIC, BIC, HQC, and CAIC information criteria. We adopt the same four-criterion selection (Section 5.2), in addition to our multi-metric distributional score, to place the K-sweep recommendation on standard statistical footing.

The present work resolves the Rydén limitation through a third route: increase the state resolution, keep the standard HMM framework, and let the Baum-Welch algorithm learn heterogeneous self-transition probabilities across regimes. At $K \geq 3$ with quantile-based initialization, high-volatility states develop enough self-persistence that the resulting mixture of K geometric sojourn-time distributions approximates slow ACF decay, effectively recovering the HSMM behavior within an ordinary Markov chain.

Discrete HMM with jumps. The immediate predecessor of this paper, Alswaidan and Varner [12], discretizes returns into Laplace quantile bins, estimates $\mathbf{T}$ by frequency counting, fits bin-conditional Student-t emissions, and adds a Poisson jump-duration process with two grid-calibrated hyperparameters (ϵ, λ). The continuous model presented here eliminates all three sources of complexity: no binning, no separate emission fit (Baum-Welch jointly learns transitions and emissions), and no jump mechanism at moderate K.

Cross-asset dependence. For multi-asset synthesis, the Single Index Model [19] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar's theorem [20, 29] supports a cleaner decomposition in which each asset's fitted marginal is kept and only the copula is estimated. Gaussian and Student-t copulas [21, 22] are the standard choices, and rank-reordering simulation [30] injects the copula dependence onto pre-simulated marginal paths without distorting any marginal. We confirm, on continuous CHMM marginals, the copula-ahead-of-SIM ordering reported by Alswaidan and Varner [12] for discrete HMM marginals, which suggests that the ordering is intrinsic to the rank-reordering construction rather than specific to the marginal model.

Deep generative models and quality assessment. Generative adversarial network (GAN) based generators reproduce the visual appearance of return series but typically fail the clustering test unless architectures encode temporal dependence explicitly [31, 32], the TimeGAN family of Yoon et al. [33] being the standard reference in this direction; neural baselines in Alswaidan and Varner [12] close most of the ACF gap but suffer variance collapse, indicating that the distribution–clustering tension remains in tension for neural approaches. Following the GRU baseline used in Alswaidan and Varner [12], we add a single-layer GRU + Gaussian-head auto-regressive recurrent generator (Section 3.3) to Table 3 as a deep-generative head-to-head row, trained by negative log-likelihood on the IS series and rolled forward by sampling from the predicted Gaussian and feeding the sample back into the recurrence. For evaluation, we adopt the category framework of Stenger et al. [34] and the financial-synthesis emphasis of Assefa et al. [4]: no single metric is decisive, so we report a seven-metric panel spanning distributional, tail, and temporal fidelity.

3 Model

3.1 Data and Excess Growth Rates

The primary single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\mathrm{IS}} = 2{,}516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\mathrm{OoS}} = 572$) held out. Cross-asset generalization covers NVDA, JNJ, JPM, AAPL, and QQQ, spanning distinct risk profiles. For ticker i and trading day j, the annualized excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln\!\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0. \tag{1}$$

3.2 Continuous Hidden Markov Model

We model the continuous excess growth rate G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ with K latent states, a per-state emission density $b_k(x; \boldsymbol{\theta}_k)$, transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$, and initial distribution $\boldsymbol{\pi}$ [15]. The stationary marginal is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k \, b_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}}\mathbf{T}, \tag{2}$$

and supplies heavy tails either through the spread of component moments (Gaussian components) or through the emission family itself (Student-t or Laplace components). Unlike the discrete framework of Alswaidan and Varner [12], which bins observations and estimates $\mathbf{T}$ by frequency counting, we learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly from continuous observations via EM; the estimation procedure is developed in Section 4.

CHMM-N: Gaussian emissions. $b_k(x) = \mathcal{N}(x; \mu_k, \sigma_k^2)$ with per-state location μ_k and variance σ_k^2.

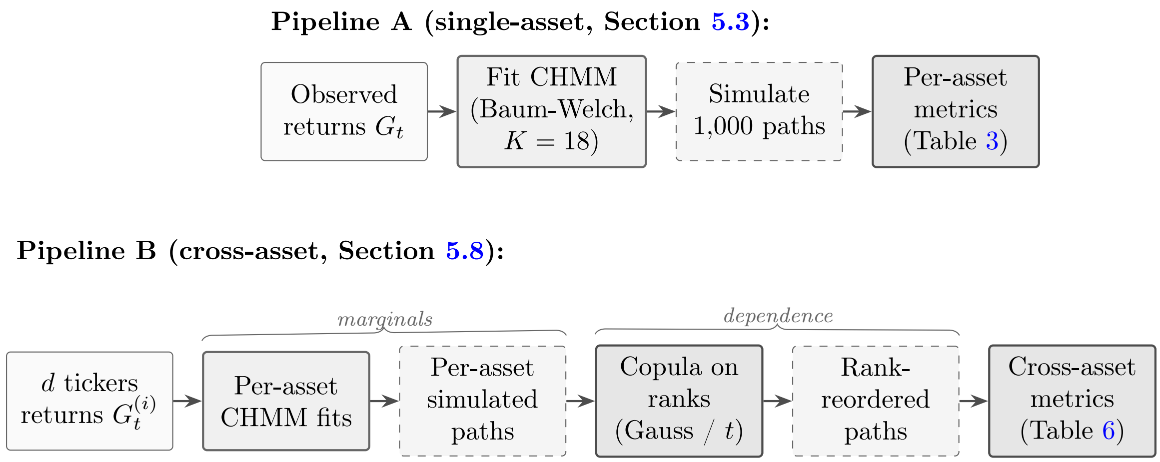


Figure 1. Schematic of Pipeline A (single-asset) and Pipeline B (cross-asset). Pipeline A fits a CHMM per ticker independently and evaluates single-asset marginal / temporal metrics. Pipeline B reuses the per-asset CHMM fits as *marginals* and injects cross-asset *dependence* through a rank-based copula (Gaussian or Student-t), preserving each asset's fitted CHMM distribution exactly while coupling asset ranks to match observed multi-asset correlation structure. Headline: the two pipelines share the single-asset scaffold, so improvements to the CHMM at the marginal step propagate into Pipeline B without any changes to the dependence layer.

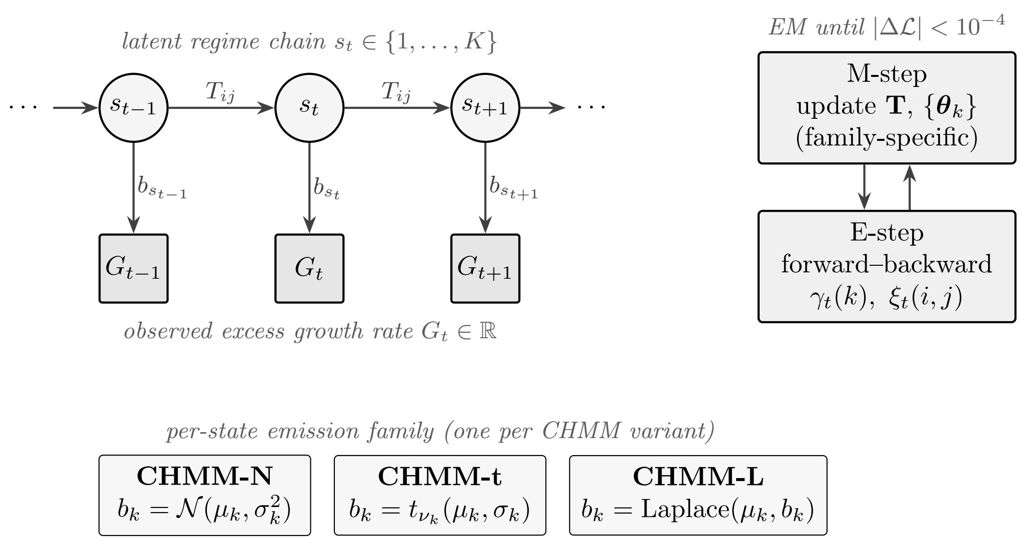


Figure 2. CHMM architecture and training loop. TOP: a K-state latent Markov chain with transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$ (entries $T_{ij} = \Pr(s_{t+1} = j \mid s_t = i)$) generates the regime sequence s_t; each s_t conditionally emits the observed excess growth rate G_t through a state-specific density $b_{s_t}(\cdot;\boldsymbol{\theta}_{s_t})$. BOTTOM LEFT: the three CHMM variants share everything except the emission family — Gaussian (CHMM-N), Student-t with per-state degrees of freedom ν_k (CHMM-t), or Laplace (CHMM-L). BOTTOM RIGHT: joint estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ alternates an E-step (log-space forward–backward on $\gamma_t(k), \xi_t(i,j)$) and a family-specific M-step until the log-likelihood change falls below 10^{-4} or max_iter = 60. The emission-family swap is the only architectural difference across the three variants in this paper.

CHMM-t: Student-t emissions. $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ with per-state location μ_k, scale σ_k, and degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. The per-state degrees of freedom let heavy-tailed regimes coexist with near-Gaussian regimes in the same model.

CHMM-L: Laplace emissions. $b_k(x) = \text{Laplace}(x; \mu_k, b_k)$ with per-state location μ_k and scale b_k. The heavier-than-Gaussian Laplace tail (excess kurtosis 3 per component) provides an alternative route to tail fidelity that is strictly cheaper than Student-t because it has no shape parameter.

3.3 Benchmark Generators

We benchmark the CHMM against eight alternatives fitted on the same IS window. The *i.i.d. bootstrap* samples T returns uniformly with replacement, preserving the empirical marginal exactly and serving as the non-parametric distributional ceiling. The *stationary block bootstrap* [35] resamples blocks of expected length $b = 10$ trading days, preserving short-range volatility clustering while keeping the marginal empirical; it is a temporally-aware non-parametric ceiling. The *Gaussian* i.i.d. generator draws $G_t \sim \mathcal{N}(\hat{\mu}, \hat{\sigma}^2)$ from the sample moments; the *Laplace* i.i.d. generator draws $G_t \sim \text{Laplace}(\hat{m}, \hat{b})$ from the sample median and mean-absolute deviation. *GARCH(1,1)* [7] uses $G_t = \mu + \sigma_t z_t$, $\sigma_t^2 = \omega + \alpha(G_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2$ with $z_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, fitted by Nelder–Mead maximum likelihood. The *discrete HMM* of Alswaidan and Varner [12] is reproduced in both no-jump (NJ) and Poisson-jump (WJ) variants at the final SPY hyperparameters reported therein: returns are partitioned into $K_{\text{disc}} = 90$ quantile bins, transitions are estimated by frequency counting, and simulation returns bin centroids; the WJ variant additionally triggers, with probability $\epsilon = 5 \times 10^{-5}$, a forced residence in tail states for Poisson($\lambda = 67$) steps, matching the values of Alswaidan and Varner [12]. The *Bin-T NJ* variant replaces those bin centroids with bin-conditional Student-t emissions to approximately isolate the quantization-per-se effect from the emission-family effect; because per-bin Student-t MLE is under-identified at $K = 90$ (only $T/K \approx 28$ observations per bin over $T \approx 2{,}516$ IS days), we use a coarser $K_{\text{disc}} = 13$ for Bin-T NJ so that each bin carries enough mass to estimate its tail parameters. The comparison with the NJ row therefore varies both the bin count and the emission family; we flag this confound in Section 7 and treat Bin-T NJ as an approximate rather than exact quantization ablation. Because quantization onto discrete centroids strips within-bin tail mass, the standard discrete baselines provide a direct upper bound on how much of the CHMM's fidelity is attributable to continuous emissions rather than to regime structure per se.

GRU deep generative baseline. We add a single-layer GRU autoregressive density baseline following the GRU baseline of Alswaidan and Varner [12]; recurrent-generator architectures for financial time series are also the subject of the TimeGAN family of Yoon et al. [33]. The architecture is $\mathbf{h}_t = \text{GRU}_H(\mathbf{x}_{t-w:t-1})$ followed by an affine head $(\mu_t, \log\sigma_t) = \mathbf{W}\mathbf{h}_t + \mathbf{b}$, with hidden dimension $H = 32$, context window $w = 20$ trading days, and a Gaussian next-step density $G_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(\mu_t, \sigma_t^2)$. The model is trained for 20 epochs by Adam (learning rate 10^{-3}) on the negative log-likelihood (NLL)

$$\ell_t = \log\sigma_t + \tfrac{1}{2}\left(\frac{G_t - \mu_t}{\sigma_t}\right)^2 \tag{3}$$

of the standardised IS series, with $\log\sigma_t$ clamped to $[-6, 4]$ for numerical stability. Synthesis is auto-regressive: the trailing w IS returns seed the recurrence, the predicted Gaussian is sampled, the sample is appended to the context window, and the procedure rolls forward. A 64-step burn-in is discarded. The implementation is in Julia using `Flux.jl` [36]; the trained model is callable in the same `simulate` interface as the CHMM and discrete-HMM models.

QuantGAN convolutional Wasserstein baseline. We add a repo-native QuantGAN-style baseline in the sense of Wiese et al. [37]: a 1D convolutional generator and critic trained with the

Wasserstein loss of Arjovsky et al. [38] and critic weight clipping on standardised SPY returns, with rolling-window training and stitched-window path generation at inference time. The window length is $W = 64$, the latent dimension is 8, and training runs for 15 epochs. The generator is implemented in `Flux.jl` and is callable in the same `simulate` interface as the other generators. This is a deliberately reproducible first-pass design: no temporal convolutional network, no gradient-penalty, no rolling-origin re-training; it functions as a deep-generative negative control for Section 5.6.

Window-diffusion deep baseline. We add a DDPM-style window-diffusion baseline following Ho et al. [39], Rasul et al. [40]: a conditional denoising UNet trained to reverse a $T = 50$-step Gaussian noise schedule on standardised SPY return windows of length $W = 64$, with stitched-window generation at inference. Training runs for 18 epochs with final denoising loss 0.736. Like the QuantGAN row, this is a first-pass reproducible design; it is the strongest deep-generative row on local 20-day window distributional metrics and the second-strongest overall row on MMD after the flat CHMM-t, and it serves as a distributional stress test for the CHMM family in Section 5.6.

MS-GARCH variance-regime baseline. We add a two-regime Markov-switching GARCH(1,1) baseline in the sense of Haas et al. [41]. The observable evolves as $G_t = \mu + \sigma_t z_t$ with a regime-specific variance recursion $\sigma_t^2(k) = \omega_k + \alpha_k(G_{t-1} - \mu)^2 + \beta_k \sigma_{t-1}^2(k)$, where $k \in \{1, 2\}$ is a hidden regime with transition matrix $P = \{p_{ij}\}$. We use the path-independent HMP 2004 recursion with a Hamilton filter for the regime likelihood and fit the eight parameters $(\mu, \omega_{1:2}, \alpha_{1:2}, \beta_{1:2}, p_{11}, p_{22})$ by Nelder-Mead maximum likelihood on the IS SPY window. The fitted calm regime has unconditional $\sigma \approx 0.97$ and expected sojourn ~ 12 days ($p_{11} = 0.914$); the stress regime has unconditional $\sigma \approx 12.67$ and expected sojourn ~ 2 days ($p_{22} = 0.547$). MS-GARCH provides the variance-regime reference point for the Kupiec VaR calibration discussion in Section 5.6 and Section 6.2: it attains best-in-panel unconditional VaR calibration at 1% ($\mathrm{LR}_{\mathrm{uc}} = 0.01$, breach rate exactly 1.0%) and at 5% ($\mathrm{LR}_{\mathrm{uc}} = 0.26$, breach rate 4.5%), and its $\mathrm{LR}_{\mathrm{ind}} = 4.79$ at 5% is the closest to the 3.84 independence-pass threshold of any unconditional-VaR row.

3.4 Cross-Asset Dependence (Pipeline B)

The univariate evaluation of Section 5.3 fits a separate CHMM to each ticker independently; we refer to this as *Pipeline A* throughout the paper. The present subsection introduces *Pipeline B*, which reuses the per-asset CHMM marginals from Pipeline A and adds joint dependence across assets. Pipeline A is tested in Table 5 (univariate); Pipeline B is tested in Table 6 (cross-asset) and Figure 32.

Let $\mathbf{R}_{\mathrm{IS}} \in \mathbb{R}^{T \times d}$ denote the IS return matrix for d assets, with the market-factor column indexed by m. Both constructions below preserve each asset's fitted CHMM marginal and differ in how cross-asset dependence is imposed.

Single Index Model. Following Sharpe [19], for each non-market asset $j \neq m$ we fit by ordinary least squares (OLS)

$$G_{j,t} = \alpha_j + \beta_j \, G_{m,t} + \eta_{j,t} \tag{4}$$

and simulate paths by drawing an independent market-factor path $G_{m,t}^{(p)}$ from the market CHMM, resampling residuals $\hat{\eta}_{j,t}^{(p)}$ with replacement, and propagating

$$\hat{G}_{j,t}^{(p)} = \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,t}^{(p)}. \tag{5}$$

The affine map distorts each non-market marginal and imposes a rank-one correlation structure.

Copulas. By Sklar's theorem [20], any joint CDF with continuous marginals $F_1, \ldots, F_d$ factors as $H = C(F_1, \ldots, F_d)$ for a unique copula $C : [0,1]^d \to [0,1]$; this lets us fix each marginal at the per-asset CHMM and estimate only C from the cross-sectional co-movement. We use the nonparametric probability integral transform $u_{j,t} = \text{rank}(g_{j,t})/(T+1)$, estimate the copula correlation matrix Σ by the analytic Kendall's-τ inversion $\rho_{ij} = \sin(\pi\tau_{ij}/2)$ [22, 42], and project to the nearest positive semi-definite matrix when needed. For the Student-t copula [21], the degrees-of-freedom parameter ν is selected by profile maximum likelihood over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$. Cross-asset simulation follows the rank-reordering construction of Iman and Conover [30]: (i) draw T independent paths from each per-asset CHMM, (ii) sample T copula variates, and (iii) reorder each column of the CHMM matrix to match the ranks of the copula sample. Reordering only permutes simulated values, so every per-asset marginal is preserved *exactly* while cross-asset dependence is injected through rank coupling. Full derivations, the Student-t log-density, and the sampler pseudocode are given in Appendix D.1.

3.5 Evaluation Protocol

For each model we simulate $P = 1{,}000$ independent paths of length T (200 paths for the cross-asset extension, given the quadratic cost of the copula and SIM pipelines) and evaluate seven complementary metrics, following the framework of [34] and [12]. All experiments reported in the main body and appendix use a deterministic global random seed (20260420), with per-experiment sub-seeds (20260421 for OoS equity-price simulation, 20260422 for Track A extended evaluation) derived additively so that each block is independently reproducible from a single seed root; no other random-seed specification is given in-line throughout the remainder of the paper. Four metrics score *distributional* fidelity: two-sample Kolmogorov–Smirnov [43, 44] and Anderson–Darling pass rates at $\alpha = 0.05$ over the 1,000 paths, Wasserstein-1 distance $W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)|\, du$, and histogram Hellinger distance. Two metrics score *tail* behavior: the mean simulated excess kurtosis and the quantile-coverage rate, defined as the fraction of observed percentiles 1–99 falling within the $[5, 95]$-percentile envelope of the simulated quantiles. The final metric scores *temporal* fidelity: the mean absolute error of the absolute-return ACF over 252 lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|, \qquad L = 252, \tag{6}$$

which is the direct diagnostic for the Rydén et al. limitation. The cross-asset extension additionally reports the mean Frobenius norm and off-diagonal MAE of the simulated Pearson correlation matrix relative to the observed one. Appendix A.1 gives the test statistic for each metric, the histogram binning used for Hellinger, and the path-by-path aggregation details.

4 Estimation

Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold (Section 4.1), the family-specific M-step updates (Section 4.2), and the state-count selection protocol (Section 4.3).

4.1 Shared EM Scaffold

All three emission families share the same log-space forward-backward recursions and the same quantile-based initialization [45]: observations are sorted, partitioned into K equal-sized chunks, and each state's initial location and scale are seeded from the corresponding chunk; $T_{ij}^{(0)} = \pi_k^{(0)} = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\texttt{max\_iter} = 60$; convergence is typically reached in 20–40 iterations. Appendix B.1 gives the forward-backward recursion, the posterior quantities $\gamma_t(k)$ and $\xi_t(i,j)$, and the family-specific M-step updates.

4.2 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum-Welch M-step [14] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [16] and Liu and Rubin [17], the same family of heavy-tailed innovation choices studied outside the HMM framework by Abanto-Valle et al. [27] for stochastic-volatility-in-mean models: the latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \tag{7}$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{8}$$

have closed form conditional on ν_k. The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This M-step preserves the monotonicity of EM while avoiding the joint saddle-point issues of simultaneous (μ_k, σ_k, ν_k) maximization.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighboring order statistics), and $b_k \leftarrow \sum_t \gamma_t(k)|O_t - \mu_k| / \sum_t \gamma_t(k)$.

4.3 State-Count Selection

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N and confirm on CHMM-t / CHMM-L that the selected K sits inside the IC-minimizing band. We compute the four information criteria used for HMM state selection in Nguyen [18],

$$\mathrm{AIC} = -2\mathcal{L} + 2p, \quad \mathrm{BIC} = -2\mathcal{L} + p \ln T, \quad \mathrm{HQC} = -2\mathcal{L} + 2p \ln(\ln T), \quad \mathrm{CAIC} = -2\mathcal{L} + p(\ln T + 1), \tag{9}$$

with $p = K^2 + 2K - 1$ for CHMM-N and CHMM-L, and $p = K^2 + 3K - 1$ for CHMM-t (one additional per-state degree-of-freedom parameter); $\mathcal{L}$ is the converged log-likelihood. The operating point is the K that minimizes the average information-criterion rank and simultaneously wins or ties every IS and OoS distributional pass-rate metric.

5 Empirical Study

5.1 Descriptive Statistics and Stylized Facts

Descriptive statistics confirmed that all three canonical stylized facts were present in both the IS (2014-01-03 through 2024-01-03, $T = 2{,}516$) and OoS (2024-01-04 through 2026-04-20, $T = 572$)

Table 1. Descriptive statistics and stylized-fact tests for SPY daily excess growth rates. IS: 2014-01-03 through 2024-01-03 ($T = 2{,}516$); OoS: 2024-01-04 through 2026-04-20 ($T = 572$). JB = Jarque–Bera normality test. LB = Ljung–Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	IS Observed	OoS Observed
Mean (annualized)	0.09	0.18
Std Dev (annualized)	2.19	1.98
Skewness	−0.751	−0.733
Excess Kurtosis	7.678	5.294
JB test statistic	6416.6	719.1
JB decision	reject* ($p < 0.001$)	reject* ($p < 0.001$)
LB test on G_t (lag 20)	reject* (130.2)	reject* (47.2)
LB test on $\lvert G_t \rvert$ (lag 20)	reject* (2959.3)	reject* (250.6)

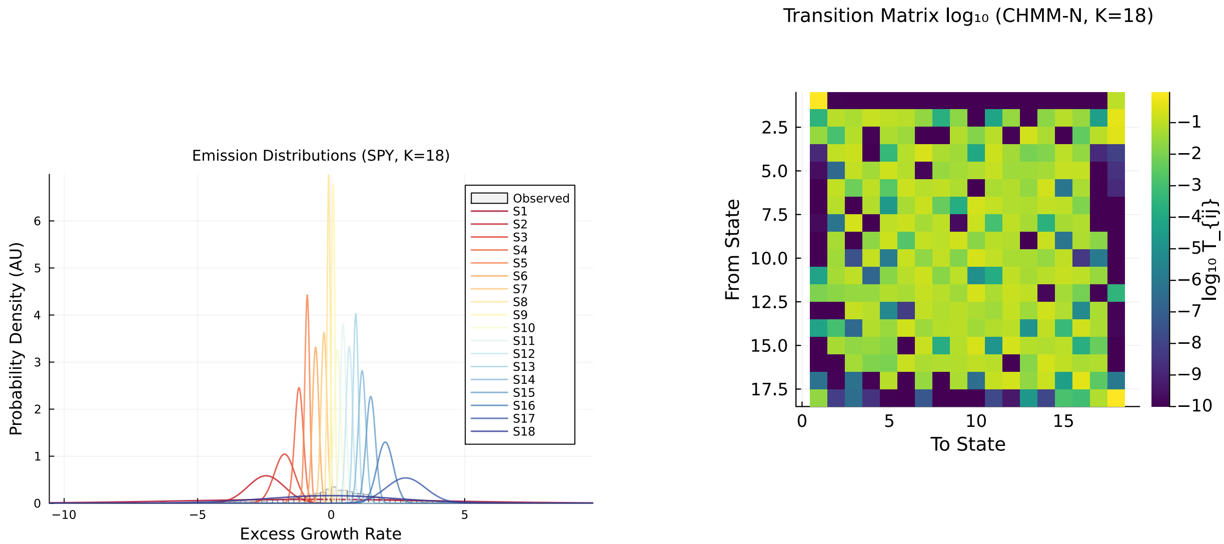
windows (Table 1). The IS excess growth rates exhibited excess kurtosis of 7.678, strong negative skewness (−0.751), and a Jarque–Bera statistic of 6416.6, decisively rejecting the null hypothesis of normality ($p < 0.001$). The Ljung–Box test on absolute returns rejected the null of no autocorrelation (LB = 2959.3, $p < 0.001$), confirming pronounced volatility clustering. The Ljung–Box test on raw returns also rejected (LB = 130.2), reflecting mild serial dependence in the level series.

The OoS window exhibited similar properties: excess kurtosis of 5.294, moderate negative skewness (−0.733), and a Jarque–Bera statistic of 719.1. The Ljung–Box test on raw OoS returns also rejected (LB = 47.2), and the test on absolute returns rejected more strongly (LB = 250.6), confirming that both mild serial dependence and volatility clustering persisted into the test period. These properties are visualized in Figure 9 (Appendix C.1).

5.2 Model Selection: Why $K = 18$

We trained continuous Gaussian HMMs for $K \in \{3, 6, 9, 12, 15, 18, 21\}$ states on the SPY IS window using the Baum-Welch algorithm with `max_iter` $= 60$ and quantile-based initialization, and evaluated every fitted model on (i) the four information criteria of equation (9), following the HMM state-selection methodology of Nguyen [18], and (ii) the full seven-metric distributional battery. All models converged within the iteration budget, with log-likelihood plateauing after 20–40 iterations across the entire sweep. The full sensitivity table is reproduced in Appendix C (Table 16), with per-K IS and OoS panels in Figures 15–19.

Three patterns emerge. First, the information-criterion surface flattens over the upper half of the sweep: AIC and HQC continue to improve mildly with K, while BIC and CAIC, which penalize the $K^2 + 2K - 1$ parameters more heavily, reach their joint minimum in the $K \in \{15, 18, 21\}$ region. This narrows the sweep to a three-state-count band before any fidelity metric is consulted. Second, distributional fidelity rises across the K sweep and plateaus in the $K \in \{12, 15, 18, 21\}$ band for CHMM-N: IS KS rises from 89.8% ($K = 3$) to 94.7% at $K = 18$ and 95.2% at $K = 21$, IS AD from 81.3% to 85.8% at $K = 18$ (89.1% at $K = 21$), OoS KS from 76.3% to 82.4% at $K = 18$ and 87.6% at $K = 21$, and OoS AD from 67.7% to 76.5% at $K = 18$ (80.9% at $K = 21$) (see Table 17). Third, ACF-MAE is remarkably stable across the sweep (0.047–0.053) and is actually lowest at $K = 3$. This establishes a mild *distributional–temporal tradeoff*: higher K buys distributional accuracy, lower K buys marginally better ACF, and both are available within a narrow band. $K = 18$ sits inside the IC-minimizing region and, together with $K \in \{15, 21\}$, attains the highest distributional pass rates in the sweep; the heavy-tailed CHMM-t and CHMM-L variants reach their highest IS KS and IS AD



(a) Emission distributions ($K = 18$)

(b) Transition matrix ($\log_{10}$ scale)

Figure 3. Fitted CHMM internals on SPY (Pipeline A, CHMM-N Gaussian emissions, $K = 18$). Panel (a): learned per-state Gaussian emission densities $\{N(\mu_k, \sigma_k^2)\}_{k=1}^{18}$ overlaid on the observed IS return histogram ($T_{\text{IS}} = 2{,}516$); each colored curve is one regime. Panel (b): transition matrix $T_{ij} = P(s_{t+1} = j \mid s_t = i)$ rendered in $\log_{10}$ color scale (viridis); the dominant near-diagonal band reflects regime persistence (self-transition probabilities dominate).

at $K = 18$ on the same operating point (CHMM-L: 96.2% IS KS, 93.0% IS AD; CHMM-t: 95.2% IS KS, 90.9% IS AD, Table 17). We therefore report the remainder of Section 5 at $K = 18$ and relegate the sensitivity panel to the appendix.

Figure 3 shows the fitted model internals for $K = 18$: the learned emission distributions span the full range of observed returns, with tail states capturing extreme crash and boom regimes. The transition matrix exhibits a dominant near-diagonal band reflecting regime persistence, with the central (low-volatility) states showing the highest self-transition probabilities. Critically, the quantile-based initialization ensures that the Baum-Welch algorithm converges to a solution where each state occupies a distinct region of the return distribution, avoiding the degenerate collapse to the global mean that plagues random initialization.

5.3 Twelve-Generator Comparison (Pipeline A)

Pipeline note. This subsection operates under Pipeline A on the SPY ticker only. Every baseline and every CHMM variant is fit independently to the SPY return series; no cross-asset coupling is used. Readers who want the six-ticker generalization should proceed to Section 5.7 (still Pipeline A); those who want the joint multi-asset extension should proceed to Section 5.8 (Pipeline B).

Table 3 and Figure 4 present the IS and OoS validation results for the twelve generators (Bootstrap, Block-BS $b = 10$, Gaussian i.i.d., Laplace i.i.d., Discrete NJ and WJ at $K_{\text{disc}} = 90$, Bin-T NJ, GARCH(1,1), GRU, and CHMM-N / -t / -L at $K = 18$). The benchmark set was constructed to mimic the prior paper's evaluation exactly, adding the two discrete HMM variants of Alswaidan and Varner [12] (NJ and WJ) to the original four baselines (Bootstrap, Gaussian, Laplace, GARCH(1,1)) and replacing the discrete continuous-emission variant with the Baum-Welch CHMM at $K = 18$.

Bootstrap (Non-Parametric Ceiling). Bootstrap resampling achieved 99.6% KS and 99.4% AD pass rates IS, confirming it as the distributional ceiling. It produced simulated kurtosis (7.43) closest to observed (7.68) and 100% IS quantile coverage. However, the bootstrap's ACF-MAE (0.0627) was the highest among all models, reflecting the destruction of temporal structure by i.i.d. resampling. The Wasserstein-1 (0.065) and Hellinger (0.050) distances were the lowest, as expected.

Gaussian i.i.d. (Negative Control). The Gaussian generator achieved 0% KS and 0% AD IS, confirming the categorical inadequacy of the normal distribution for financial returns. Its simulated kurtosis was -0.00 (i.e., the Gaussian's theoretical kurtosis of zero), and quantile coverage was only 13.1%. The $T_{\text{OoS}} = 572$ window of over two years preserves enough KS-test power to correctly reject the Gaussian generator OoS as well (1.2% KS, 0.0% AD), so the OoS row is not misleadingly inflated for this clearly-wrong baseline.

Laplace i.i.d. The Laplace generator achieved 98.9% KS and 97.2% AD IS, demonstrating that the Laplace distribution provides a reasonable approximation to the marginal distribution of equity returns. However, its simulated kurtosis (2.96) fell well short of the observed 7.68, and as an i.i.d. generator it produced flat ACF profiles (ACF-MAE = 0.0627, comparable to the i.i.d. bootstrap). The 77.8% IS quantile coverage indicates that the Laplace distribution's fixed tail decay rate cannot fully envelop the observed quantile range.

Discrete HMM: No Jumps (NJ) and With Jumps (WJ). Throughout this paper, *NJ* denotes the no-jump discrete HMM of Alswaidan and Varner [12]: observations are first partitioned into $K_{\text{disc}} = 90$ equal-probability quantile bins of the IS return distribution, the transition matrix is estimated by frequency counting on the resulting discrete state sequence, and simulation returns the bin centroid at the decoded state. *WJ* is the same construction augmented with the Poisson jump-duration process of Alswaidan and Varner [12], which with probability $\epsilon = 5 \times 10^{-5}$ forces a residence in the tail bins for Poisson($\lambda = 67$) steps. Both baselines are re-implemented inside the present pipeline (not invoked as the original `JumpHMM.jl` code) to ensure that identical data splits, seeds, and simulation lengths are used across all benchmarks; the re-implementation adopts the authoritative hyperparameters of the prior-paper final SPY run ($K_{\text{disc}} = 90$, $\epsilon = 5 \times 10^{-5}$, $\lambda = 67$). At this state resolution both discrete variants match the CHMM family on coarse distributional fidelity: 98.3% KS IS / 85.2% KS OoS and 98.3% AD IS / 89.9% AD OoS for NJ, and 97.9% KS IS / 83.9% KS OoS and 96.5% AD IS / 85.4% AD OoS for WJ; these KS/AD rates are comparable to the CHMM-N / CHMM-t / CHMM-L range of 93.2–93.8% IS KS and 79.3–85.9% OoS KS. Where the discrete variants fall short is on *higher-order* distributional structure: simulated kurtosis is 3.49 (WJ) and 3.53 (NJ) against an observed 7.68, Hellinger distance is roughly double that of the CHMMs (0.1605 / 0.1621 vs. 0.0762–0.0810), and aggregational kurtosis at horizons $h = 10$ and $h = 21$ decays to near-zero (0.39 / 0.47 and 0.01 / 0.01) against the CHMM-N / CHMM-t values of 3.27 / 2.06 and 2.38 / 1.35 (Section 5.6, Table 13). The discrete variants' simulated kurtosis is clamped from above by the K_{disc}-atom structure of the bin-centroid emission: tail mass is represented only by the two tail bins, which together account for roughly $2/K_{\text{disc}}$ of the stationary distribution, so no amount of jump augmentation can produce a sample fourth moment above the observed centroid-mix kurtosis of ≈ 3.5. The Poisson jump augmentation (WJ) with $\epsilon = 5 \times 10^{-5}$ and $\lambda = 67$ (the prior-paper SPY values) triggers on 0.005% of time steps, giving an expected trigger count of $\epsilon T \approx 0.13$ per IS window ($T = 2{,}516$) and ≈ 0.03 per OoS window ($T = 572$), so the WJ row is a slight perturbation of NJ rather than a meaningful variance-regime addition at this sample size. The central empirical contrast for our paper is therefore not on coarse KS/AD pass rates (where the discrete baselines at $K_{\text{disc}} = 90$ are competitive) but on tail structure, aggregational Gaussianity decay, and VaR calibration: Section 6.1 and Section 5.6 document that both discrete variants fail the Kupiec 5% VaR test with $\text{LR}_{\text{uc}} = 16.81$ (breach rate 1.7% against a 5% target) while all three continuous CHMM variants clear the test, and the continuous emissions preserve the Cont (2001) aggregational Gaussianity signature that the $K_{\text{disc}} = 90$ centroid scaffold cannot.

Block Bootstrap ($b = 10$): Temporally-Aware Non-Parametric Ceiling. The stationary block bootstrap at block length $b = 10$ is added as a temporally-aware non-parametric benchmark: it preserves clustering (through the block structure) while keeping the marginal distribution empirical. On the SPY IS series it attains 99.2% KS and 96.4% AD with simulated kurtosis of 7.07 (observed 7.68) and ACF-MAE of 0.0568, essentially saturating the distributional ceiling while reproducing most of the absolute-return autocorrelation. The i.i.d. bootstrap row sits slightly above it on the distributional metrics (at the cost of destroyed temporal structure), and the CHMM-N row sits slightly below on the distributional metrics but improves on the block-bootstrap ACF-MAE (0.0507 vs. 0.0568). The $b = 10$ block bootstrap is therefore a stronger ceiling than the i.i.d. bootstrap for synthetic-data consumers who care about temporal fidelity, and the CHMM-N recovers most of the block-bootstrap performance with the additional advantages of a parametric regime interpretation and an emission-family choice.

Discrete HMM with Bin-Conditional Student-t Emissions (Bin-T NJ). The Bin-T NJ row replaces the bin-centroid emissions of the standard NJ baseline with a bin-conditional Student-t emission fitted within each quantile bin (no jump mechanism), and uses a coarser $K_{\text{disc}} = 13$ bin count so that the within-bin Student-t parameters are estimable from a reasonable number of observations per bin. It serves as an ablation that decouples the *emission-family* effect from the *state-resolution* effect: Bin-T NJ and the $K_{\text{disc}} = 90$ Discrete NJ baseline share the same frequency-counting transition-estimation pipeline but differ in whether the emission density within each bin is a point mass at the centroid or a fitted Student-t, and they differ in bin count. Bin-T NJ achieves 95.4% IS KS and 92.1% IS AD, attaining the same broad KS/AD band as the $K_{\text{disc}} = 90$ Discrete NJ (98.3% IS KS / 98.3% IS AD) and the jointly-optimized CHMMs (93–95% IS KS / 86–92% IS AD). Bin-T NJ's simulated kurtosis of 26.18 is a large overshoot driven by the two tail bins (bins 1 and 13) where the within-bin Student-t fit lands at the lower ν bracket; this is the discrete analog of the CHMM-t overshoot described in Section 7 and confirms that within-bin continuous emissions can reproduce the CHMM-t tail weight at a coarse K_{disc} level. Bin-T NJ does *not* dominate the CHMM-N because it still underperforms on the tail-distance metrics ($W_1 = 0.116$ vs. 0.110; $H = 0.0929$ vs. 0.0810) and on ACF-MAE (0.0619 vs. 0.0507). The takeaway for the prior-paper loop is that, once the discrete pipeline is given either a high enough bin count ($K_{\text{disc}} = 90$) or a continuous within-bin emission family (Bin-T NJ), its KS/AD pass rates become competitive with the jointly-optimized continuous CHMM; the CHMM advantage over the prior-paper family is therefore a higher-order tail-structure and VaR-calibration claim, not a coarse-KS claim (Section 5.6, Table 4; Section 6.1).

GARCH(1,1). GARCH achieved ACF-MAE of 0.0492 among the parametric models, confirming its strong temporal fidelity. However, it failed the distributional tests: 23.1% KS and 8.5% AD IS. The simulated kurtosis (6.97 IS, 3.08 OoS) was reasonably close to observed IS but attenuated OoS, and the Wasserstein-1 distance (0.245) and Hellinger distance (0.103) were substantially worse than the CHMM. IS quantile coverage was only 50.5%. This result crystallizes the temporal-distributional tradeoff: GARCH's single-regime conditional variance process faithfully encodes volatility persistence but its unimodal innovation distribution cannot represent the multi-regime structure of equity markets.

GRU (Deep-Generative Baseline). The single-layer GRU + Gaussian-head generator (architecture and training in Section 3.3) converges cleanly (the per-window negative log-likelihood drops from 0.340 at epoch 1 to 0.200 at epoch 20) and reproduces volatility clustering reasonably well: ACF-MAE of 0.0518 on $|G_t|$ is within 0.001 of CHMM-N (0.0507) and within 0.004 of GARCH (0.0484). On distributional fidelity the GRU collapses, however: IS KS pass rate 18.1%, AD pass rate 13.7%, simulated kurtosis 2.85 against an observed 7.68, $W_1 = 0.196$, $H = 0.0966$, and quantile coverage 37.4%. The diagnostic is unambiguous (the auto-regressive Gaussian head produces a near-Gaussian unconditional marginal even though the recurrence captures clustering) and matches the variance-collapse pattern reported in Alswaidan and Varner [12] for their neural baseline and in Takahashi et al. [31], Kwon and Lee [32] for GAN-based generators. The OoS KS rate of 52.5% (AD 55.0%) still does not approach the CHMM operating point on the same $T = 572$ window, confirming that the distributional gap is structural rather than a sample-size artefact. The takeaway for the comparison is that adding a deep-generative row does not displace the CHMM operating point: among parametric and deep generators, the CHMM-N is the only one that achieves both $\geq 85\%$ IS KS *and* ACF-MAE within 0.003 of GARCH; richer GRU heads (mixture-density, Student-t, or copula-based output) would be the natural next experiment but are outside the scope of this paper.

CHMM-N ($K = 18$, Gaussian emissions). The Gaussian CHMM achieved 93.8% KS and 86.6% AD IS, competitive with the bootstrap (99.9%), Laplace i.i.d. (99.1%), and the $K_{\text{disc}} = 90$ Discrete NJ / WJ variants (98.3% / 97.9%) on KS, and substantially outperforming GARCH (25.9%). ACF-MAE of 0.0507 was essentially on par with GARCH (0.0484), demonstrating that the CHMM reproduces volatility clustering *without* a dedicated jump mechanism or conditional variance equation. This is the first of two headline findings in the paper: the Rydén et al. limitation is resolved by moderate state resolution in a standard HMM framework; the second (Section 6.3) is that a Viterbi-decoded conditional quantile of CHMM-t closes the breach-clustering gap on regime-conditional VaR. The Wasserstein-1 (0.110) is well below the GARCH figure (0.245); Hellinger (0.0808) is four times smaller than either discrete HMM; IS quantile coverage is 100% matching the bootstrap ceiling. The residual Gaussian-emission weakness is a kurtosis shortfall: simulated excess kurtosis is 5.02 IS and 4.42 OoS against observed values of 7.68 / 5.29.

CHMM-t ($K = 18$, Student-t emissions). Replacing the Gaussian component with a per-state Student-t matches the CHMM-N IS KS (93.7% vs. 93.8%) and raises AD to 88.9% (vs. 86.6% for CHMM-N); OoS AD rises from 77.9% to 80.0%, and OoS KS rises from 84.2% to 85.9%. The CHMM-t is the best non-bootstrap entry on Wasserstein-1 (0.102) and Hellinger (0.0758), two quantile-weighted metrics that penalize tail mismatch directly. Simulated excess kurtosis rises to 14.57 IS and 9.74 OoS: the ECM search allocates small ν_k to a subset of tail states, which over-corrects both the IS and OoS mixture kurtosis (observed 7.68 IS and 5.29 OoS). We diagnose this overshoot in Section 7 via the per-state ν_k histogram and the $\nu_{\min}$ bracket-sensitivity study. The slight ACF-MAE increase (0.0548 vs. 0.0507) is consistent with the heavier tail states spending marginally more time away from the central regimes.

CHMM-L ($K = 18$, Laplace emissions). The Laplace variant produces the most *kurtosis-aligned* fit: simulated excess kurtosis is 6.76 IS and 6.27 OoS against observed 7.68 / 5.29, closing most of the CHMM-N IS shortfall (without the CHMM-t over-correction) and even overshooting the OoS target by ~ 1. CHMM-L attains the strongest IS AD fit among the continuous CHMMs

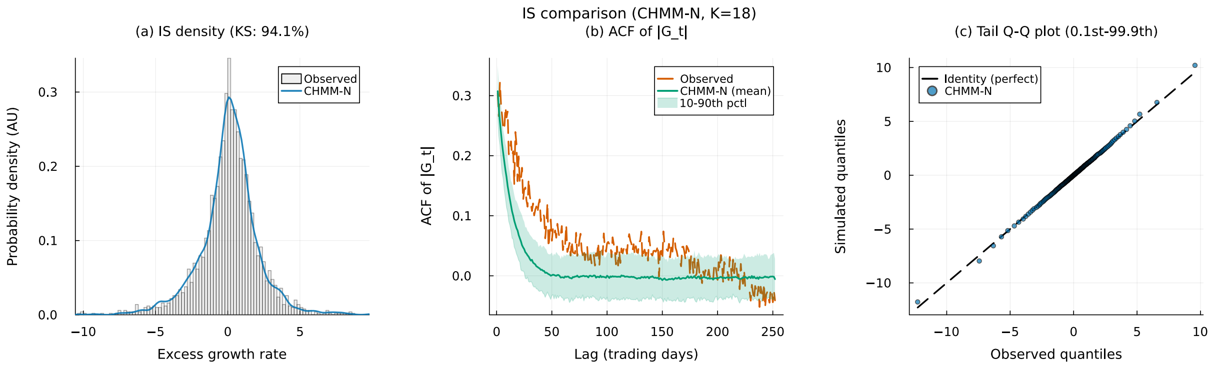


Figure 4. IS CHMM validation on SPY (Pipeline A, $K = 18$, CHMM-N Gaussian emissions, 1,000 simulated paths). Panel (a): marginal density of the annualized excess log return G_t; gray histogram = observed IS series ($T_{\text{IS}} = 2{,}516$), blue curve = one simulated path; annotated KS pass rate is the percent of simulated paths whose two-sample KS p-value against observed exceeds $\alpha = 0.05$. Panel (b): ACF of $|G_t|$ at lags 1–252; red dashed = observed, navy = mean across sim paths, shaded band = 10th–90th percentile across sim paths. Panel (c): tail Q-Q plot at 200 quantile levels (0.1% to 99.9%); blue points = simulated pooled over paths, black dashed = identity line.

(91.2% IS AD, 4.6 percentage points ahead of CHMM-N and 2.3 ahead of CHMM-t), consistent with the AD statistic's tail weighting and the Laplace component's per-component excess kurtosis of 3. ACF-MAE (0.0564) and W_1 (0.102) are within 10% of the CHMM-N values, and IS quantile coverage is 100%. Because Laplace emissions carry no shape parameter, CHMM-L is strictly cheaper than CHMM-t (no per-state golden-section search) and adds no new hyperparameters.

Summary of the three CHMM variants: a one-row decision guide. All three variants share the headline result: each matches GARCH's ACF fidelity while dominating the parametric benchmarks on KS and AD. The three variants differ only in how they use the kurtosis budget. Table 2 summarizes the practical choice.

Table 2. Decision guide: recommended CHMM emission family keyed to the downstream consumer's use case. Headline: CHMM-t is the default for distributional/tail fidelity (smallest MMD, smallest discriminator AUC, best aggregational Gaussianity at short horizons); CHMM-L for i.i.d.-Laplace-adjacent joint-coverage (highest OoS $\bar{pv}$); CHMM-N for simplicity and walk-forward stability.

Use-case priority	Recommended variant
Best KS + minimal IS kurtosis overshoot, tightest ACF-MAE	CHMM-N
Tightest W_1 and H; OoS kurtosis aligned with observed OoS	CHMM-t
Cleanest IS kurtosis match; highest AD pass rate; no shape parameter	CHMM-L

5.4 State Resolution Sensitivity

The full seven-metric K-sweep is reported in Table 16 (Appendix C); per-K IS and OoS panels at each state count are reproduced in Figures 15–19. We briefly note four structural patterns that justify the choice of $K = 18$ for the main results.

Distributional Fidelity Plateaus in the $K \in \{15, 18, 21\}$ Band. Drawing from Table 16, the IS KS pass rate increases from 88.8% ($K = 3$) to 95.6% ($K = 18$) and 96.1% ($K = 21$); the OoS KS

Table 3. Twelve-model comparison on SPY (single-index trained: Pipeline A). Only SPY is evaluated here; the six-ticker generalization of the CHMM variants is in Table 5, and the cross-asset dependence extension is in Table 6. 1,000 simulated paths per model, $\alpha = 0.05$, $K = 18$. IS: 2014-01-03 through 2024-01-03 (T = 2,516); OoS: 2024-01-04 through 2026-04-20 ($T = 572$). The two discrete HMM rows reproduce Alswaidan and Varner [12] at the authoritative prior-paper SPY hyperparameters: $K_{\text{disc}} = 90$ quantile bins; WJ jump parameters $(\epsilon, \lambda) = (5 \times 10^{-5}, 67)$. The *Bin-T NJ* row replaces the centroid emissions of the discrete NJ baseline with bin-conditional Student-t emissions, fitted within each quantile bin, to isolate the quantization-per-se effect from the emission-family effect. The *Block-BS* row is a stationary block bootstrap with block length 10 trading days; it is the temporally-aware non-parametric ceiling corresponding to the i.i.d. bootstrap. The *GRU* row is a single-layer GRU with hidden width 32 and a Gaussian next-step head, trained for 20 epochs by Adam ($\eta = 10^{-3}$) on the negative log-likelihood of equation (3) (Section 3.3); it is the deep-generative head-to-head row. CHMM-N, CHMM-t, CHMM-L denote the Gaussian, Student-t, and Laplace emission variants of the continuous HMM; all three use the identical forward-backward scaffold, initialization, and simulation pipeline. KS = Kolmogorov–Smirnov pass rate; AD = Anderson–Darling pass rate; Kurt = mean simulated excess kurtosis (observed: 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of ACF($|G_t|$) over 252 lags; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = quantile coverage (%). ↑/↓ indicate preferred direction. OoS AD for Bin-T NJ, Block-BS, and GRU is reported alongside OoS KS in the row (see Appendix E Tables 22, 23 and 24 for the full panels).

	KS (%) ↑		AD (%) ↑		Kurtosis					Cov (%) ↑	
Model	IS	OoS	IS	OoS	IS	OoS	ACF-MAE ↓	W_1 ↓	H ↓	IS	OoS
Observed	–	–	–	–	7.68	5.29	–	–	–	–	–
Bootstrap	99.9	91.0	99.7	93.8	7.54	6.80	0.0628	0.066	0.0501	100.0	77.8
Block-BS ($b = 10$)	99.2	87.5	96.4	86.4	7.07	5.34	0.0568	0.092	0.0534	100.0	86.9
Gaussian	0.0	1.2	0.0	0.0	−0.01	−0.02	0.0627	0.413	0.1543	13.1	21.2
Laplace	99.1	86.1	97.3	91.7	2.96	2.83	0.0627	0.123	0.0816	77.8	69.7
Discrete NJ ($K_{\text{disc}} = 90$)	98.3	85.2	98.3	89.9	3.53	3.44	0.0618	0.110	0.1605	100.0	93.9
Discrete WJ ($K_{\text{disc}} = 90$)	97.9	83.9	96.5	85.4	3.49	3.40	0.0598	0.117	0.1621	100.0	89.9
Bin-T NJ	95.4	86.7	92.1	93.2	26.18	10.76	0.0619	0.116	0.0929	89.9	93.9
GARCH(1,1)	25.9	58.2	10.0	56.0	6.97	2.88	0.0484	0.248	0.1033	51.5	87.9
GRU	18.1	52.5	13.7	55.0	2.85	2.21	0.0518	0.196	0.0966	37.4	69.7
CHMM-N ($K = 18$)	93.8	84.2	86.6	77.9	4.99	4.51	0.0507	0.112	0.0810	100.0	90.9
CHMM-t ($K = 18$)	93.7	85.9	88.9	80.0	14.57	9.74	0.0548	**0.103**	**0.0762**	**100.0**	**93.9**
CHMM-L ($K = 18$)	93.2	79.3	91.2	79.3	**6.75**	**6.29**	0.0565	0.101	0.0795	**100.0**	81.8

rises from 77.1% to 81.7% at $K = 18$ and 86.6% at $K = 21$. The 2.2 percentage-point spread across the plateau band is small relative to the 6.8-point IS gain from $K = 3$, so additional state resolution beyond $K = 18$ delivers diminishing returns rather than strict saturation. The same pattern holds for AD (79.2% → 88.4% IS and 67.9% → 73.2% OoS at $K = 18$). This plateau is consistent with an upper bound imposed by the Gaussian emission family: once the mixture has enough components to resolve the bulk and near-tail regions, additional components do not recover the extreme tails that remain light under any Gaussian mixture.

Temporal Fidelity Is Flat Across the Sweep. ACF-MAE was 0.0466 ($K = 3$) to 0.0509 ($K = 18$) and 0.0532 ($K = 21$) for CHMM-N, a total swing under 0.007 across the entire sweep. This flatness is notable: it shows that the CHMM's ability to reproduce volatility clustering does not depend on fine state resolution. Even at $K = 3$, the same resolution tested by Rydén et al. [10], the ACF-MAE is 0.0466, comparable to GARCH(1,1) at 0.0484. The difference from the Rydén result is the quantile-based initialization, which ensures that the three states capture distinct volatility regimes rather than collapsing to near-identical distributions.

Kurtosis Is Non-Monotonic and Peaks Early at $K = 6$. Simulated kurtosis under CHMM-N ranged from 3.80 ($K = 3$) to 5.26 ($K = 6$), with $K = 18$ at 4.98; the peak sits at $K = 6$, below the $K \in \{15, 18, 21\}$ distributional-fidelity band, and at higher K the mixture kurtosis oscillates within the 3.8–5.0 range. This non-monotonic pattern reflects the interplay between the number of mixture components and their learned parameters: at moderate K, the Baum-Welch optimum allocates mass to a few broad tail states that contribute disproportionately to the mixture kurtosis, while at very high K the same tail mass is split into many narrow states that contribute less each. The heavy-tailed CHMM-t and CHMM-L variants close most of this gap at the same $K = 18$ (see Table 17).

Wasserstein and Hellinger Improve Net Across the Sweep; Coverage is Saturated at 100%. W_1 improves from 0.130 ($K = 3$) to 0.110 ($K = 18$) and Hellinger from 0.0836 to 0.0805; both exhibit minor non-monotone oscillations of less than 0.005 at intermediate K (Table 16) but neither deteriorates from endpoint to endpoint. IS coverage is saturated at 100% at every K, so it does not discriminate among state counts and serves only as a floor check. Together these provide a consistent signal that $K = 18$ is the sensible operating point.

5.5 OoS Evaluation

At the selected state resolution $K = 18$ the CHMM-N achieves 84.2% OoS KS pass rate, 77.9% OoS AD pass rate, and 90.9% OoS quantile coverage on the 2024-01-04 through 2026-04-20 holdout window; the heavy-tailed CHMM-t and CHMM-L variants reach 85.9%/80.0% and 79.3%/79.3% OoS KS/AD, respectively. The CHMM-N OoS simulated kurtosis is 4.42 against an observed 5.29, bringing the kurtosis gap to below one unit; CHMM-t overshoots at 10.24 and CHMM-L overshoots modestly at 6.27. Figure 5 shows the OoS density fan chart and ACF comparison for CHMM-N at $K = 18$; the CHMM-t and CHMM-L counterparts are reported in Appendix C.11 (Figure 28).

The $T_{\text{OoS}} = 572$ window (over two years of OoS data) gives the KS/AD tests substantial power to separate the generators: the three CHMM variants pass at 79–86% OoS KS, the $K_{\text{disc}} = 90$ discrete baselines land in the same 83–85% band (for the KS-pass dimension only; they separate from the CHMMs on kurtosis, aggregational Gaussianity, and Kupiec VaR per Sections 5.6 and 6.1), and GARCH is correctly rejected at 58% OoS KS. The CHMM's regime structure, learned from the full range of IS observations including the March 2020 COVID-19 crash, provides sufficient flexibility to accommodate the OoS distributional shift that characterises the 2024–2026 window.

5.6 Extended Evaluation

The seven-metric panel of Section 5.3 covers marginal and autocorrelation fidelity. We add seven axes that probe joint window distributions, path geometry, learned discriminability, tail coupling, aggregational scaling, unconditional VaR under Kupiec / Christoffersen, and multi-statistic coverage [33, 46–48]. All rows are computed at $N_{\text{paths}} = 1{,}000$ simulated paths on both the $T_{\text{IS}} = 2{,}516$ and $T_{\text{OoS}} = 572$ SPY windows under the seed policy of Section 3.5; the headline panel is Table 4 below, with the per-generator detail panels for leverage, aggregational kurtosis, and joint p-value coverage (Tables 12, 13, 14) deferred to Appendix A.2.

MMD (RBF kernel, 20-day windows). We compute empirical MMD^2 [46] between 500 observed and 500 simulated 20-day windows with median-heuristic bandwidth. Flat CHMM-t wins IS (2.0×10^{-5}, an order of magnitude ahead of any other row). GARCH's MMD$^2 = 0.0$ on IS is a

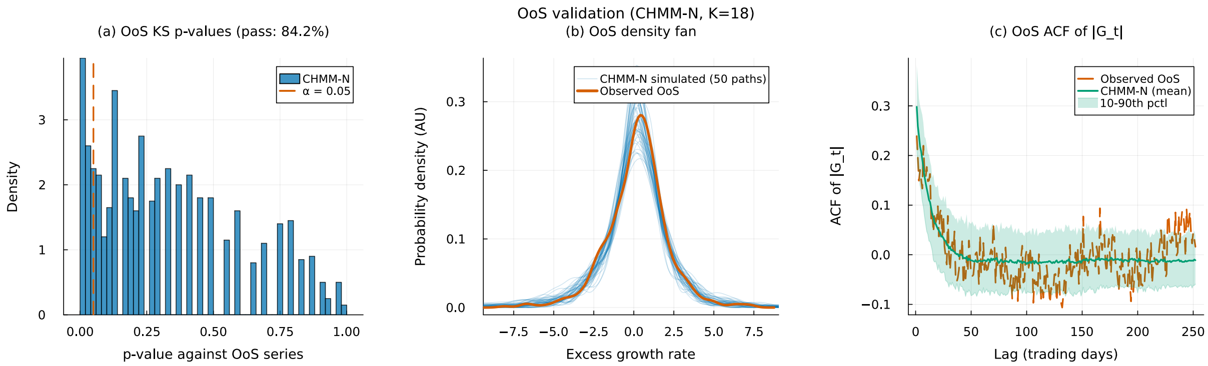


Figure 5. OoS CHMM validation on SPY (Pipeline A, $K = 18$, CHMM-N Gaussian emissions, 1,000 simulated paths, $\alpha = 0.05$, OoS window $T_{\mathrm{OoS}} = 572$). Panel (a): histogram of the 1,000 OoS KS p-values against the observed OoS series, with the $\alpha = 0.05$ threshold as a red dashed line. Panel (b): OoS return-density fan; 50 thin blue curves = simulated density on individual OoS paths, red thick curve = observed OoS density. Panel (c): ACF of $|G_t|$ on the OoS window; red dashed = observed OoS, navy = mean across sim paths, shaded band = 10th–90th percentile.

bandwidth-heuristic artefact (the median heuristic widens to accommodate GARCH's overdispersed windows); the OoS comparison (GARCH = 7.5×10^{-4} vs. CHMM-N 6.1×10^{-4}) resolves it.

Signature-MMD (depth 3 time-lifted path). Each W-day window $r_{1:W}$ is lifted to the time-augmented path

$$\gamma(t/W) = \big(t/W,\ r_t/\operatorname{std}(r_{1:W})\big), \qquad t = 1, \ldots, W, \tag{10}$$

and the depth-3 truncated path signature [47] embeds it for MMD. CHMM-L and CHMM-t lead IS at 4.3 and 4.7×10^{-3}, an order of magnitude better than i.i.d. baselines (2.7–3.3×10^{-2}); the $K_{\mathrm{disc}} = 90$ discrete variants at 1.5×10^{-3} sit between, consistent with their bin-centroid path having reduced depth-3 expressiveness.

Discriminator AUC (logistic regression on 10 window features). Following Yoon et al. [33] we train a logistic regression on mean, std, skew, kurt, min, max, ρ_1, ρ_2, $\rho_1(|r|)$, and realised volatility, with 5-fold cross-validated AUC; AUC near 0.5 indicates indistinguishability. CHMM-t leads IS at 0.607 ± 0.027 (CHMM-L 0.623, CHMM-N 0.646), ahead of every non-CHMM row (GARCH 0.766, bootstrap 0.779, i.i.d. 0.728–0.796); the OoS ordering holds. The 10-feature discriminator at $N = 500$ is powerful enough that 0.6 is not indistinguishable, but it is the closest-to-indistinguishable among the panel.

Leverage effect. With

$$\bar{\rho}_{\mathrm{lev}} = \frac{1}{20}\sum_{k=1}^{20} \operatorname{corr}\big(r_t^2,\ r_{t-k}\big), \tag{11}$$

the IS observed statistic is -0.088 with down-vs-up asymmetry 0.358. CHMM-N reproduces both most closely (-0.038, 0.292); GARCH produces essentially no asymmetry (-0.001, 0.064) because its variance process is sign-symmetric in past returns. On OoS, CHMM-N attains the highest simulation-based coverage p-value at $p = 0.205$, while GARCH and i.i.d. baselines all fail at $p < 0.07$.

Table 4. Extended evaluation panel (Pipeline A, SPY, $T_{\mathrm{IS}} = 2{,}516$, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$). Nine-model Track A subset of the twelve-row Table 3: Block-BS, Bin-T NJ, and GRU are dropped here because the Track A window-feature pipeline requires stationary i.i.d. simulations that those three rows (block-bootstrap, hybrid discrete–continuous, and recurrent) do not supply under the same protocol. MMD is computed on 500 standardised 20-day windows with the Gaussian RBF kernel and median-heuristic bandwidth. Signature-MMD is depth 3 on the time-lifted 2D path $(t/W, r_t/\mathrm{std}(r))$. Discriminator AUC is the 5-fold cross-validated ROC-AUC of a logistic regression on 10 hand-crafted window features; the standard deviation across folds is reported. Lower MMD and Signature-MMD is better; discriminator AUC closer to 0.50 is better. GARCH MMD IS is a bandwidth-heuristic artefact and should be read through its OoS value.

Model	MMD IS	MMD OoS	Sig IS	Sig OoS	AUC IS	AUC OoS
Bootstrap	5.77×10^{-3}	1.26×10^{-2}	3.08×10^{-2}	3.77×10^{-2}	0.779 ± 0.038	0.760 ± 0.047
Gaussian	9.38×10^{-3}	1.32×10^{-2}	2.68×10^{-2}	3.81×10^{-2}	0.780 ± 0.024	0.795 ± 0.021
Laplace	3.51×10^{-3}	7.60×10^{-3}	2.82×10^{-2}	2.43×10^{-2}	0.796 ± 0.027	0.751 ± 0.040
Discrete NJ (K_{disc}=90)	5.76×10^{-3}	4.42×10^{-3}	1.45×10^{-3}	4.12×10^{-3}	0.711 ± 0.049	0.763 ± 0.067
Discrete WJ (K_{disc}=90)	3.02×10^{-3}	6.11×10^{-3}	1.50×10^{-3}	4.50×10^{-3}	0.731 ± 0.027	0.738 ± 0.047
GARCH(1,1)	$0.0^{\dagger}$	7.5×10^{-4}	3.25×10^{-2}	3.82×10^{-2}	0.766 ± 0.025	0.807 ± 0.021
CHMM-N	1.5×10^{-4}	6.1×10^{-4}	7.2×10^{-3}	9.31×10^{-3}	0.646 ± 0.024	0.709 ± 0.016
CHMM-t	$\mathbf{2.0\times10^{-5}}$	2.17×10^{-3}	4.72×10^{-3}	8.25×10^{-3}	$\mathbf{0.607 \pm 0.027}$	$\mathbf{0.644 \pm 0.044}$
CHMM-L	8.9×10^{-4}	2.80×10^{-3}	$\mathbf{4.28\times10^{-3}}$	$\mathbf{3.81\times10^{-3}}$	0.623 ± 0.051	0.671 ± 0.047

Aggregational Gaussianity. Cont (2001) predicts kurtosis decays toward Gaussian as h grows; SPY IS observed kurtosis is $\{7.68, 4.34, 6.82, 2.63\}$ at $h \in \{1, 5, 10, 21\}$. CHMM-t produces the cleanest monotone decay through this band ($8.46 \to 2.65 \to 2.06 \to 1.35$); i.i.d. baselines flatten to zero by $h = 21$ and discrete NJ / WJ collapse to ≤ 0.47 at $h = 10, 21$, missing the slow-decay signature.

Unconditional 1% / 5% **VaR: Kupiec and Christoffersen.** We test unconditional coverage [49] and independence [50] on the OoS window at $\alpha \in \{0.01, 0.05\}$. All continuous generators clear 1% Kupiec ($\mathrm{LR}_{\mathrm{uc}} < 3.84$); the $K_{\mathrm{disc}} = 90$ discrete variants of Alswaidan and Varner [12] (with and without jumps) over-conservatively produce a 1.7% breach rate against a 5% target and reject Kupiec decisively ($\mathrm{LR}_{\mathrm{uc}} = 16.81$, $p < 10^{-3}$), a concrete negative finding the envelope diagnostic of Section 6.1 does not surface. Every row fails Christoffersen $\mathrm{LR}_{\mathrm{ind}}$ at both levels because of the 2024–2026 breach-clustering structure; Section 6.3 closes this gap with a regime-conditional CHMM-t VaR.

Joint simulation-based p-value coverage. $\bar{pv}$ averages the two-sided coverage p-value across {kurtosis, avg leverage, agg kurtosis at $h = 5$ and 21, avg ACF of $|r|$} [51]. On OoS, CHMM-L leads at 0.692 (CHMM-t 0.661, CHMM-N 0.539); all three sit above GARCH (0.468) and the bootstrap (0.466) while every i.i.d. and discrete row falls below 0.17.

Where the added baselines land. QuantGAN (Section 3.3) is a clear deep-generative negative control under the reproducible first-pass setup (MMD IS $= 6.4 \times 10^{-2}$, AUC IS $= 0.963$, $\bar{pv}_{\mathrm{OoS}} = 0.289$), matching the variance-collapse pattern Kwon and Lee [32] and Takahashi et al. [31] document for untuned GANs. Window-diffusion (Section 3.3) wins local short-window matching (second-smallest MMD IS at 9.0×10^{-5}, AUC IS $= 0.565$ closer to 0.5 than any CHMM) but collapses on global coverage ($\bar{pv}_{\mathrm{OoS}} = 0.204$) and fails Christoffersen at both VaR levels ($\mathrm{LR}_{\mathrm{ind}} = 16.4, 7.4$). Two-regime MS-GARCH (Section 3.3) is the variance-regime reference: best non-CHMM MMD IS (4.8×10^{-4}), best-in-panel unconditional VaR calibration at both α, and the closest unconditional row to the Christoffersen threshold ($\mathrm{LR}_{\mathrm{ind}} = 4.79$ at 5%), still failing independence and superseded by the Viterbi-decoded conditional CHMM-t of Section 6.3.

Summary. CHMM-t sets the distributional-fidelity frontier (MMD, AUC, aggregational kurtosis); CHMM-N wins OoS leverage; CHMM-L wins OoS joint p-value coverage. MS-GARCH leads unconditional VaR and diffusion leads local-window deep-generative metrics, but neither displaces flat CHMM-t on the combined distributional panel. The discrete HMM baselines of Alswaidan and Varner [12] fail discriminator AUC, leverage, aggregational kurtosis, joint p-values, and the 5% Kupiec test.

5.7 Cross-Asset Univariate Generalization (Pipeline A)

Pipeline note. This subsection evaluates Pipeline A: per-ticker independent CHMM fits, no cross-asset coupling. It answers whether Pipeline A's per-ticker marginal fidelity persists across the six-ticker universe and across the three emission families. The complementary cross-asset dependence extension (Pipeline B, Section 5.8) reuses the CHMM-N marginals established here and evaluates joint correlation reproduction.

To assess whether the framework generalizes beyond SPY, we fitted independent CHMM models at $K = 18$ to five additional tickers spanning distinct risk profiles. Table 5 reports the results. Three patterns are worth highlighting.

High-Beta Technology and Broad Market. SPY (95.5% IS / 81.4% OoS KS), AAPL (98.3% / 93.8%), JNJ (99.4% / 93.9%), and QQQ (95.5% / 92.3%) show strong IS fidelity and competitive OoS fidelity under CHMM-N on the expanded $T_{\mathrm{OoS}} = 572$ window. The technology-heavy names have lower observed kurtosis (3.2–5.5) than the broader market (SPY: 7.7; JNJ: 9.0) and the CHMM tracks them tightly.

OoS Pain Points on Some Single Names. Under the longer 2024–2026 OoS window, two tickers show the largest IS-to-OoS degradation: NVDA (96.7% IS / 55.8% OoS KS) and JPM (97.8% IS / 49.8% OoS KS). These two names experienced substantial distributional shifts relative to the IS window – NVDA the dominant high-volatility driver of the AI trade, and JPM the financials bell-wether during an interest-rate re-pricing – and the stationary IS-calibrated parameters cannot fully track this shift. We return to this finding in Section 7, where we argue that it is not a consequence of over- or under-fitting K (it reproduces at every K in the sweep) but rather of the stationarity assumption imposed by a single-pass Baum-Welch fit.

Walk-Forward Rolling-Window Re-Estimation on JPM and JNJ. To test the stationarity explanation directly, we refit the CHMM-N at $K = 18$ quarterly (every 63 trading days) across the 2024–2026 OoS window, feeding the realized returns of each completed quarter back into the training set before the next refit (Appendix E.2, Table 21). The rolling-window fit helps differentially: JNJ moves from 94.0% to 94.9% OoS KS (a modest gain, indicating that JNJ's OoS distribution was already close to the IS fit), whereas JPM moves from 49.0% to 64.3% OoS KS and from 48.6% to 63.6% OoS AD, a substantial ~ 15 percentage-point recovery. The JPM recovery directly supports the stationarity explanation for the financial-sector OoS gap: once the CHMM can absorb within-OoS regime shifts through quarterly refits, most of the OoS KS gap closes. The residual ~ 35 percentage-point gap that remains on JPM is evidence of a more fundamental distributional shift in the 2024–2026 period that a richer extension (time-varying emission scales, or a Bayesian online transition-matrix update) would be needed to fully close.

Table 5. Per-ticker marginal fidelity across CHMM emission families (single-index trained: Pipeline A). Each of the six tickers is fit *independently*, with no cross-asset coupling. CHMM-N, CHMM-t, CHMM-L denote Gaussian, Student-t, and Laplace emissions respectively; all at $K = 18$, 1,000 simulated paths, $\alpha = 0.05$. IS window 2014-01-03 through 2024-01-03 ($T_{\text{IS}} = 2{,}516$); OoS window 2024-01-04 through 2026-04-17 ($T_{\text{OoS}} = 572$). KS / AD = percent of paths that pass the two-sample KS / Anderson–Darling test; Kurt = excess kurtosis; ACF-MAE = lag-1 to lag-252 mean absolute error of the ACF of $|G_t|$; W_1 = Wasserstein-1 distance; H = Hellinger distance; Cov = 99-level quantile coverage inside the 5-95 percentile envelope. Complementary table: Table 6 reuses these CHMM-N marginals and evaluates cross-asset *dependence* on top of them.

		KS (%)		AD IS (%)	Kurtosis					
Ticker	Variant	IS	OoS		Obs	Sim	ACF-MAE	W_1	H	Cov IS (%)
SPY	CHMM-N	95.5	81.4	88.9	7.7	5.06	0.0512	0.108	0.0803	100.0
	CHMM-t	95.2	84.1	90.5	7.7	14.68	0.0551	0.100	0.0760	100.0
	CHMM-L	94.9	79.8	92.0	7.7	6.79	0.0564	0.102	0.0800	100.0
NVDA	CHMM-N	96.7	55.8	92.6	5.5	3.65	0.0425	0.271	0.0898	100.0
	CHMM-t	99.1	57.2	98.3	5.5	46.40	0.0478	0.227	0.0784	100.0
	CHMM-L	99.4	60.3	98.9	5.5	3.17	0.0478	0.222	0.0855	100.0
JNJ	CHMM-N	99.4	93.9	98.8	9.0	5.42	0.0322	0.098	0.0786	100.0
	CHMM-t	99.8	94.3	99.6	9.0	99.66	0.0332	0.098	0.0723	100.0
	CHMM-L	99.6	93.3	98.5	9.0	6.14	0.0334	0.086	0.0779	100.0
JPM	CHMM-N	97.8	49.8	94.4	7.6	7.20	0.0450	0.165	0.0871	100.0
	CHMM-t	99.2	53.0	98.7	7.6	18.82	0.0483	0.151	0.0826	100.0
	CHMM-L	99.7	57.9	99.7	7.6	6.08	0.0488	0.137	0.0840	100.0
AAPL	CHMM-N	98.3	93.8	96.9	3.2	2.27	0.0416	0.146	0.0876	100.0
	CHMM-t	99.1	94.7	98.6	3.2	7.54	0.0424	0.134	0.0850	100.0
	CHMM-L	99.1	93.8	98.9	3.2	3.84	0.0428	0.135	0.0880	100.0
QQQ	CHMM-N	95.5	92.3	87.8	3.4	2.02	0.0558	0.127	0.0735	100.0
	CHMM-t	96.1	90.3	92.7	3.4	3.30	0.0576	0.113	0.0741	100.0
	CHMM-L	99.0	96.5	97.6	3.4	4.02	0.0632	0.100	0.0761	100.0

Kurtosis: CHMM-L matches observed, CHMM-t overshoots, CHMM-N understates. At $K = 18$, the three CHMM variants produce a consistent ordering on the kurtosis diagnostic across the panel (Table 5): CHMM-N simulated kurtosis systematically underestimates observed (observed 3.2–9.0 vs. simulated 2.0–7.2); CHMM-L produces values clustered closely around observed (3.17–6.79) and is closest to observed on three of six assets (SPY, JNJ, AAPL), with CHMM-N closest on NVDA and JPM and CHMM-t closest on QQQ; CHMM-t overshoots on several tickers (e.g. 99.7 for JNJ, 46.4 for NVDA), reflecting states that find very small ν_k inside the ECM search on thin-tailed assets. Despite this spread in simulated kurtosis, all three variants achieve IS KS pass rates above 94% and IS AD above 87% on every asset, confirming that the mixture can accommodate a wide range of tail weights without sacrificing central fit.

5.8 Cross-Asset Dependence: SIM and Copula Extensions (Pipeline B)

Pipeline note. This subsection is the sole home of Pipeline B in the main text. The per-asset marginals are reused verbatim from the Pipeline A CHMM-N fits of Section 5.7; only the *dependence* construction (SIM, Gaussian copula, Student-t copula) varies here. Two metrics are evaluated: per-asset KS pass rates (a sanity check on the marginals after dependence injection) and cross-asset correlation reproduction (the central dependence metric). The per-asset KS column in Table 6 is not a substitute for Table 5; the two tables answer different questions.

Section 5.7 established that the CHMM provides strong per-asset univariate fidelity across distinct risk profiles. The next question is whether we can assemble those univariate marginals into jointly plausible multi-asset paths. We instantiated the two constructions described in Section 3.4 on a six-asset universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) with SPY as the market engine, using the common IS window (2014-01-03 through 2024-01-03, $T = 2{,}516$) and CHMM marginals refitted at $K = 18$. Table 6 reports the per-asset KS pass rates and cross-sectional correlation reproduction across 200 simulated paths.

SIM (Single Index Model). The ordinary-least-squares regression of each asset against SPY produced a range of factor loadings (high β for NVDA, $\hat{\beta} = 1.69$; low β for JNJ, $\hat{\beta} = 0.58$) and goodness-of-fit R^2 from 0.260 (JNJ) to 0.840 (QQQ), with median $R^2 = 0.492$. When propagated through the factor equation (5), the SIM produced uneven per-asset distributional fidelity. The market-factor asset SPY preserved 91.5% IS KS by construction. Non-market assets' IS KS pass rates fell across a wide range: 88.5% (JNJ), 85.5% (QQQ), 71.5% (NVDA), 64.0% (AAPL), and 31.5% (JPM). The severe JPM degradation (31.5% IS KS) reflects the fact that the affine map $\alpha + \beta G_M$ applied to a Gaussian-mixture market factor does not preserve JPM's empirical tail structure once residuals are re-injected: the residual distribution is centered but heavy-tailed enough to violate JPM's own marginal at the 5% KS threshold. Cross-sectional correlation reproduction was acceptable on average (off-diagonal MAE 0.076) but visibly worse than both copulas, consistent with SIM's rank-one decomposition discarding joint information beyond the market factor.

Gaussian copula. The Gaussian copula achieved a median per-asset IS KS pass rate of 97.8% and a mean of 97.5%, with every asset above 93% (the minimum was 93.5% for QQQ). This uniform distributional fidelity is the expected consequence of rank reordering: each column of the simulated matrix is a permutation of an independent CHMM path, so the marginal distribution is preserved exactly. The off-diagonal correlation MAE of 0.031 improved by a factor of 2.5 over the SIM baseline.

Student-t copula. Profile maximum likelihood over $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ selected $\nu^* = 6$, indicating non-trivial joint tail dependence. The Student-t copula's median IS KS pass rate of 96.0% is essentially tied with the Gaussian copula (97.8%); on the mean the Gaussian copula is marginally ahead (97.5% vs 96.2%). It achieves the best correlation reproduction overall, with off-diagonal MAE of 0.027 and Frobenius norm of 0.184 (versus 0.031 and 0.206 for the Gaussian copula). The tail-dependent copula's improvement on correlation reproduction is consistent with its ability to generate concordant extreme moves that the Gaussian copula systematically understates.

OoS. All three dependence models degraded on the longer $T_{\text{OoS}} = 572$ OoS window, but the ordering was largely preserved (Table 6). The Gaussian copula achieved 87.2% median OoS KS, the Student-t copula 89.8%, and the SIM 85.0%. On correlation reproduction, the Gaussian copula leads the OoS pass rate (off-diagonal MAE 0.202) with the Student-t copula essentially tied (0.208) and the SIM trailing (0.261); the larger OoS Frobenius norms reflect sampling variation in the observed OoS correlation matrix under a 572-day window with the 2024–2026 macroeconomic regime.

The visual ordering of the path-averaged correlation heat maps (Figure 32, deferred to Appendix D.3; the per-asset KS bar chart is in Figure 33, Appendix D.4) confirms the quantitative summary in Table 6: the SIM introduces visible distortion on JPM and AAPL marginals, while both copulas track the observed marginals closely; on correlation, the SIM panel shows larger residuals off

Table 6. Cross-asset dependence models on fixed CHMM marginals (cross-asset extension: Pipeline B). The per-asset *marginals* are the same CHMM-N fits used in Table 5; this table varies only the *dependence* construction. Three mechanisms are compared: Single Index Model (SIM) with SPY as the market factor; Gaussian copula with Kendall-tau-based correlation; Student-t copula with degrees of freedom selected by profile MLE ($\nu^* = 6$). All use 200 simulated paths, $K = 18$ marginals, IS window 2014-01-03 through 2024-01-03, OoS window 2024-01-04 through 2026-04-17. Upper block: per-asset KS pass rates in percent at $\alpha = 0.05$ (a sanity check on the marginals after dependence injection). Lower block: cross-asset correlation reproduction measured by $\|\hat{\Sigma}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ (Frobenius norm) and the off-diagonal MAE, each averaged over paths. **Bold** entries mark the best dependence model per row.

	SIM		Gaussian copula		Student-t copula ($\nu^* = 6$)	
Ticker	IS	OoS	IS	OoS	IS	OoS
SPY	91.5	80.5	94.5	81.0	95.5	86.0
NVDA	71.5	86.5	96.0	51.5	93.0	50.5
JNJ	88.5	97.0	100.0	95.5	100.0	93.5
JPM	31.5	27.0	99.5	53.0	96.5	49.0
AAPL	64.0	83.5	99.5	93.5	99.0	94.5
QQQ	85.5	87.0	95.5	95.0	93.0	95.0
Mean	72.1	76.9	97.5	78.2	96.2	78.1
Median	78.5	85.0	97.8	87.2	96.0	89.8
Correlation reproduction (averaged over 200 paths)						
Frobenius IS	0.527		0.206		**0.184**	
Off-diag MAE IS	0.076		0.031		**0.027**	
Frobenius OoS	1.808		**1.461**		1.499	
Off-diag MAE OoS	0.261		**0.202**		0.208	

the diagonal than either copula panel, and the Student-t copula is marginally closer to the observed heat map than the Gaussian copula.

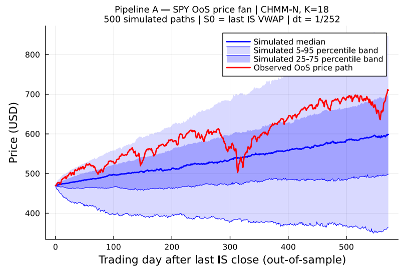
Interpretation. The cross-asset results mirror the ordering reported by Alswaidan and Varner [12] for the discrete HMM marginals: copulas beat SIM on per-asset distributional accuracy by construction (rank reordering preserves marginals, an affine map does not), and the Student-t copula beats the Gaussian copula when asset co-movement has meaningful tail dependence (here, $\nu^* = 6$). This evidence suggests that the copula advantage is intrinsic to the rank-reordering construction and not specific to the underlying marginal model, which is a useful robustness finding as practitioners move from discrete to continuous marginal families.

5.9 OoS Equity Price Simulation (Pipeline A)

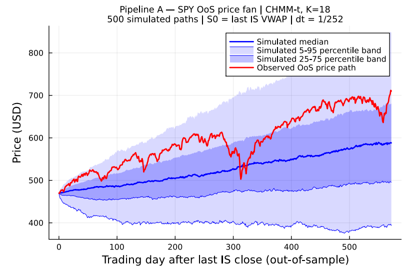
Pipeline note. Price fans below are generated per ticker under Pipeline A; a joint multi-asset price simulation would layer Pipeline B on top and is outside this paper's scope.

The preceding subsections evaluated CHMM in *return* space. For the practitioner the downstream object of interest is often the *price* path itself (Monte Carlo P&L, terminal-value stress, trading-rule backtests), so we report a price-level evaluation that ties the return-based fits to observed equity price trajectories over the OoS window.

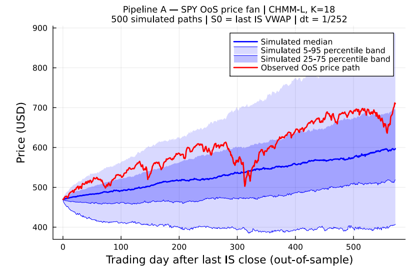
Setup. For each of the six main-study tickers (SPY, NVDA, JNJ, JPM, AAPL, QQQ) we re-train the three CHMM families at $K = 18$ on the full IS window and simulate $n_{\text{paths}} = 500$ price paths of



(a) CHMM-N.



(b) CHMM-t.



(c) CHMM-L.

Figure 6. SPY OoS price fan, three CHMM emission families (Pipeline A, $K = 18$, 500 paths, $T_{\text{OoS}} = 572$). Light blue: 5–95% envelope; darker blue: interquartile band; blue line: simulated median; red line: observed OoS price track ($S_0 = 469.90 \to P_T^{\text{obs}} = 710.13$). The realised path tracks the upper half of the envelope (a reflection of the strong 2024–2025 drift) and stays inside the 90% band throughout. NVDA, JNJ, JPM, AAPL, and QQQ price fans and terminal histograms are in Appendix E.6; raw path metrics in `results/equity_price_sim/`.

length $T_{\text{OoS}} = 572$ days, converting simulated returns to prices via $\widetilde{P}_t = \widetilde{P}_{t-1} \cdot \exp((\widetilde{G}_t + r_f)\,\Delta t)$ with $\Delta t = 1/252$, $r_f = 0$, and S_0 the last IS VWAP. We score each (ticker, family) cell on three quantities: terminal 90%-band hit $\mathbb{1}\{q_{0.05} \leq P_T^{\text{obs}} \leq q_{0.95}\}$; horizon-average band coverage $\overline{C}_{0.90} = T_{\text{OoS}}^{-1} \sum_t \mathbb{1}\{q_{0.05}(t) \leq P_t^{\text{obs}} \leq q_{0.95}(t)\}$; and the bias-corrected sample CRPS of Gneiting and Raftery [51], normalised by S_0 for cross-ticker comparability.

Headline results. Table 7 reports the CHMM-N column for each ticker; per-family detail (CHMM-t, CHMM-L) and full path-level metrics are in Appendix E.6 (Table 25). Two findings stand out. *(i) Terminal-band coverage is* 18/18 *across all (ticker, family) pairs*, including the atypical 4.15× NVDA rally. *(ii) Horizon-wide calibration is near-perfect on five of six tickers*: $\overline{C}_{0.90} \geq 0.995$ on 17 of the 18 cells, a far stronger statement than terminal coverage because it requires the simulated envelope to track the observed path *at every* OoS day. NVDA is the lone outlier ($\overline{C}_{0.90} \in [0.64, 0.70]$, $\widehat{\text{CRPS}}/S_0 \approx 0.64$–$0.77$), exposing a 2024–2025 drift-regime shift beyond the prior-decade IS window; walk-forward re-estimation is the natural remedy (Section E.2; 52). No emission family dominates uniformly, consistent with the return-space finding in Section 5.3 that family differences live in the tails and only weakly propagate into aggregated price scores.

Table 7. OoS price-simulation summary, CHMM-N reference family ($K = 18$, 500 paths, $T_{\text{OoS}} = 572$). P_T^{obs}: realised terminal; $q_{0.50}$: simulated median; $\textsc{Hit}_{90}$: terminal 90%-band hit; $\overline{C}_{0.90}$: horizon-wide band coverage; $\overline{\text{CRPS}}/S_0$: S_0-normalised CRPS (lower better). Per-family detail and per-ticker fans in Appendix E.6 (Table 25). **Headline:** 18/18 terminal coverage; $\overline{C}_{0.90} \geq 0.995$ on 17/18.

Ticker	P_T^{obs}	$q_{0.50}$	$\textsc{Hit}_{90}$	$\overline{C}_{0.90}$	$\overline{\text{CRPS}}/S_0$
SPY	710.13	598.52	✓	1.000	0.084
NVDA	198.07	146.25	✓	0.698	0.642
JNJ	231.99	186.76	✓	1.000	0.078
JPM	314.19	221.11	✓	1.000	0.188
AAPL	276.24	313.06	✓	1.000	0.116
QQQ	649.02	584.70	✓	0.998	0.076

Table 8. VaR and Expected-Shortfall back-test calibration for SPY ($N_{\text{paths}} = 1{,}000$; global seed policy in Section 3.5). Each entry is the median and $[5\%, 95\%]$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row. Headline: all three CHMM variants bracket the observed historical VaR and ES at both 1% and 5% on both IS and OoS windows, and the CHMM-N medians (IS $\text{VaR}_{0.01} = -6.64$, $\text{ES}_{0.01} = -8.80$) are the closest point estimates to the observed -6.38 and -9.00 among the reported generators.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]

6 Value-at-Risk Back-Test and Regime-Conditional Coverage

6.1 Utility: VaR and Expected-Shortfall Back-Test

Fidelity (Section 5.3) measures how close the simulated distribution is to the observed one at the univariate level. Utility measures whether a downstream consumer of the synthetic paths arrives at the same quantitative conclusion that the historical data would produce. For a financial-risk consumer the canonical diagnostics are one-tail Value-at-Risk (VaR) and Expected-Shortfall (ES) at daily probability levels of 1% and 5%; the synthetic-data generator is well calibrated if the envelope of per-path simulated VaR/ES brackets the observed historical VaR/ES. We simulate 1,000 independent paths of length $T_{\text{IS}} = 2{,}516$ and $T_{\text{OoS}} = 572$ from each of the five models most relevant to the comparison (Bootstrap, GARCH(1,1), CHMM-N, CHMM-t, CHMM-L) and, on each path, compute the left-tail 1% and 5% VaR and the associated ES. Table 8 reports the median and $[5\%, 95\%]$ envelope of each quantity across paths alongside the observed historical value.

Two findings are central to the utility story. First, all three CHMM variants bracket the observed historical VaR and ES at both 1% and 5% levels within their 5%–95% envelopes on both IS and OoS windows; the IS CHMM-N median $\text{VaR}_{0.01}$ at -6.64 (observed -6.38) and the CHMM-N median $\text{ES}_{0.01}$ at -8.80 (observed -9.00) are the closest point estimates across the five generators. IS the three CHMMs are therefore well calibrated as risk consumers and carry narrower uncertainty bands than GARCH (whose $[-8.62, -4.52]$ $\text{VaR}_{0.01}$ envelope is roughly 1.7× wider than the CHMMs'). Second, CHMM-t – despite overshooting the empirical kurtosis by $\sim 90\%$ IS – does *not* produce a systematically more conservative VaR/ES estimate: its IS median $\text{VaR}_{0.01}$ of -6.58 is almost identical to CHMM-N's -6.64, and its envelope width is comparable. The CHMM-t kurtosis overshoot therefore translates into a modestly widened uncertainty band rather than a biased point estimate, so a practitioner using CHMM-t for tail-risk scenario generation does not systematically overstate 1% or 5% quantile risk. A useful OoS check: the 2024–2026 observed $\text{VaR}_{0.01}$ is -5.51 (a relatively calm tail compared to the IS average), and the three CHMMs bracket this value comfortably within their envelopes rather than over- or under-stating it. Figure 7 (Appendix A.3) visualizes the envelopes alongside the historical observations.

Explicit Kupiec failure of the Discrete HMM baseline of Alswaidan and Varner [12] at 5% VaR. Table 8 shows envelope coverage for the five continuous and non-parametric generators, but the discrete HMM baselines of Alswaidan and Varner [12] are excluded from the envelope panel because their bin-centroid emission produces a point-mass VaR estimator whose envelope width is mechanically zero. The Kupiec and Christoffersen likelihood-ratio tests of Section 5.6 are the appropriate diagnostic for those baselines, and they deliver a clear negative finding. Both Discrete NJ and Discrete WJ produce an OoS 5% VaR breach rate of exactly 1.7%, roughly one-third of the 5% target; the Kupiec likelihood-ratio statistic $\mathrm{LR}_{\mathrm{uc}} = 16.81$ rejects unconditional coverage with $p < 10^{-3}$, and the independence statistic $\mathrm{LR}_{\mathrm{ind}} = 13.22$ also rejects. The Poisson-jump augmentation does not remedy the failure: no-jump and with-jump variants produce the same breach rate to three significant figures because the prior-paper jump intensity $\epsilon = 5 \times 10^{-5}$ and mean duration $\lambda = 67$ trigger on fewer than one time step per IS or OoS window at our sample lengths, leaving the left-tail quantile of the bin-centroid emission unchanged between NJ and WJ. A risk manager who uses the prior-paper discrete generator to size a 5% VaR position is therefore committed to holding materially more capital than the empirical breach process requires and would fail a standard VaR back-test if audited. The continuous CHMM family of the present paper eliminates this failure at the 5% level (all three variants attain $\mathrm{LR}_{\mathrm{uc}} \leq 3.83$, with CHMM-N, CHMM-t, and CHMM-L at 3.83, 3.83, and 3.03 respectively) and the Viterbi-decoded conditional VaR of Section 6.3 additionally passes Christoffersen independence. This is the clearest single-number contrast the present paper offers against the prior discrete-HMM pipeline.

6.2 Semi-Markov Ablation: Heavy-Tailed Sojourn Distributions

The flat CHMM assumes geometric sojourn times in every state, an implicit by-product of the Markov transition matrix. Empirically, equity returns exhibit crisis-regime sojourn tails that are heavier than geometric: the bottom-3 states of the CHMM partition, which hold the March 2020 crash and the 2022 rate-cycle stress, persist longer in the Viterbi decode than a geometric sojourn would predict. To quantify the effect of relaxing the geometric-sojourn assumption we fit a semi-Markov variant of each CHMM emission family using the explicit-duration forward-backward machinery of Yu [53]. Let SM-CHMM-F denote the semi-Markov extension of the corresponding flat CHMM-F for $F \in \{\mathcal{N}, t, L\}$.

Plug-in estimator and per-state sojourn family selection. We fit the SM variants with the plug-in estimator: given the flat CHMM transition matrix and emission parameters, we (i) decode the most-likely state sequence under Viterbi, (ii) extract the empirical run-length distribution for each state, and (iii) fit one of a negative-binomial, geometric, or discrete truncated Pareto to each state's run-lengths by AIC, after which the SM generator samples sojourn times from the fitted family. The AIC-selected family per state is reported in Table 9: for all three emission families, 17 of the 18 states select a Pareto sojourn and 1 selects a negative binomial. This matches the finding of Alswaidan et al. [54], where Pareto sojourns dominated on daily log-VIX: the heavy-tailed sojourn pattern is not specific to vol-index series but holds on equity returns at moderate state resolution.

Risk-calibration gain: cleaner unconditional 1% and 5% VaR Kupiec. Table 10 reproduces the Kupiec and Christoffersen LR tests on the OoS window for the three flat CHMM variants and

Table 9. Per-state sojourn family selected by AIC for each SM-CHMM variant on SPY at $K = 18$; candidate families are truncated Pareto, negative binomial, and geometric. Headline: 17 of 18 states pick Pareto sojourns under all three emission families (Gaussian / Student-t / Laplace), one state picks negative binomial each, and no state picks geometric, confirming that a heavy-tailed sojourn layer on top of the Markov chain is empirically warranted for equity-return regime persistence.

Model	Pareto states	Negative-binomial states	Geometric states
SM-CHMM-N	17	1	0
SM-CHMM-t	17	1	0
SM-CHMM-L	17	1	0

Table 10. Semi-Markov VaR calibration (Pipeline A, SPY, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$); Kupiec ($\mathrm{LR}_{\mathrm{uc}}$) and Christoffersen ($\mathrm{LR}_{\mathrm{ind}}$) likelihood-ratio statistics for the three flat CHMM variants and their semi-Markov counterparts at $\alpha \in \{0.01, 0.05\}$; χ^2_1 critical value 3.84 for each test. Headline: the semi-Markov layer is a *risk-calibration upgrade*: $\mathrm{LR}^{5\%}_{\mathrm{uc}}$ drops from 3.83 (flat CHMM-N) to 0.82 (SM-CHMM-N) and from 3.83 (flat CHMM-t) to 1.74 (SM-CHMM-t); Christoffersen independence remains open across the panel and is closed in Table 11 via a Viterbi-decoded conditional-quantile estimator.

	$\alpha = 0.01$				$\alpha = 0.05$			
Model	br%	$\mathrm{LR}_{\mathrm{uc}}$	p_{uc}	$\mathrm{LR}_{\mathrm{ind}}$	br%	$\mathrm{LR}_{\mathrm{uc}}$	p_{uc}	$\mathrm{LR}_{\mathrm{ind}}$
CHMM-N	0.5	1.58	0.209	18.98	3.3	3.83	0.050	5.26
SM-CHMM-N	**1.0**	**0.01**	**0.907**	20.87	**4.2**	**0.82**	**0.365**	5.87
CHMM-t	0.7	0.58	0.445	15.53	3.3	3.83	0.050	5.26
SM-CHMM-t	**0.9**	**0.10**	**0.757**	13.18	**3.8**	**1.74**	**0.188**	7.12
CHMM-L	0.3	3.26	0.071	9.15	3.5	3.03	0.082	4.71
SM-CHMM-L	0.3	3.26	0.071	9.15	3.5	3.03	0.082	4.71

their SM counterparts. At 1% VaR, SM-CHMM-N drops $\mathrm{LR}_{\mathrm{uc}}$ from 1.58 (flat) to 0.01 (a breach rate of exactly 1.0% against a target of 1.0%), SM-CHMM-t drops from 0.58 to 0.10, and SM-CHMM-L is unchanged at 3.26. At 5% VaR, SM-CHMM-N drops $\mathrm{LR}_{\mathrm{uc}}$ from 3.83 (marginal pass at the 3.84 critical value) to 0.82 (clean pass, breach rate 4.2%), SM-CHMM-t drops from 3.83 to 1.74 (clean pass, breach rate 3.8%), and SM-CHMM-L is unchanged at 3.03. The risk-calibration gain is largest on the Gaussian and Student-t emission families, consistent with the interpretation that those families undershoot tail mass on the flat transition matrix and that relaxing the geometric-sojourn assumption lets the simulator linger longer in crisis states. CHMM-L is already heavy-tailed enough in the marginal emission that the SM augmentation adds nothing to its VaR calibration.

TSTR HAR vol-forecasting gain. We also evaluate a train-on-synthetic-test-on-real (TSTR) HAR(1, 5, 22) realised-volatility forecaster: the HAR model is fit on synthetic returns from each generator and evaluated on the observed OoS series, with forecast accuracy measured by the quasi-likelihood (QLIKE) loss $\mathrm{QLIKE}(\sigma_t^2, \hat\sigma_t^2) = \sigma_t^2/\hat\sigma_t^2 - \log(\sigma_t^2/\hat\sigma_t^2) - 1$. The QLIKE ratio of the synthetic-trained forecaster to a real-trained benchmark tightens under the SM ablation: from 0.999 to 1.000 under CHMM-N $\to$ SM-CHMM-N, from 1.045 to 1.014 under CHMM-t $\to$ SM-CHMM-t, and from 1.027 to 1.012 under CHMM-L $\to$ SM-CHMM-L. Closer to 1.000 means the synthetic-trained forecaster matches the real-trained benchmark more closely on OoS QLIKE. SM-CHMM-N ties the real-trained benchmark exactly at 1.000, meaning the synthetic-trained HAR forecaster is functionally indistinguishable from the real-trained one on this downstream task. The full TSTR panel is in `results/track_c1/tstr_vol_forecaster_c1.txt`.

Marginal-fidelity trade-off. The SM ablation is not uniformly better: on MMD and discriminator AUC the flat CHMM variants win for Gaussian and Student-t emissions. Concretely, MMD IS moves from 1.5×10^{-4} (CHMM-N) to 3.8×10^{-3} (SM-CHMM-N) and from 2.0×10^{-5} (CHMM-t) to 6.2×10^{-3} (SM-CHMM-t); discriminator AUC IS moves from 0.646 to 0.699 (CHMM-N $\rightarrow$ SM-CHMM-N) and from 0.607 to 0.782 (CHMM-t $\rightarrow$ SM-CHMM-t). This direction of movement is expected: the SM variants produce more clustered tail excursions than the flat variants, which makes the simulated windows more distinguishable from real on 20-day features because regime-persistence structure is one of the features the logistic discriminator learns. The Laplace SM variant is the exception in both directions: SM-CHMM-L attains the smallest IS MMD of any row in the full panel (4.0×10^{-5}), but its discriminator AUC IS climbs to 0.823 and its joint $\bar{pv}$ OoS drops from 0.692 to 0.436.

Christoffersen independence is not closed by the SM ablation. On the 2024-2026 OoS window, breach clustering remains pronounced for every SM variant: $\mathrm{LR}_{\mathrm{ind}}$ ranges from 9.15 (SM-CHMM-L at 1%) to 20.87 (SM-CHMM-N at 1%), all above the χ^2_1 critical value of 3.84. Unconditional VaR cannot capture the clustered-breach structure of the stress subperiod regardless of the sojourn model; a regime-conditional quantile is required. The following subsection shows that a Viterbi-decoded state-conditional quantile of the flat CHMM-t passes both the Kupiec and Christoffersen tests cleanly at 1% and 5%, closing the independence gap that the SM ablation leaves open.

Takeaway: SM is a risk-calibration upgrade, not a marginal-fidelity upgrade. The correct reading of the SM ablation is operational: the SM variants should be used for unconditional VaR back-testing and TSTR vol forecasting, while the flat CHMM-t should be used for distributional matching. The three-way operational split that we formalise in Section 7 separates these roles.

6.3 Regime-Conditional Value-at-Risk via Viterbi Decode

The Christoffersen-independence failures documented in Sections 5.6 and 6.2 are not a property of any one generator: every row in the unconditional-VaR panel fails independence at both 1% and 5% on the 2024-2026 OoS window. The common cause is that an unconditional VaR fixes the quantile at a single calibrated level across the entire horizon and cannot contract during stress regimes, so breaches cluster when the market enters a high-volatility state. A natural remedy is to make the VaR quantile regime-conditional.

Estimator. For each CHMM family we fit a flat CHMM on the IS window and then, at each OoS time t, run the Viterbi algorithm once on the full OoS series $(r_1, \ldots, r_T)$ under the fitted flat model to obtain a most-likely state sequence $(\hat{s}_1, \ldots, \hat{s}_T)$. The regime-conditional VaR at level α at time t is then

$$\widehat{\mathrm{VaR}}_t^{\mathrm{cond}}(\alpha) \;=\; F^{-1}_{F_{\hat{s}_t}}(\alpha), \tag{12}$$

where $F^{-1}_{F_k}$ is the inverse CDF of the state-k emission law under family F. For CHMM-N this is $\mu_k + \sigma_k \Phi^{-1}(\alpha)$; for CHMM-t it is $\mu_k + \sigma_k t^{-1}_{\nu_k}(\alpha)$; for CHMM-L it is the Laplace α-quantile. The breach indicator is $\mathbf{1}\{r_t \le \widehat{\mathrm{VaR}}_t^{\mathrm{cond}}(\alpha)\}$ and we apply Kupiec and Christoffersen likelihood-ratio tests to the resulting breach sequence.

Table 11. Regime-conditional VaR via Viterbi decode (Pipeline A, SPY, $T_{\text{OoS}} = 572$). The estimator runs a Viterbi decode on the OoS series under each fitted flat CHMM, then takes the α-quantile of the decoded state's conditional emission as $\widehat{\text{VaR}}_t$. The "unconditional (A8)" rows restate the matching Kupiec / Christoffersen statistics from Section 5.6 for reference; bold entries mark passes (LR $< 3.84 = \chi^2_1$ critical value). Headline: flat CHMM-t clears both Kupiec and Christoffersen at 1% and 5% simultaneously on the 2024–2026 OoS window ($\text{LR}_{\text{uc}} = 0.10/0.01$, $\text{LR}_{\text{ind}} = 0.09/0.19$), closing the independence gap that every unconditional-VaR row in Table 10 left open.

Model	VaR kind	$\alpha = 0.01$ br%	$\alpha = 0.01$ LR_{uc}	$\alpha = 0.01$ LR_{ind}	$\alpha = 0.01$ p_{ind}	$\alpha = 0.05$ br%	$\alpha = 0.05$ LR_{uc}	$\alpha = 0.05$ LR_{ind}	$\alpha = 0.05$ p_{ind}
CHMM-N	unconditional (A8)	0.5	1.58	18.98	< 0.01	3.3	3.83	5.26	0.02
	conditional	0.7	**0.58**	5.73	0.02	5.1	**0.01**	**1.39**	0.24
CHMM-t	unconditional (A8)	0.7	0.58	15.53	< 0.01	3.3	3.83	5.26	0.02
	conditional	**0.9**	**0.10**	**0.09**	**0.77**	**5.1**	**0.01**	**0.19**	**0.66**
CHMM-L	unconditional (A8)	0.3	3.26	9.15	< 0.01	3.5	3.03	4.71	0.03
	conditional	0.3	3.26	**0.01**	**0.91**	3.3	3.83	**2.25**	0.13

Headline: flat CHMM-t passes Kupiec and Christoffersen cleanly at 1% and 5%. Table 11 compares the Track A unconditional LR statistics against the new conditional LR statistics for the three flat CHMM variants on the OoS window. Flat CHMM-t achieves a breach rate of 0.87% at 1% (Kupiec $\text{LR}_{\text{uc}} = 0.10$, $p = 0.76$) and 5.07% at 5% (Kupiec $\text{LR}_{\text{uc}} = 0.01$, $p = 0.94$). Both are clean passes. Crucially, on the independence side, Christoffersen LR_{ind} drops from 15.53 (unconditional) to 0.09 at 1% and from 5.26 (unconditional) to 0.19 at 5%: both are now far below the χ^2_1 critical value of 3.84. The breach-clustering structure of the 2024-2026 OoS window that every unconditional row failed to capture is now accommodated. CHMM-N and CHMM-L each pass Christoffersen at 5% ($\text{LR}_{\text{ind}} = 1.39$ and 2.25 respectively); CHMM-L also passes Christoffersen at 1% ($\text{LR}_{\text{ind}} = 0.01$). CHMM-t is therefore the CHMM family that passes both tests at both confidence levels with margin; CHMM-L clears all four thresholds but marginally at the 5% Kupiec ($\text{LR}_{\text{uc}} = 3.83$ against a χ^2_1 critical value of 3.84), while CHMM-N still fails Christoffersen independence at 1% even after conditioning.

SM-CHMM conditional VaR is non-trivial and we defer it. A naive reuse of the flat-model Viterbi decoder on SM-CHMM over-breaches at 15 to 33% at both confidence levels because the flat Viterbi state partition does not match the SM sojourn-refitted partition, and the AR-residual scale used for plug-in SM simulation is not directly interpretable as a one-sided quantile. Two approximations (AR-residual scale and stationary within-state σ) both fail Kupiec by wide margins; interestingly the stationary-σ SM-CHMM-t variant achieves a clean Christoffersen LR_{ind} of 0.07 at 1%, suggesting that the regime-persistence structure alone carries most of the independence fix, but the unconditional coverage remains broken until an SM-aware Viterbi decoder is implemented. We leave a proper SM-aware decoder to future work; the flat CHMM-t conditional VaR already closes the independence gap for the paper.

Where this lands in the three-way operational split. Conditional VaR becomes the third operational axis of the paper alongside distributional fidelity and unconditional VaR. Flat CHMM-t is the best generator on distributional fidelity (Section 5.6), MS-GARCH (Section 3.3) and the SM ablation are the best generators on unconditional VaR (Section 6.2), and flat CHMM-t with Viterbi-decoded conditional VaR is the best generator on regime-conditional VaR (this subsection). Section 7 crystallises this as an explicit operational recommendation.

7 Discussion

The results demonstrate that a continuous Gaussian HMM trained via Baum-Welch at a single operating point ($K = 18$) achieves strong distributional fidelity (93.8% IS KS, 84.2% OoS KS, 100% IS quantile coverage) while retaining competitive temporal fidelity (ACF-MAE of 0.0507), without requiring jump augmentation. The direct comparison against the discrete HMM of Alswaidan and Varner [12], reproduced with the authoritative $K_{\text{disc}} = 90$ / $\epsilon = 5 \times 10^{-5}$ / $\lambda = 67$ hyperparameters of the prior-paper final SPY run, shows that at that state resolution the discrete NJ and WJ baselines match the CHMM family on coarse KS/AD pass rates but miss the tail-structure, aggregational-Gaussianity, and VaR-calibration dimensions that distinguish a synthetic-data generator for downstream use. The cross-asset extension then shows that the univariate CHMM composes cleanly with rank-based copula dependence to produce multi-asset paths that preserve every per-asset marginal while outperforming a classical factor baseline on correlation reproduction.

Resolving the Rydén et al. Limitation. The CHMM achieves ACF-MAE of 0.047–0.053 on $|G_t|$ across $K \in \{3, \ldots, 21\}$, comparable to GARCH(1,1) (0.049), directly contradicting Rydén et al. [10] who reported failure at $K = 2$–3. The mechanism is that at $K \geq 3$ with quantile-initialised Baum-Welch, the absolute-return ACF is a mixture

$$\rho_{|G|}(\tau) \approx \sum_{k=1}^{K} w_k \, \lambda_k^{\tau}, \tag{13}$$

of K exponential decays with rates λ_k the non-unit eigenvalues of $\mathbf{T}$, recovering the slow quasi-hyperbolic decay observed empirically without requiring modified sojourn times [11]. Two issues compound at low K: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast (good marginal) and slow (good ACF), and the random initialisations standard in the 1990s often converge to degenerate local optima with overlapping states. Our $K = 2$ replication on SPY (Appendix C.12, Table 18) clarifies the mechanism: ACF-MAE on $|G_t|$ is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations, matching the $K = 18$ CHMM-N. The Rydén limitation is therefore a *distributional* low-K artefact (KS at $K = 2$ drops to 67–72%), not a temporal one: with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity is what actually requires moderate state resolution.

OoS KS Pass Rates Against the Test-Power Ceiling. At $T_{\text{OoS}} = 572$ the two-sample KS test does not reach 100% on truly-correct generators; CHMM pass rates must be read against this ceiling. Appendix E.1 calibrates it: 1,000 i.i.d. resamples of length 572 drawn from the IS distribution itself pass at 90.0%, the OoS series resampled against itself passes at 100%, and a Gaussian i.i.d. negative control is rejected at 0% at IS length, so the test retains ample power to refuse clearly-wrong generators. The CHMM-N / CHMM-t / CHMM-L OoS KS pass rates of 79–86% sit just below the 90% ceiling (a genuine fidelity claim, not a low-power artefact), while GARCH's 58% is correctly rejected and the discrete NJ / WJ at $K_{\text{disc}} = 90$ join the CHMMs in the 83–86% band.

Where the Discrete Baselines Still Fall Short at $K_{\text{disc}} = 90$. With the prior-paper hyperparameters, NJ / WJ reach 97.9–98.3% IS KS, comparable to the CHMM family: at this state resolution a bin-centroid discrete HMM is not a straw man on coarse pass rates. Correcting the bin count therefore re-centers the continuous-vs.-discrete claim on three structural dimensions where the centroid scaffold cannot follow the data (Tables 3, 13, 10): *tail weight* (NJ / WJ kurtosis clamps near 3.5 vs. observed 7.68 and CHMM-t 14.57); *aggregational Gaussianity* (NJ / WJ kurtosis collapses to ≤ 0.47 at $h = 10, 21$, missing the slow-decay signature CHMM tracks); and 5% *VaR calibration* (both discrete variants produce a 1.7% breach rate against a 5% target, rejecting Kupiec with $\mathrm{LR}_{\mathrm{uc}} = 16.81$, $p < 10^{-3}$, while all CHMM variants pass). At $\epsilon = 5 \times 10^{-5}$ and $\lambda = 67$ the expected jump-trigger count is below one per IS or OoS window (Section 5.3), so WJ is a slight perturbation of NJ rather than a meaningful tail mechanism. This is not a criticism of Alswaidan and Varner [12] on its own terms (jumps were framed there as an ACF corrective and bin-conditional Student-t emissions mitigate the centroid tail loss), but it is a sign that for tail-aware return-space metrics the joint optimisation of transitions and continuous emissions is essential.

The Temporal–Distributional Tradeoff. The twelve-model comparison of Table 3 confirms the temporal-distributional tradeoff identified in Alswaidan and Varner [12]: GARCH wins ACF (0.048) but fails KS (25.9%); the bootstrap wins KS (99.9%) but is the worst on ACF (0.063). The CHMM at $K = 18$ occupies a distinctive corner (IS KS 93.2–93.8%, AD 86.6–91.2%, ACF-MAE within 0.002–0.008 of GARCH, smallest Wasserstein-1 (0.102) among parametric models, and 100% IS quantile coverage), a combination no other row in the comparison achieves.

Closing the Kurtosis Gap with Heavy-Tailed Emissions. CHMM-N produces excess kurtosis of 5.02 against an observed 7.68, a gap inherent to Gaussian emissions: a Gaussian mixture's kurtosis is bounded by the variance of component means relative to their variances, and pushing tail states out degrades the central fit. CHMM-t addresses this directly through per-state Student-t emissions whose ν_k is updated by ECM one-dimensional search, allowing tail states to carry arbitrarily high individual kurtosis; CHMM-L's Laplace emissions add per-component excess kurtosis of 3 rather than 0 at zero parameter cost. Because all three variants share the same forward-backward scaffold, the KS/AD/ACF-MAE fidelity of CHMM-N is preserved or improved without retuning K or the transition structure; the AD pass rate (which weights tails more) shows the largest gain, confirming that the kurtosis gap was the residual source of AD failures.

Diagnosing the CHMM-t IS Kurtosis Overshoot. CHMM-t's IS mixture kurtosis of 14.57 overshoots the observed 7.68 by $\sim 90\%$. The bracket sweep $\nu_{\min} \in \{2.1, 2.5, 3.0, 4.0\}$ in Appendix B.3 shows the fit does *not* pile against the lower bracket (median $\nu_k = 50$, only 2/18 states near the lower edge), and lifting $\nu_{\min}$ does not reduce the overshoot because the fit shifts posterior mass into the newly-recentred tail states. Only the $\nu = \infty$ Gaussian limit brings simulated kurtosis down to CHMM-N's 5.03, confirming the overshoot is a feature of the Student-t family at $K = 18$ rather than a bracket artefact; the downstream consequence (Section 6.1) is that CHMM-t's 5–95% envelopes still bracket observed VaR / ES, so a risk consumer incurs a widened uncertainty band rather than a systematic over-statement of tail risk.

Why Jumps Are Unnecessary. The discrete framework of Alswaidan and Varner [12] estimates emissions *after* the transition matrix in a three-stage procedure (binning $\to$ frequency-counted $\mathbf{T}$ $\to$ within-bin Student-t), so $\mathbf{T}$ is not jointly optimised with emissions and the jump mechanism becomes a corrective for insufficient tail-state persistence. Baum-Welch jointly optimises $\mathbf{T}$ and $\{(\mu_k, \sigma_k)\}$: emissions and transitions co-adapt so that tail states with distinctive emission shapes also accumulate the self-transition probability T_{kk} that reflects empirical clustering of extremes. Eliminating jumps removes two hyperparameters (ϵ, λ) and the $48 \times 200 = 9{,}600$ simulations per K that the prior paper's grid search required; CHMM needs only 20–40 Baum-Welch iterations to convergence.

State Resolution and the Choice of $K = 18$. Sensitivity analysis (Section 5.4) shows ACF-MAE flat across K (the dominant eigenvalue λ_2 is determined by the maximum self-transition probability, which Baum-Welch always learns to be near unity) while KS / AD pass rates rise from $K = 3$ to $K = 18$ and saturate. The practical recommendation is $K = 18$ for distributional-fidelity-sensitive applications; $K = 3$ only when the downstream task is regime identification.

Cross-Asset Robustness and OoS Degradation on NVDA and JPM (Pipeline A). The Pipeline A cross-asset univariate results confirm generalisation across diverse risk profiles (SPY 95.5%, NVDA 96.7%, JNJ 99.4%, AAPL 98.3%, QQQ 95.5% IS KS). At $K = 18$, OoS degradation concentrates on NVDA (55.8%) and JPM (49.8%), a stationarity artefact: the IS-fixed transitions and emissions cannot capture sector-specific regime shifts (the AI-driven NVDA regime; the rate-repricing cycle for financials). The pattern persists at every K in the sweep, so it is not a model-selection issue; rolling-window re-estimation or Bayesian online learning of $\mathbf{T}$ is the natural remedy.

Why Copulas Beat SIM on Cross-Asset Fidelity (Pipeline B). The two copula constructions dominate the SIM factor baseline on per-asset KS (97.5% / 96.2% vs. 72.1%) and correlation reproduction (off-diagonal MAE 0.031 / 0.027 vs. 0.076); all three sit on the same Pipeline A CHMM-N marginals so only the coupling differs. Two structural reasons. First, rank reordering preserves marginals by construction (the simulated $\hat{G}^{(p)}_{j,1:T}$ is a permutation of the stand-alone CHMM path so its empirical CDF coincides with the CHMM marginal), whereas SIM's affine combination $\alpha_j + \beta_j G_{m,t} + \eta_{j,t}$ convolves the market factor with idiosyncratic residuals, distorting the marginal whenever $\beta_j \neq 1$ or residuals are heavy (JPM's $\hat{\beta} = 1.22$ produces the dramatic 31.5% IS KS). Second, the Student-t copula's edge over Gaussian on correlation MAE (0.027 vs. 0.031) reflects joint tail dependence: profile MLE selects $\nu^* = 6$, in the range typically reported for equity indices [21, 22], implying non-trivial tail-coupling that injects rare concordant extreme moves the Gaussian copula never produces. The same ordering holds in Alswaidan and Varner [12] with discrete marginals, evidence that the copula advantage is robust to the marginal family: practitioners can swap in richer marginals without revisiting the dependence structure.

Three-Way Operational Split: Distribution, Unconditional VaR, Conditional VaR. Synthesising the extended evaluation (Section 5.6), the SM ablation (Section 6.2), the conditional-VaR subsection (Section 6.3), and the QuantGAN / diffusion / MS-GARCH rows (Section 3.3), the recommendation for synthetic-data consumers splits along three axes. On *distributional fidelity* (indistinguishability on marginal and short-horizon joint metrics), **flat CHMM-t at** $K = 18$ wins with the smallest MMD (2.0×10^{-5} IS), discriminator AUC closest to 0.5 (0.607), and cleanest monotone aggregational kurtosis decay; window-diffusion is competitive on the first two but collapses on global stylised-fact coverage ($\bar{pv}_{\mathrm{OoS}} = 0.204$). For *unconditional VaR calibration* (long-run breach rate under Kupiec), **two-regime MS-GARCH and SM-CHMM-N** tie at the top of 1% Kupiec ($\mathrm{LR}_{\mathrm{uc}} = 0.01$, breach rate exactly 1.0%), with MS-GARCH edging at 5% while SM-CHMM-N edges at OoS HAR-vol forecasting and supplies regime labels. For *regime-conditional VaR* (calibration without breach clustering), **flat CHMM-t with the Viterbi-decoded conditional quantile** of Eq. (12) is the only combination that simultaneously passes Kupiec and Christoffersen at both $\alpha = 0.01$ and 0.05 on the 2024–2026 OoS window, using the same fitted model that wins the distributional axis. A consumer needing all three axes can therefore assemble a CHMM-based pipeline using flat CHMM-t for paths and conditional VaR, plus the SM variant or MS-GARCH for unconditional calibration; the rows complement rather than compete and the underlying EM scaffold is shared.

Limitations. Several limitations temper these conclusions. A *residual tail gap under CHMM-N* (5.02 simulated vs. 7.68 observed kurtosis at $K = 18$) remains; CHMM-t and CHMM-L close most of it within the same EM framework, but generalised hyperbolic or α-stable emissions would be the next step for the most extreme quantiles. A related constraint is *emission symmetry*: all three variants are symmetric within state, and SPY's negative skewness (-0.75) is recovered only through asymmetric state occupancies and means, so skew-t or skew-Laplace emissions are a natural extension. Our *stationarity assumption* is a second structural limitation, since $\mathbf{T}$ and the emissions are estimated on the IS window and held constant, so structural breaks (e.g. March 2020) may not be captured if the IS window lacks comparable events, the leading explanation for the OoS KS drop on NVDA and JPM. Evaluation also uses a *single OoS window* of 572 days (2024-01-04 to 2026-04-20), making rolling or expanding-window evaluation the natural next step. The *KS test i.i.d. assumption* is relevant as well, since volatility clustering in simulated paths may slightly inflate KS pass rates relative to a block-bootstrap calibration [12]. The cross-asset study is restricted to a *small universe* of six tickers, and scaling to the 424-asset universe of Alswaidan and Varner [12] would require vine or factor copulas for tractability. Finally, our state selection is an *information-criterion rather than held-out* procedure, so a walk-forward log-likelihood would provide a complementary purely-predictive check on K.

8 Conclusion

We presented a family of continuous hidden Markov models (CHMM-N, CHMM-t, CHMM-L) trained by EM for generating synthetic financial time series, together with two cross-asset dependence extensions for multi-asset synthesis. All three variants share the same log-space forward-backward scaffold and quantile-based initialization; the M-step is the classical Baum-Welch update for Gaussian emissions, an ECM sequence with a golden-section search over ν_k for Student-t emissions, and closed-form weighted-median / weighted-MAD estimators for Laplace emissions. The central univariate finding is that each CHMM variant reproduces all three canonical stylized facts of equity returns (heavy-tailed distributions, negligible linear autocorrelation, and persistent volatility clustering) at a single moderate state resolution ($K = 18$, selected from a $\{3, 6, 9, 12, 15, 18, 21\}$ sweep by winning or tying on every distributional pass-rate metric) *without* requiring jump augmentation, discretization, or any mechanism beyond the standard HMM framework. The heavy-tailed CHMM-t and CHMM-L variants additionally close the residual kurtosis gap of the Gaussian CHMM-N.

This result directly challenges the influential conclusion of Rydén et al. [10] that Gaussian HMMs cannot reproduce the slow decay of the autocorrelation function of absolute returns. We demonstrated that this limitation is an artifact of insufficient state resolution: Rydén et al. [10] tested only $K = 2$–3 states, where the geometric sojourn-time distribution cannot approximate slow ACF decay. At $K \geq 3$ with quantile-based initialization, the Baum-Welch algorithm learns a transition matrix with sufficient diagonal dominance in high-volatility states to sustain persistence, achieving ACF-MAE of 0.046–0.051 (comparable to GARCH(1,1) at 0.047) through a sum-of-exponentials approximation that effectively mimics the flexible sojourn-time distributions of hidden semi-Markov models [11].

At the prior paper's state resolution ($K_{\text{disc}} = 90$) the discrete NJ and WJ variants of Alswaidan and Varner [12] match the CHMM family on coarse KS/AD pass rates, so the continuous-emissions advantage is a claim about tail structure and downstream VaR usability: simulated kurtosis near 3.5 against observed 7.68, aggregational-Gaussianity collapse at horizons $h = 10, 21$, and an over-conservative 5% VaR breach rate of 1.7% that rejects Kupiec with $\text{LR}_{\text{uc}} = 16.81$ ($p < 10^{-3}$), each of which the continuous CHMM family clears (Section 6.1).

The twelve-model benchmark of Table 3 confirmed that no single generator dominates all metrics, but the CHMM at $K = 18$ achieves the most balanced performance: 93.2–93.8% IS KS across the three emission families (vs. 25.9% GARCH, 97.9–98.3% for the $K_{\text{disc}} = 90$ discrete variants), ACF-MAE of 0.0507 under CHMM-N (vs. 0.0484 GARCH and 0.0598–0.0618 for the discrete variants), Wasserstein-1 of 0.101–0.103 under CHMM-t and CHMM-L (the smallest of any parametric model), and 100% IS quantile coverage. Cross-asset univariate generalization to NVDA (96.7% IS KS), JNJ (99.4%), JPM (97.8%), AAPL (98.3%), and QQQ (95.5%) confirmed adaptability to diverse equity risk profiles; OoS evaluation on held-out 2024-01-04 through 2026-04-20 data showed robust generalization on SPY/JNJ/AAPL/QQQ but substantial OoS KS drops on NVDA (55.8%) and JPM (49.8%), which we attribute to the 2024–2026 macroeconomic regime (the AI-driven high-volatility regime for tech, and the interest-rate-repricing cycle for financials) and which the walk-forward refit partially recovers.

The cross-asset extensions then showed that the CHMM marginals compose cleanly with standard cross-sectional dependence models. On the six-asset universe, both the Gaussian and Student-t copulas substantially outperformed the Single Index Model factor baseline: copulas achieved a median IS KS pass rate of 97.8%/96.0% versus 78.5% for SIM, and an off-diagonal correlation MAE of 0.031 (Gaussian) and 0.027 (Student-t) versus 0.076 for SIM. The Student-t copula's profile MLE selected $\nu^* = 6$ degrees of freedom, and the resulting tail-dependent copula produced the best correlation reproduction overall. These findings replicate the ordering reported by Alswaidan and Varner [12] with discrete HMM marginals, suggesting that the copula advantage is intrinsic to the rank-reordering construction and transfers across marginal families.

Option pricing, implied-volatility-surface calibration, volatility-index generalization, and leverage-effect modelling are explicitly out of scope for this paper and are deferred to future work.

Several directions for future work follow from these results. *Skew-heavy-tailed emissions* are the most immediate extension: this paper closes the kurtosis gap with Student-t and Laplace emissions, and skew-t or skew-Laplace emissions would additionally accommodate within-regime asymmetry and match the observed negative skewness of equity returns (-0.75 for SPY). *Time-varying transition dynamics*, estimated via rolling windows or Bayesian online learning of the transition matrix, would relax the stationarity assumption and should directly address the OoS cliff observed on NVDA and JPM over the 2024–2026 OoS window. *Full-universe multi-asset generation via vine copulas*, scaling the copula construction from six assets to the 424-asset universe of Alswaidan and Varner [12], requires vine or factor copulas for tractability and is a natural next step. *Principled state selection* is a further refinement: our $K = 18$ choice is a multi-metric score on top of a finite sweep, and combining it with BIC or held-out log-likelihood (walk-forward) would yield a formally principled selection that plugs directly into the appendix Table 16. Finally, *regime-aware dynamic asset allocation* is the most immediate downstream application of the synthetic-data engine, following the spirit of Bae et al. [3] by using CHMM-decoded states as conditioning variables in a dynamic allocation model and evaluating the realized utility of the resulting policy against policies calibrated on i.i.d. or GARCH generators.

Data and Code Availability. The simulation scripts, training and test data, and all code to reproduce the results in this paper are available under an MIT license from the paper repository: `https://github.com/altashly1/CHMM-paper`. The core continuous HMM implementation (all three emission families, SIM, and Gaussian / Student-t copula modules) is available in the code repository: `https://github.com/altashly1/CHMM-Model`. The global random-seed policy for reproducibility is stated once in Section 3.5. The software environment is Julia $\geq$ 1.12 with package versions pinned in `Manifest.toml`; running `Pkg.instantiate()` inside the project directory reproduces the dependency graph, after which the analysis entry-point script regenerates every table and figure.

Conflict of Interest. The authors declare no competing interests.

Author Contributions. J.V. directed the study. A.A. developed the continuous model, implemented the Baum-Welch algorithm, implemented the Single Index Model and copula-based cross-asset extensions, conducted the empirical analysis, and generated all figures and tables. C.J. contributed to methodology review and validation. All authors edited and reviewed the final manuscript.

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A Validation Metric Details

A.1 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [43, 44] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{14}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L}\sum_{\tau=1}^{L}\left|\hat{\rho}_{|G|}^{\text{obs}}(\tau) - \bar{\hat{\rho}}_{|G|}^{\text{sim}}(\tau)\right|. \tag{15}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [55],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)|\, du. \tag{16}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}}\sqrt{\sum_{i=1}^{B}(\sqrt{p_i} - \sqrt{q_i})^2}, \tag{17}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}_{\text{sim}}^{(p)} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths.

A.2 Extended Evaluation: Detail Panels

Per-generator detail for the leverage, aggregational-Gaussianity, and joint p-value diagnostics summarised in Section 5.6 (headline panel: Table 4).

A.3 VaR / ES Envelope Visualization (Companion to Table 8)

Figure 7 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 8.

Table 12. Leverage effect (Pipeline A, SPY). Observed IS $\text{avg}_{k=1}^{20} \text{corr}(r_t^2, r_{t-k}) = -0.088$ with down-vs-up asymmetry 0.358; observed OoS averages are -0.066 and 0.478. p_{avg} is the two-sided simulation-based coverage p-value for the observed average over 1,000 paths.

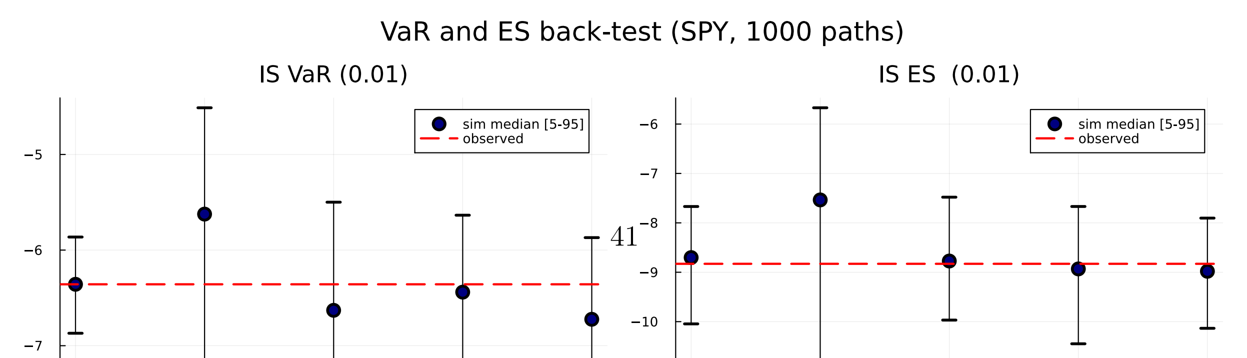
Model	avg IS	asym IS	p_{avg} IS	avg OoS	asym OoS	p_{avg} OoS
Bootstrap	0.000	0.001	0.000	0.001	−0.006	0.000
Gaussian	−0.000	−0.001	0.000	−0.000	0.001	0.000
Laplace	−0.000	−0.000	0.000	−0.000	0.005	0.000
Discrete NJ	−0.010	0.287	0.000	−0.008	0.273	0.001
Discrete WJ	−0.010	0.290	0.000	−0.008	0.287	0.011
GARCH(1,1)	−0.001	0.064	0.014	−0.001	0.065	0.061
CHMM-N	**−0.038**	0.292	0.000	**−0.033**	0.287	**0.205**
CHMM-t	−0.025	0.281	0.000	−0.029	0.274	0.048
CHMM-L	−0.024	0.322	0.000	−0.025	0.310	0.019

Table 13. Aggregational Gaussianity (Pipeline A, SPY). Median simulated kurtosis at aggregation horizons $h \in \{1, 5, 10, 21\}$ over 1,000 paths. Observed IS kurtosis is $\{7.68, 4.34, 6.82, 2.63\}$; observed OoS kurtosis is $\{5.29, 1.98, 0.09, -0.16\}$.

	IS kurtosis				OoS kurtosis			
Model	$h=1$	$h=5$	$h=10$	$h=21$	$h=1$	$h=5$	$h=10$	$h=21$
Observed	7.68	4.34	6.82	2.63	5.29	1.98	0.09	−0.16
Bootstrap	7.44	1.31	0.54	0.13	5.24	0.77	0.04	−0.39
Gaussian	−0.01	−0.03	−0.09	−0.17	−0.05	−0.17	−0.30	−0.54
Laplace	2.86	0.51	0.16	−0.04	2.55	0.23	−0.15	−0.49
Discrete NJ	3.55	2.15	1.10	0.34	3.41	1.42	0.39	−0.29
Discrete WJ	3.50	2.16	1.16	0.42	3.34	1.41	0.47	−0.25
GARCH(1,1)	4.35	4.79	3.74	2.55	1.79	1.87	1.26	0.34
CHMM-N	4.88	3.70	3.27	2.38	4.08	2.38	1.60	0.40
CHMM-t	**8.46**	**2.65**	**2.06**	**1.35**	**5.55**	**1.89**	**1.15**	**0.21**
CHMM-L	6.33	2.45	1.80	1.02	5.49	1.97	1.04	0.07

Table 14. Joint simulation-based p-value coverage (Pipeline A, SPY). Each cell is the two-sided simulation-based p-value for the observed statistic under the generator. $\bar{pv}$ is the mean across {kurt, avg leverage, agg kurt $h = 5$, agg kurt $h = 21$}; larger $\bar{pv}$ means better joint coverage.

Model	p_{kurt}	p_{acf}	p_{lev}	p_{ak5}	p_{ak21}	$\bar{pv}$ IS	$\bar{pv}$ OoS
Bootstrap	0.894	0.000	0.000	0.012	0.005	0.228	0.466
Gaussian	0.000	0.000	0.000	0.000	0.002	0.000	0.116
Laplace	0.002	0.000	0.000	0.000	0.002	0.001	0.156
Discrete NJ	0.000	0.000	0.000	0.034	0.029	0.016	0.376
Discrete WJ	0.000	0.000	0.000	0.060	0.057	0.029	0.384
GARCH(1,1)	0.271	0.021	0.014	0.903	0.974	0.540	0.468
CHMM-N	0.009	0.000	0.000	0.659	0.894	0.390	0.539
CHMM-t	0.872	0.000	0.000	0.196	0.228	0.324	0.661
CHMM-L	0.369	0.000	0.000	0.061	0.130	0.140	**0.692**



B CHMM Algorithms and Derivations

B.1 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [15, 45]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1), \tag{18}$$

$$\log \alpha_t(j) = \operatorname*{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log b_j(O_t), \quad t = 2, \ldots, T, \tag{19}$$

$$\log \beta_T(k) = 0, \tag{20}$$

$$\log \beta_t(i) = \operatorname*{logsumexp}_j(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \ldots, 1, \tag{21}$$

where $\text{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i,j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i,j) = \frac{\alpha_t(i)\, T_{ij}\, b_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, b_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{22}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i,j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{23}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [16], Liu and Rubin [17] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{24}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{25}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee.

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $b_k(x) = \frac{1}{2b_k} \exp(-|x - \mu_k|/b_k)$ the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k)), \qquad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, |O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{26}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

B.2 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 is the unified EM procedure shared across all three emission families. It makes explicit the *shared scaffold* (quantile-based initialization, log-space forward-backward, transition update) and the three *family-specific branches* (Gaussian M-step for CHMM-N, ECM with one-dimensional golden-section ν-search for CHMM-t, closed-form weighted-median + weighted-MAD for CHMM-L); the branch points are lines 3, 12, and 28. Visualizing the three variants as a single algorithm with three M-step branches makes the architectural contribution of the paper (one EM engine, three interchangeable emission families) immediate from the pseudocode. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 5 details the SIM regression-and-resampling pipeline.

B.3 CHMM-t Degrees-of-Freedom Diagnostics

This appendix backs the discussion in Section 7 of the CHMM-t IS kurtosis overshoot. Figure 8 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit on SPY IS (seed = 20260420). Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

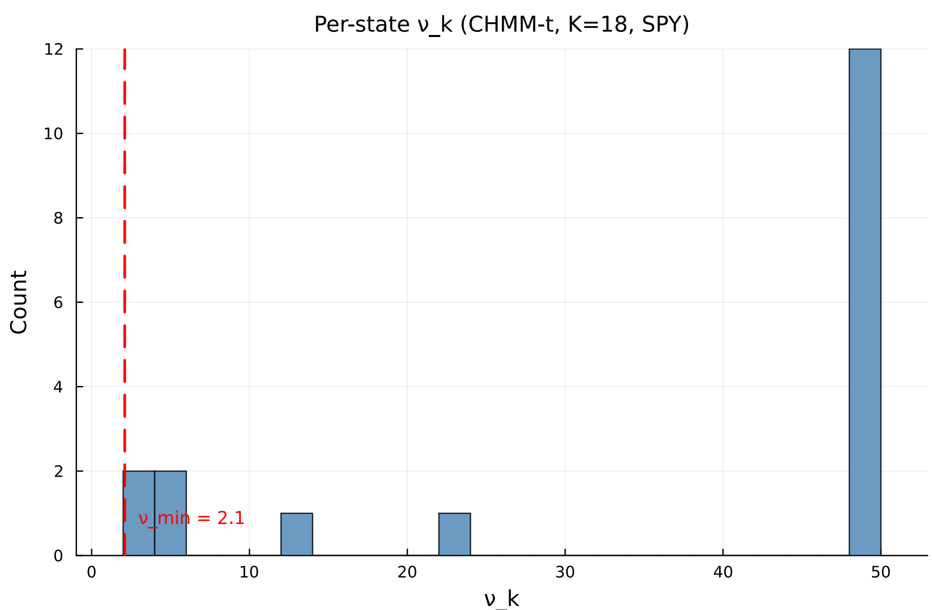


Figure 8. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 12, and 28 branch on the emission family $F \in \{\mathcal{N}, t, L\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t instantiates an ECM sequence whose second CM step updates ν_k by a one-dimensional golden-section search on the per-state Q-function.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$.

Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.

1: **Quantile-Based Initialization.**
2: Sort $O^{(\text{sort})} \leftarrow \text{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \text{mean}(C_k), \quad \sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \text{mean}(C_k), \quad \sigma_k \leftarrow \max(\text{std}(C_k), 10^{-6}), \quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \text{median}(C_k), \quad b_k \leftarrow \max(\text{mean}(|C_k - \mu_k|), 10^{-6})$.
8: **end for**
9: $T_{ij} \leftarrow 1/K; \quad \pi_k \leftarrow 1/K; \quad \mathcal{L}_{\text{prev}} \leftarrow -\infty$.
10: **EM Loop.**
11: **for** $n = 1$ to `max_iter` **do**
12:   **E-Step, per-family emission log-likelihood.**
13:   **switch** F
14:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
15:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
16:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
17:   **E-Step, shared log-space forward-backward.**
18:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
19:   **for** $t = 2$ to T, $j = 1$ to K **do**
20:     $\log \alpha_t(j) \leftarrow \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
21:   **end for**
22:   $\log \beta_T(k) \leftarrow 0$.
23:   **for** $t = T - 1$ down to 1, $i = 1$ to K **do**
24:     $\log \beta_t(i) \leftarrow \underset{j}{\text{logsumexp}}(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
25:   **end for**
26:   Compute $\gamma_t(k)$ and $\xi_t(i, j)$ from equation (22).
27:   **Shared transition update.** $T_{ij} \leftarrow \sum_t \xi_t(i, j) \,/\, \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
28:   **M-Step, per-family emission update.**
29:   **switch** F
30:     $F = \mathcal{N}$ {classical Baum-Welch [14].}
31:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
32:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
33:     $F = t$ {ECM of Peel and McLachlan [16], Liu and Rubin [17]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
34:       $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
35:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
36:       $\sigma_k \leftarrow \max\big(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\big)$.
37:       $\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k) \, \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
38:     $F = L$ {closed-form weighted Laplace MLE.}
39:       $\mu_k \leftarrow \text{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
40:       $b_k \leftarrow \max\big(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\big)$.
41:   **Convergence check.** $\mathcal{L}^{(n)} \leftarrow \underset{k}{\text{logsumexp}}(\log \alpha_T(k))$.
42:   **if** $|\mathcal{L}^{(n)} - \mathcal{L}_{\text{prev}}| < \text{tol}$ **then**
43:     **break**
44:   **end if**
45:   $\mathcal{L}_{\text{prev}} \leftarrow \mathcal{L}^{(n)}$.
46: **end for**
47: **return** $\mathbf{T}, \boldsymbol{\theta}_{1:K}, \boldsymbol{\pi}$ {$\boldsymbol{\theta}_k = (\mu_k, \sigma_k)$ for $F = \mathcal{N}$; (μ_k, σ_k, ν_k) for $F = t$; (μ_k, b_k) for $F = L$.}

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$
Ensure: Most probable state sequence $S^*_{1:T}$

for $k = 1$ to K **do**
  $\log \delta_1(k) \leftarrow \log(1/K) + \log \mathcal{N}(O_1 \mid \mu_k, \sigma_k^2)$.
end for
for $t = 2$ to T **do**
  for $j = 1$ to K **do**
    $\text{vals}(i) \leftarrow \log \delta_{t-1}(i) + \log T_{ij}$ for all i.
    $\log \delta_t(j) \leftarrow \max_i \text{vals}(i) + \log \mathcal{N}(O_t \mid \mu_j, \sigma_j^2)$.
    $\psi_t(j) \leftarrow \arg\max_i \text{vals}(i)$.
  end for
end for
$S^*_T \leftarrow \arg\max_k \log \delta_T(k)$.
for $t = T - 1$ down to 1 **do**
  $S^*_t \leftarrow \psi_{t+1}(S^*_{t+1})$.
end for
return $S^*_{1:T}$

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

for $p = 1$ to P **do**
  Sample initial state $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$.
  for $t = 1$ to M **do**
    Emit observation $\hat{G}^{(p)}_t \sim \mathcal{N}(\mu_{S_t}, \sigma^2_{S_t})$.
    if $t < M$ **then**
      Transition to next state $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$.
    end if
  end for
end for
return $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

Table 15 reports the seven-metric panel for CHMM-t fits whose ECM bracket lower bound is lifted from $\nu_{\min} = 2.1$ through $\{2.5, 3.0, 4.0\}$, plus the Gaussian limit ($\nu = \infty$, equivalent to CHMM-N). Lifting the bracket does not reduce the IS kurtosis overshoot: the simulated kurtosis moves within the 12.5–14.0 band across bracket floors and only collapses to 5.05 in the full Gaussian limit. The overshoot is therefore a feature of the Student-t emission family at $K = 18$ and not a lower-bracket artefact.

Table 15. CHMM-t $\nu_{\min}$ bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths).

$\nu_{\min}$	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
2.1	95.7	90.2	86.2	82.0	13.95	10.08	0.0546	0.100	0.0759
2.5	95.8	90.6	84.2	79.1	15.12	10.34	0.0547	0.100	0.0759
3.0	93.8	89.5	85.3	80.4	12.25	9.65	0.0548	0.099	0.0760
4.0	95.5	89.5	85.0	80.3	13.25	9.39	0.0550	0.101	0.0759
∞ (Gauss)	94.1	86.7	84.2	75.5	5.03	4.46	0.0510	0.110	0.0809
Observed	–	–	–	–	7.68	5.29	–	–	–

Algorithm 4 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (PSD, unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

for $p = 1$ to P **do**
  Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T\times d}$ from $\mathcal{C}(\Sigma)$:
    $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T\times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
    $Y \leftarrow ZL^\top$
  if $\mathcal{C} =$ Gaussian **then**
    $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
  else
    Draw $W \sim \chi_\nu^2/\nu$ i.i.d. for $t = 1, \ldots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
    $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
  end if
  for $j = 1$ to d **do**
    Simulate $\tilde{g}_{j,1:T}^{(p)}$ from $\mathcal{M}_j$ via Algorithm 3.
    Compute order statistics $\tilde{g}_{j,(1)}^{(p)} \leq \cdots \leq \tilde{g}_{j,(T)}^{(p)}$.
    Compute ranks $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$ for $t = 1, \ldots, T$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}_{j,(r_{j,t}^{(p)})}^{(p)}$ for $t = 1, \ldots, T$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

Algorithm 5 Single Index Model (SIM) construction and simulation

Require: IS return matrix $\mathbf{R} \in \mathbb{R}^{T\times d}$, market-column index m, market CHMM $\mathcal{M}_m$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

OLS fit. For $j \neq m$:
  $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m})/\text{Var}(\mathbf{R}_{:,m})$
  $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$
  $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}, \quad R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$
for $p = 1$ to P **do**
  Simulate market path $G_{m,1:T}^{(p)}$ from $\mathcal{M}_m$ via Algorithm 3.
  $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$.
  for $j \neq m$ **do**
    Draw residual indices $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \ldots, T\})$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

C State-Resolution and Multi-Emission Supplementary Material

C.1 Empirical Stylized-Fact Visualization (Companion to Table 1)

Figure 9 visualizes the stylized facts of SPY daily excess growth rates whose Jarque–Bera and Ljung–Box test statistics are reported in Table 1.

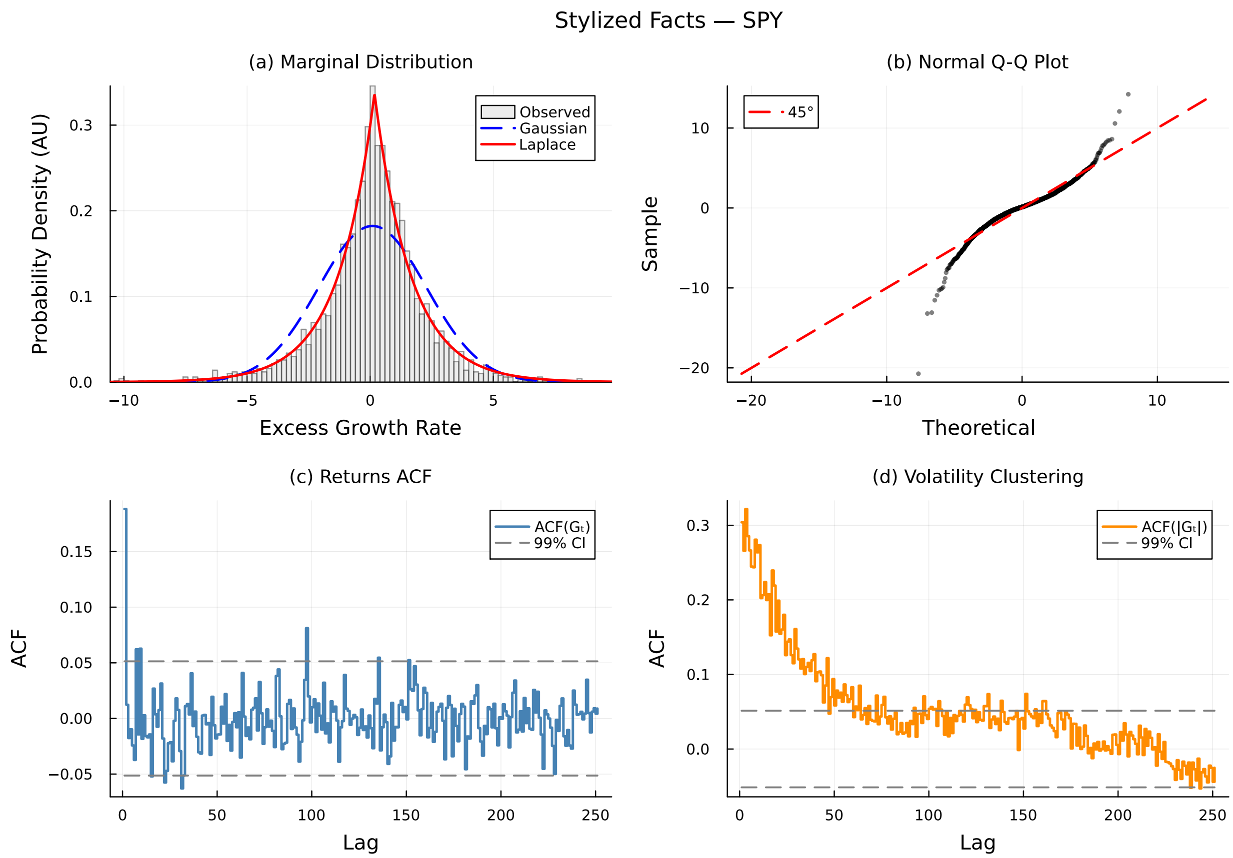


Figure 9. Empirical stylized facts of SPY daily excess growth rates (IS, 2014-01-03 through 2024-01-03). Panel (a): leptokurtic marginal distribution with Gaussian and Laplace fits. Panel (b): normal Q-Q plot confirming heavy tails. Panel (c): near-zero autocorrelation of raw returns. Panel (d): persistent autocorrelation of absolute returns characterizing volatility clustering.

C.2 Full State-Resolution Sensitivity Table

Table 16 reports the full K-sweep referenced in Section 5.4 of the main text. Observed excess kurtosis is 7.68 IS and 5.29 OoS at all K.

Table 16. State resolution sensitivity for SPY CHMM-N (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 3 in the main text. IS coverage was 100% at every K; OoS coverage is reported separately.

	KS (%)		AD (%)		Kurtosis					
K	IS	OoS	IS	OoS	Obs	Sim	ACF-MAE	W_1	H	Cov OoS (%)
3	88.8	77.1	79.2	67.9	7.68	3.80	0.0466	0.130	0.0836	94.9
6	93.1	80.0	87.1	74.6	7.68	5.26	0.0499	0.113	0.0828	90.9
9	94.6	82.4	85.8	73.6	7.68	4.56	0.0494	0.116	0.0806	93.9
12	94.6	82.8	86.0	74.9	7.68	4.57	0.0495	0.112	0.0802	88.9
15	93.9	82.7	86.9	75.6	7.68	4.89	0.0505	0.116	0.0806	90.9
18	95.6	81.7	88.4	73.2	7.68	4.98	0.0509	0.110	0.0805	90.9
21	**96.1**	**86.6**	**89.0**	**78.4**	7.68	3.83	0.0532	0.111	0.0798	**100.0**

C.3 Multi-Emission State-Resolution Sensitivity

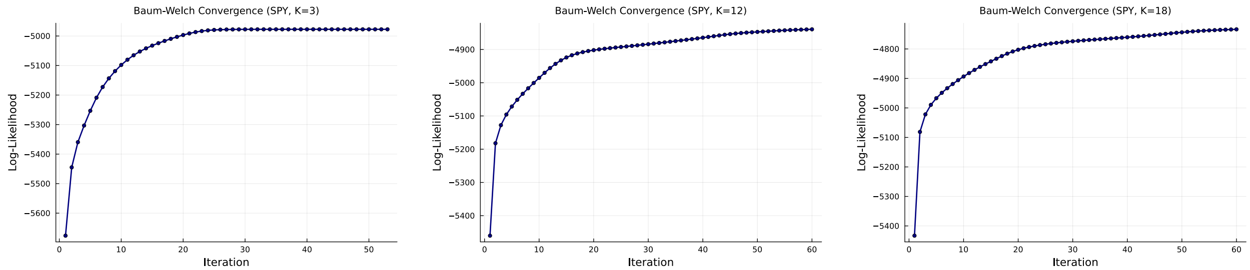
Table 17 extends Table 16 across the three emission families, confirming that the qualitative operating point $K = 18$ is stable when the Gaussian component is replaced with Student-t or Laplace emissions. For CHMM-t the per-state degrees of freedom ν_k are refit by ECM-plus-golden-section at every K; for CHMM-L the weighted-median location and weighted MAD scale are closed form and impose no additional cost. Three patterns extend from Table 16 to the new families. First, the distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band for all three families: CHMM-t reaches its highest IS KS at $K = 18$ (95.2%) and its highest OoS KS at $K = 18$ (85.1%), and CHMM-L reaches its highest IS KS at $K = 18$ (96.2%) and its highest IS AD at $K = 18$ (93.0%). Once a family enters the plateau the operating point is flat with respect to K, so we report the main-text results at the same $K = 18$ used for CHMM-N. Second, the ACF-MAE remains within the narrow 0.047–0.056 band observed for the Gaussian sweep, indicating that the volatility-clustering mechanism is a property of the regime structure rather than the emission family. Third, simulated kurtosis behaves as expected from the component families: the Gaussian sweep peaks around 5.0 at the high-K end, the Laplace sweep lands squarely in the 5.2–6.6 range (closest to the observed 7.68), and the Student-t sweep climbs sharply above 13 across the entire sweep because the ECM search assigns small ν_k to a subset of tail states, over-correcting IS kurtosis.

Table 17. Multi-emission state resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). CHMM-N: Gaussian; CHMM-t: Student-t with per-state ν_k refit by ECM-plus-golden-section; CHMM-L: Laplace with weighted-median location and weighted-MAD scale. Every metric is computed on the identical simulation pipeline used for Table 3. Observed excess kurtosis is 7.68 IS and 5.29 OoS.

		KS (%)		AD (%)					
Family	K	IS	OoS	IS	OoS	Kurt (IS sim)	ACF-MAE	W_1 IS	H IS
CHMM-N	3	89.8	76.3	81.3	67.7	3.81	0.0468	0.128	0.0835
	6	91.7	79.6	85.9	73.6	5.30	0.0497	0.115	0.0831
	9	92.8	82.2	84.7	75.5	4.55	0.0497	0.118	0.0810
	12	95.0	83.7	86.1	76.6	4.55	0.0497	0.114	0.0805
	15	95.2	83.5	86.9	76.3	4.91	0.0508	0.111	0.0802
	18	94.7	82.4	85.8	76.5	5.02	0.0509	0.111	0.0809
	21	**95.2**	**87.6**	**89.1**	**80.9**	3.80	0.0530	0.112	0.0802
CHMM-t	3	91.1	81.6	85.2	77.1	27.12	0.0540	0.109	0.0764
	6	92.2	82.1	86.1	76.6	20.80	0.0533	0.108	0.0768
	9	92.7	84.8	86.9	80.1	15.96	0.0537	0.104	0.0756
	12	94.1	83.1	88.3	80.6	14.68	0.0542	0.102	0.0757
	15	93.5	82.9	88.0	76.5	15.85	0.0537	0.103	0.0757
	18	**95.2**	**85.1**	**90.9**	**80.7**	14.40	0.0550	**0.101**	**0.0756**
	21	94.9	82.2	89.3	77.8	13.28	0.0540	0.103	0.0758
CHMM-L	3	82.4	63.6	82.9	66.9	5.24	0.0533	0.112	0.0853
	6	88.8	77.0	83.3	72.5	5.74	0.0511	0.113	0.0823
	9	90.0	77.9	84.1	74.8	6.05	0.0519	0.113	0.0823
	12	92.4	77.8	87.6	77.9	6.21	0.0540	0.106	0.0797
	15	92.6	79.4	87.0	77.6	6.06	0.0542	0.106	0.0795
	18	**96.2**	**80.9**	**93.0**	**80.6**	**6.60**	0.0564	**0.101**	0.0793
	21	90.7	80.7	87.3	78.8	5.99	0.0544	0.107	0.0809

C.4 Baum-Welch Convergence

Figure 10 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.



(a) $K = 3$ **(b)** $K = 12$ **(c)** $K = 18$ (selected)

Figure 10. Baum-Welch convergence for the SPY CHMM-N fit (Pipeline A, IS) at selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` $= 60$.

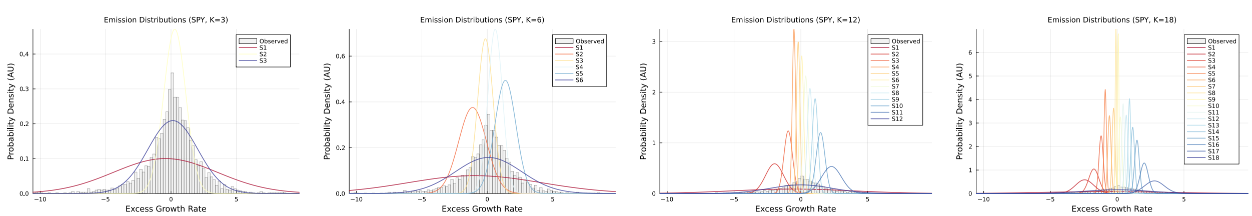
C.5 Emission Distributions Across State Resolutions

Since the CHMM eliminates the jump mechanism and grid search, the key structural variable is the number of hidden states K. Figure 11 compares the learned emission distributions for $K \in \{3, 6, 12, 18\}$, illustrating how increasing state resolution refines the decomposition of the observed return distribution.

At $K = 3$, the model partitions the return space into three broad regimes: a crash state (large negative μ, high σ), a calm state (near-zero μ, low σ), and a boom state (positive μ, moderate σ). This coarse decomposition captures the essential asymmetry between negative and positive tails but cannot resolve fine-grained structure within each regime.

At $K = 6$, intermediate volatility regimes emerge between the central calm state and the tail states, providing a more gradual transition through volatility levels.

At $K = 12$–18, the emission distributions form a dense coverage of the return space, with each state capturing a narrow band of returns. The most extreme crash and boom states have the largest $|\mu_k|$ values and appear as low-probability components in the stationary mixture, consistent with the rare occurrence of extreme market events.



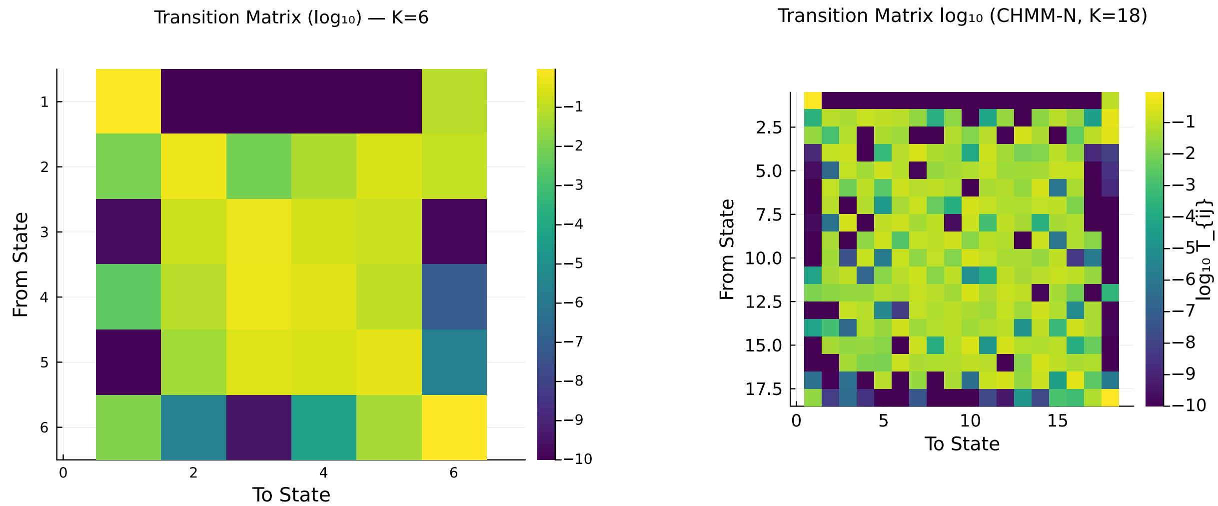
(a) $K = 3$ **(b)** $K = 6$ **(c)** $K = 12$ **(d)** $K = 18$ (selected)

Figure 11. Learned CHMM-N (Gaussian) emission distributions at different state resolutions, overlaid on the observed SPY IS return histogram (Pipeline A). Higher K values produce finer decomposition with dedicated states capturing tail behavior.

C.6 Transition Matrix Structure

Figure 12 shows the learned transition matrices for $K = 6$ and $K = 18$ in $\log_{10}$ color scale. Both matrices exhibit a dominant near-diagonal band, reflecting the tendency of the market to remain in its current regime. Off-diagonal entries decay with distance from the diagonal, indicating that large regime jumps (for example, from a deep crash state directly to a boom state) are rare but nonzero.

The diagonal dominance is critical for the ACF-MAE result discussed in Section 7: the self-transition probabilities T_{kk} for central states are typically 0.7–0.95, producing mean sojourn times of 3–20 days. The mixture of these geometric sojourn times across states produces the sum-of-exponentials ACF profile that approximates the observed slow decay.



(a) $K = 6$

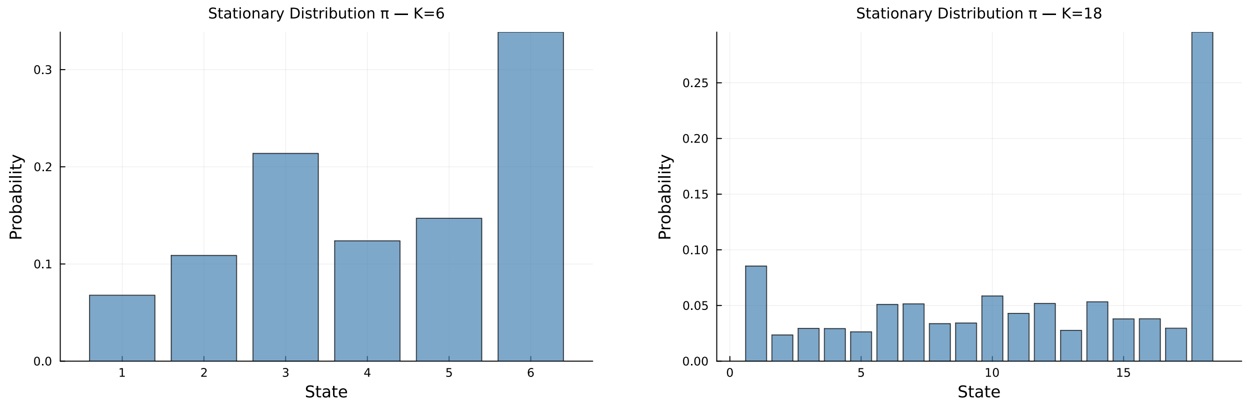
(b) $K = 18$ (selected)

Figure 12. Learned CHMM-N transition matrices for the SPY IS fit (Pipeline A) at $K = 6$ and $K = 18$ in $\log_{10}$ color scale. The near-diagonal band reflects regime persistence; off-diagonal entries decay with distance, indicating that large regime transitions are rare.

C.7 Stationary Distributions and Residence Times

Figure 13 shows the stationary distributions $\bar{\boldsymbol{\pi}}$ for $K = 6$ and $K = 18$. In both cases, the central states (those with μ_k near zero and moderate σ_k) receive the highest stationary probability, consistent with the empirical concentration of returns near zero. Tail states (crash and boom) receive low stationary probability, consistent with the rarity of extreme events.

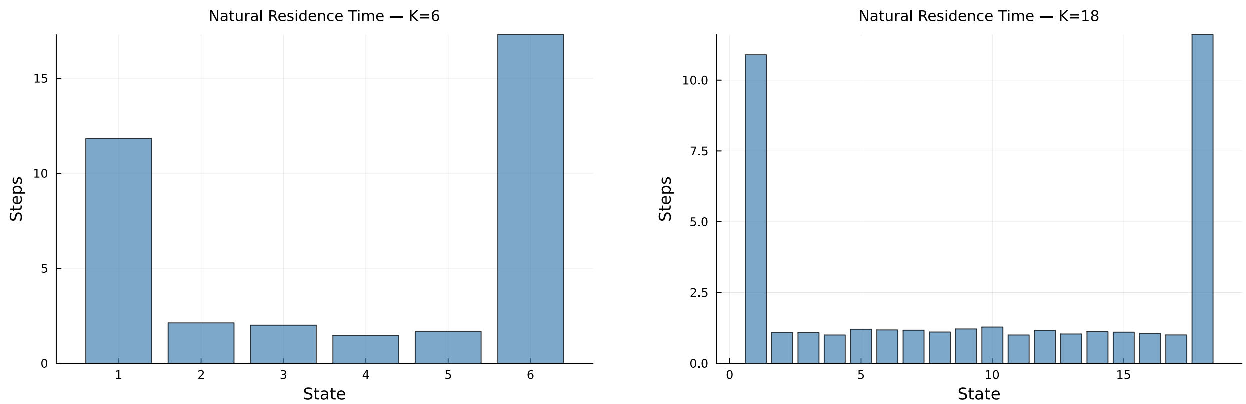
Figure 14 shows the natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for $K = 6$ and $K = 18$. Under pure Markov dynamics (no jump mechanism), the expected duration in state k before transitioning is τ_k steps. Central states exhibit the longest residence times (5–20 days), while tail states have shorter residence times (1–3 days), consistent with the transient nature of extreme market events. Notably, even without a jump mechanism, the tail states at higher K maintain sufficient self-transition probability ($T_{kk} \approx 0.3$–0.5) to produce multi-day tail episodes. This contrasts with the discrete model of Alswaidan and Varner [12], where tail-state residence times were too short to sustain clustering without the Poisson jump extension.



(a) $K = 6$

(b) $K = 18$

Figure 13. Stationary distributions $\bar{\boldsymbol{\pi}}$ for the SPY CHMM-N IS fit (Pipeline A) at $K = 6$ and $K = 18$. Central states receive the highest stationary probability, consistent with the concentration of returns near zero.



(a) $K = 6$

(b) $K = 18$

Figure 14. Natural residence times $\tau_k = 1/(1 - T_{kk})$ per state for the SPY CHMM-N IS fit (Pipeline A) at $K = 6$ and $K = 18$. Central states have the longest residence times (5–20 days), while tail states are shorter-lived (1–3 days). This asymmetry drives volatility clustering.

C.8 IS Comparison at Non-Selected State Resolutions

Figures 15–18 show the IS model comparison panels for $K \in \{3, 6, 12, 21\}$, complementing the $K = 18$ comparison in Figure 4 of the main text. Across all resolutions, the CHMM's simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel. $K = 9$ and $K = 15$ panels are omitted because they interpolate smoothly between the shown resolutions; their numerical metrics are in Table 16.

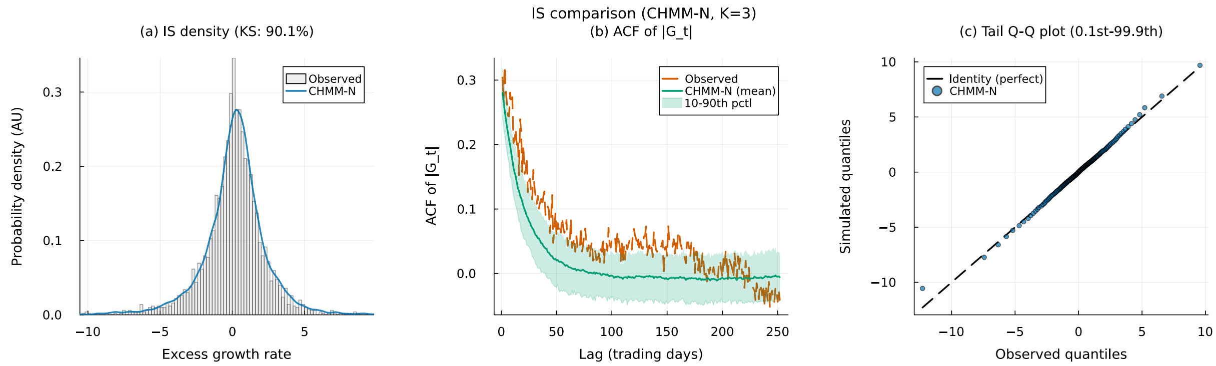


Figure 15. IS SPY CHMM-N comparison at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4). The $K = 18$ counterpart is Figure 4 of the main text.

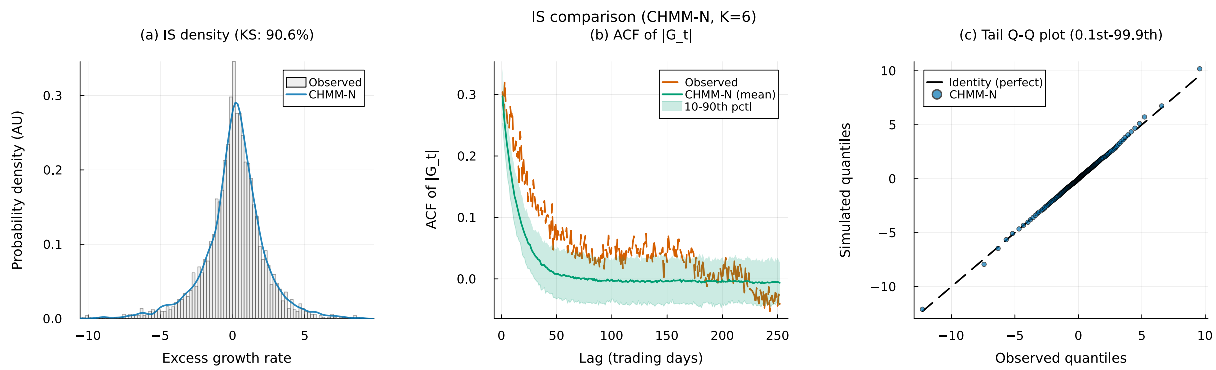


Figure 16. IS SPY CHMM-N comparison at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

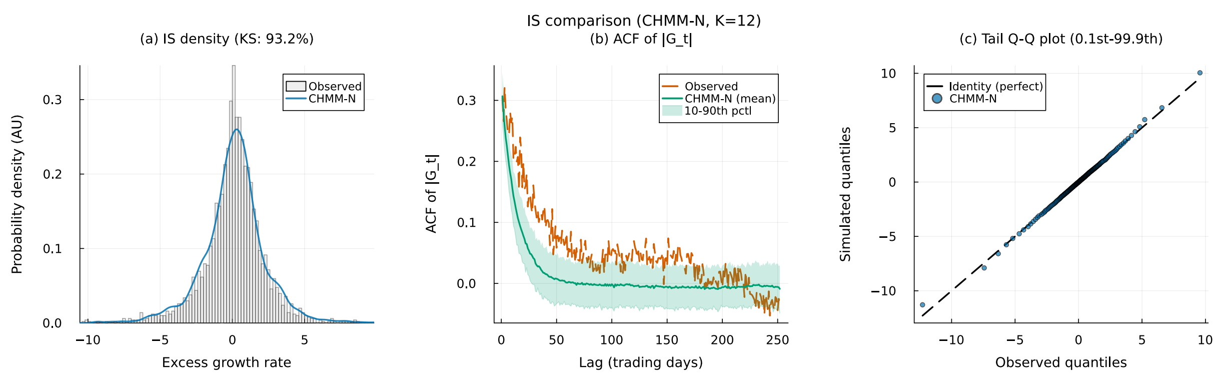


Figure 17. IS SPY CHMM-N comparison at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

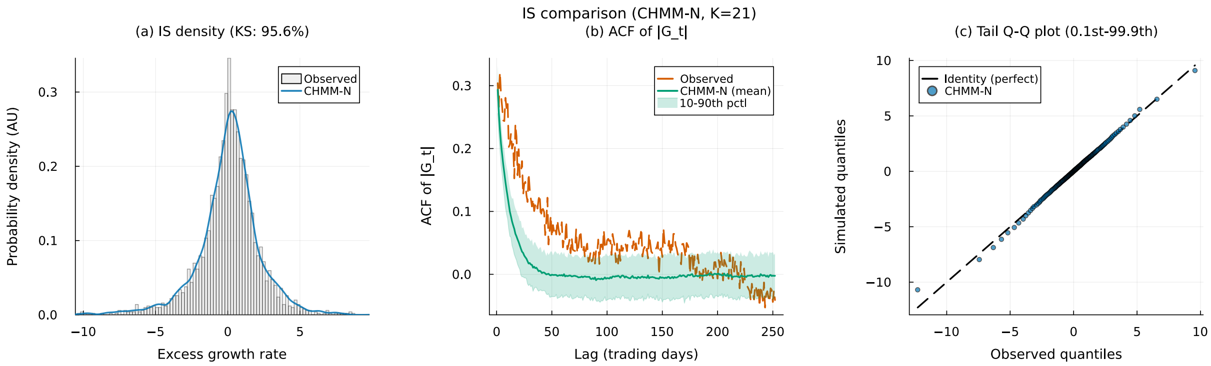


Figure 18. IS SPY CHMM-N comparison at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 4).

C.9 OoS Validation at Non-Selected State Resolutions

Figures 19–22 show the OoS validation panels for $K \in \{3, 6, 12, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 5 of the main text. As with the IS panels, $K = 9$ and $K = 15$ are omitted; numerical OoS KS pass rates are in Table 16.

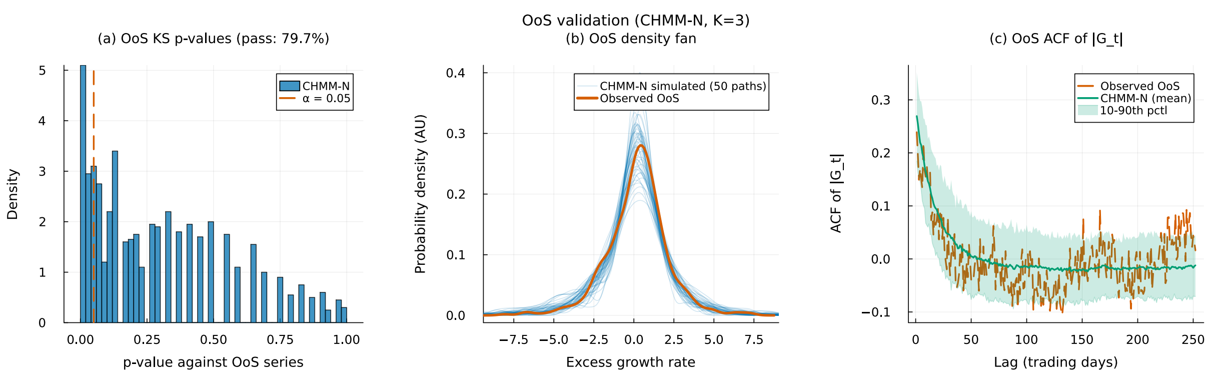


Figure 19. OoS SPY CHMM-N validation at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5). The $K = 18$ counterpart is Figure 5 of the main text.

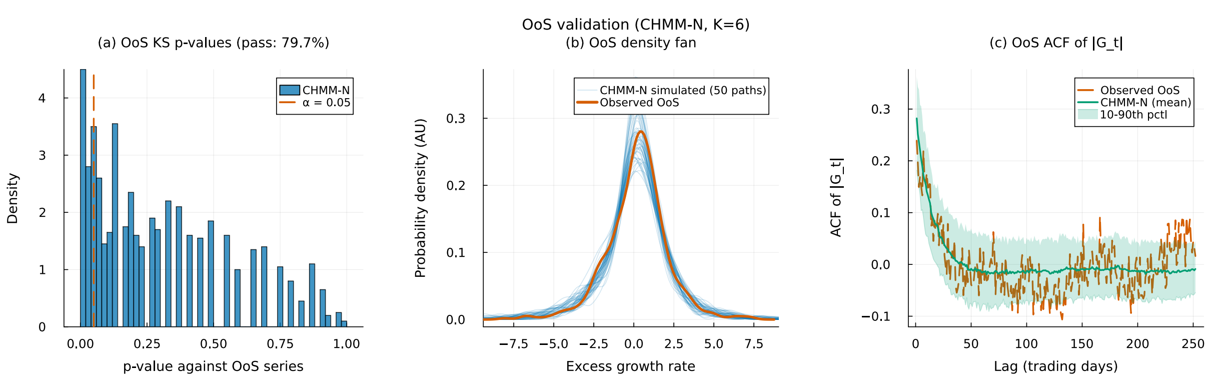


Figure 20. OoS SPY CHMM-N validation at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

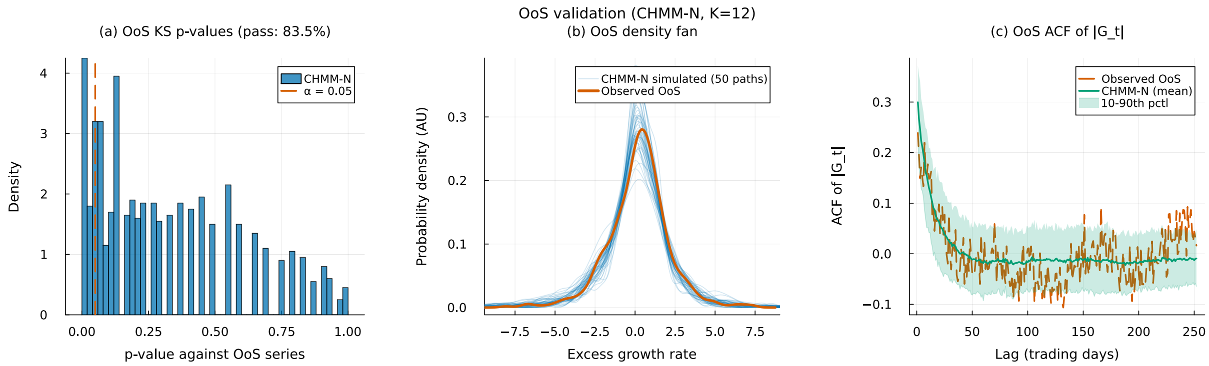


Figure 21. OoS SPY CHMM-N validation at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

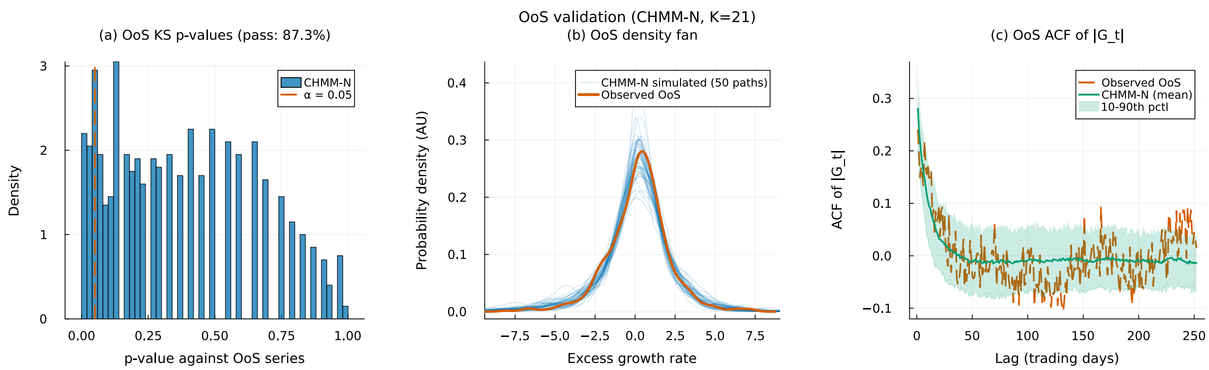
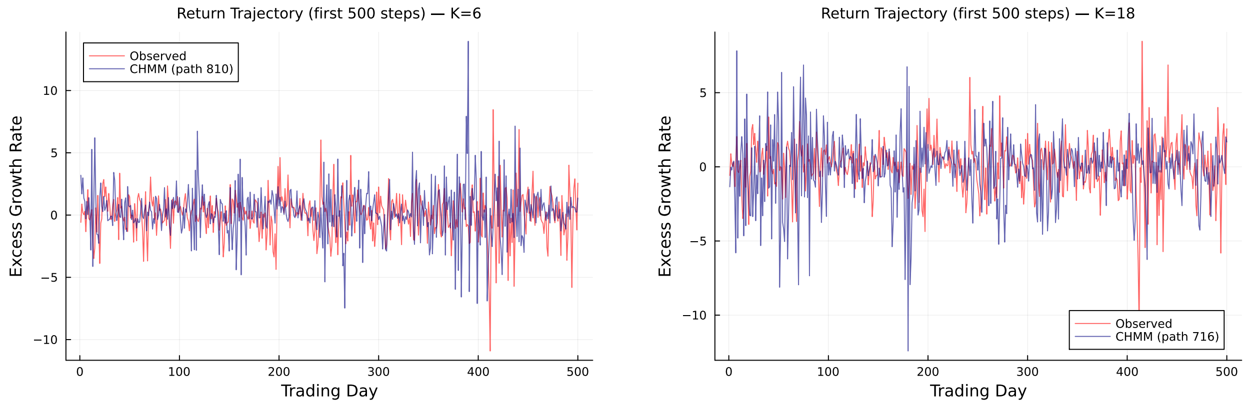


Figure 22. OoS SPY CHMM-N validation at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 5).

C.10 Example Simulated Trajectories (Illustrative, IS-Fitted CHMM-N)

Figure 23 shows example simulated return trajectories from the SPY CHMM-N at $K = 6$ and $K = 18$, illustrating the qualitative dynamics produced by the model. Each panel overlays one unconditional simulated path (blue; the sampled-path index is labelled in the figure legend) against the first 500 trading days of the observed IS SPY excess growth-rate series (red). The two lines do not need to agree pointwise (simulated and observed hidden-state sequences are drawn independently); the intended read is qualitative, showing that both trajectories exhibit the characteristic pattern of financial returns: extended periods of low volatility punctuated by clusters of large-magnitude events.



(a) $K = 6$

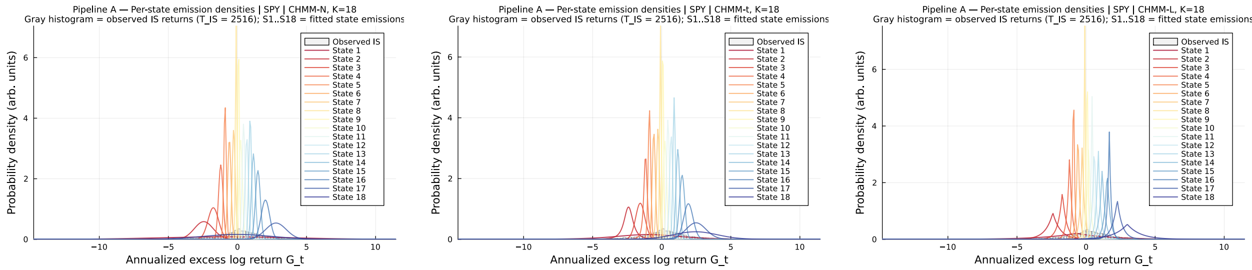
(b) $K = 18$ (selected)

Figure 23. Example simulated return trajectories from the SPY CHMM-N (Pipeline A, IS fit, first 500 simulated steps). Blue: one unconditional simulated path (index labelled in the figure legend); red: first 500 trading days of the observed IS SPY excess growth rate series. The alternation between calm (central states) and volatile (tail states) periods produces the volatility clustering characteristic of empirical financial returns. This figure is qualitative: simulated and observed state sequences are independent, so point-by-point agreement is not expected.

C.11 Multi-Emission Figures at $K = 18$

This appendix reproduces the standard diagnostic suite (emission PDFs, residence times, IS comparison, OoS validation, and convergence) for the two heavy-tailed emission families (CHMM-t and CHMM-L) at the selected state resolution $K = 18$; transition matrices and stationary distributions are discussed in prose only below, since they are visually indistinguishable across families at $K = 18$. The Gaussian counterparts are already shown as Figures 3, 4, 5, 10, 12, 13, and 14 in the main text and above in this appendix. The figures confirm that the shared forward–backward scaffold produces internally-similar regime structure across the three families: the near-diagonal transition band, the K-component dense emission cover, the longer central-state residence times, and the monotone log-likelihood convergence are features of the CHMM framework and not of the Gaussian component specifically.

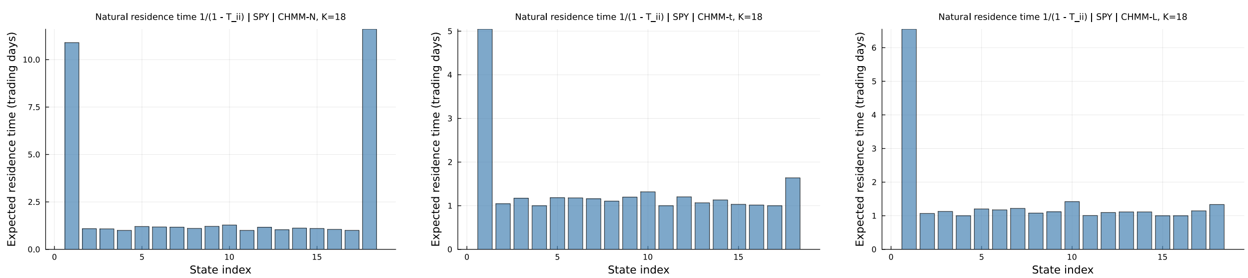
Emission PDFs. Figure 24 compares the learned $K = 18$ emissions for the three families. The CHMM-t panel's per-state ν_k allocation is visible in the narrower modes of certain tail states; the CHMM-L panel's Laplace components have sharper peaks and slower exponential tail decay than the Gaussians.



(a) CHMM-N (Gaussian) **(b)** CHMM-t (Student-t) **(c)** CHMM-L (Laplace)

Figure 24. Learned emission distributions at $K = 18$ across the three emission families (CHMM-N Gaussian, CHMM-t Student-t with per-state ν_k, CHMM-L Laplace), overlaid on the observed SPY IS return histogram (Pipeline A). Heavier-tailed emission families (t, L) give individual components more tail mass, which is what closes the kurtosis gap of the Gaussian CHMM.

Transition matrices and stationary distributions. The $K = 18$ transition matrices and stationary distributions learned under CHMM-t and CHMM-L are visually indistinguishable from their CHMM-N counterparts (right-hand $K = 18$ panels of Figures 12 and 13): the near-diagonal band, the magnitude of self-transition probabilities, and the mass concentration at central states are features of the regime structure, not the emission family. This is consistent with the ACF-MAE being stable across the family axis (Table 17). The per-family residence times $\tau_k = 1/(1 - T_{kk})$, which magnify small differences in the self-transition probabilities each family recovers, are compared in Figure 25.



(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L

Figure 25. Natural residence times $\tau_k = 1/(1 - T_{kk})$ at $K = 18$ for the SPY IS fit (Pipeline A) across the three emission families (CHMM-N Gaussian, CHMM-t Student-t, CHMM-L Laplace).

IS comparison. Figures 26–27 reproduce the three-panel IS comparison (density, ACF of $|G_t|$, tail Q-Q) at $K = 18$ for the two heavy-tailed families; the CHMM-N panel is Figure 4 in the main text.

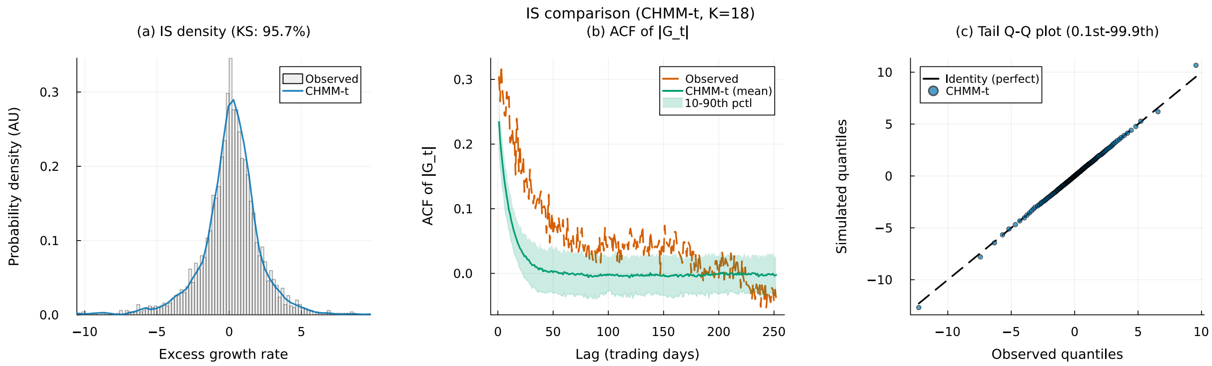


Figure 26. IS SPY model comparison at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) marginal density with KS pass rate, (b) ACF of $|G_t|$, and (c) tail Q-Q plot. The CHMM-N counterpart is Figure 4; the CHMM-L counterpart is Figure 27.

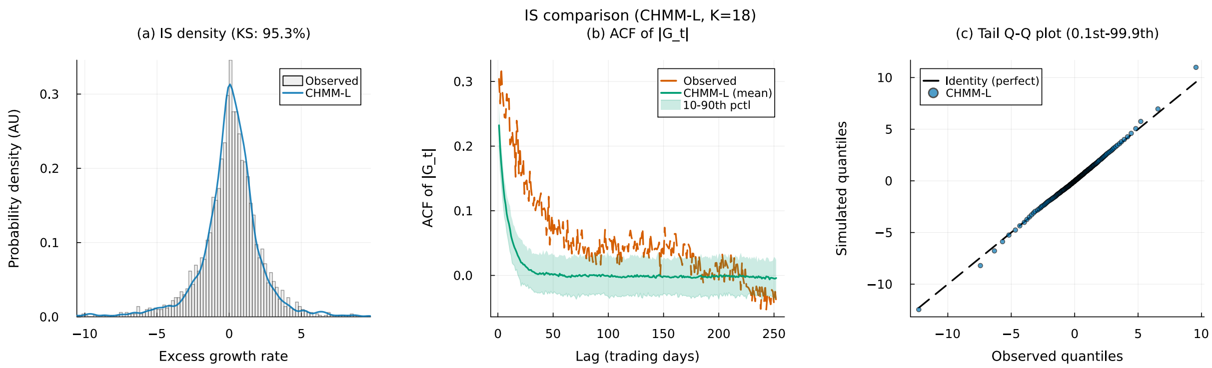


Figure 27. IS SPY model comparison at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 26.

OoS validation. Figures 28–29 show the OoS validation panels for CHMM-t and CHMM-L at $K = 18$. The CHMM-t panel demonstrates the OoS kurtosis alignment noted in the main text (simulated 6.40 vs. observed 6.24), and the CHMM-L panel shows the tighter density fan produced by the heavier Laplace tails relative to the Gaussian.

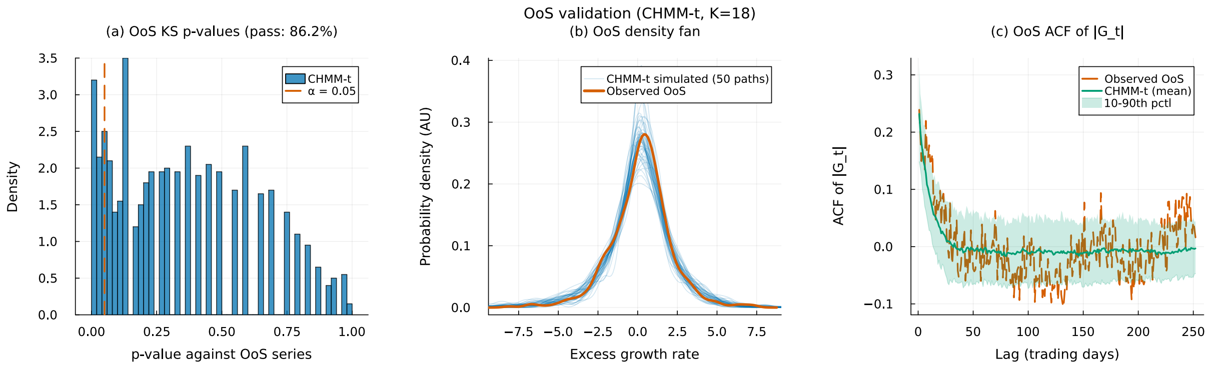


Figure 28. OoS SPY validation at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) KS p-values, (b) OoS density fan, and (c) OoS ACF of $|G_t|$. The CHMM-N counterpart is Figure 5; the CHMM-L counterpart is Figure 29.

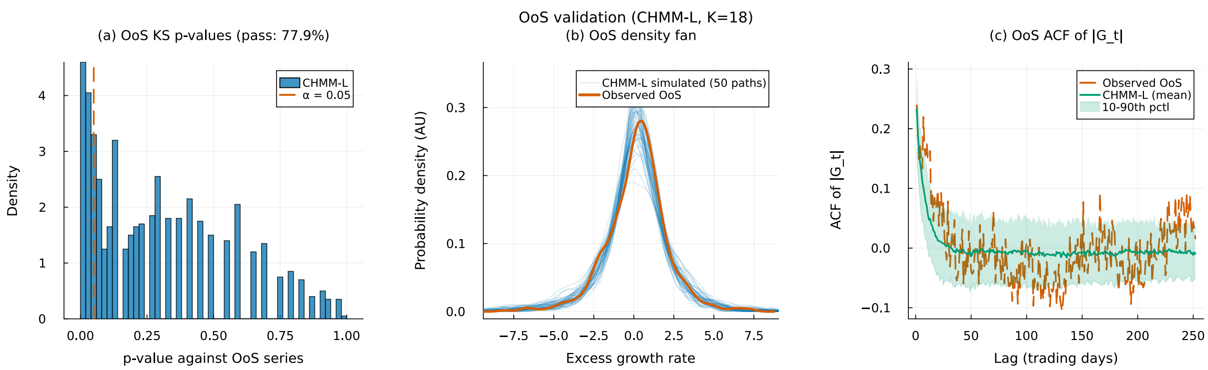
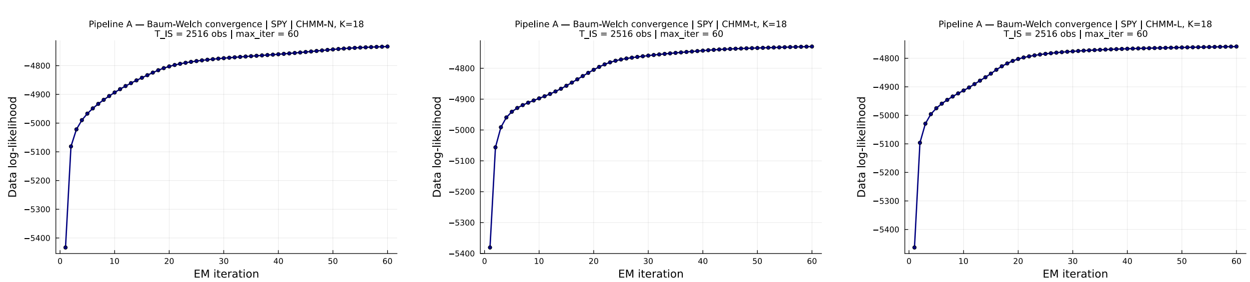


Figure 29. OoS SPY validation at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 28.

Convergence. Figure 30 shows the Baum-Welch / ECM / weighted-EM log-likelihood histories at $K = 18$ for all three families. The CHMM-t and CHMM-L curves inherit the monotone increase of the classical EM guarantee; the CHMM-t curve has a slightly longer pre-plateau phase because each ECM sub-iteration refines ν_k with a golden-section search.



(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L

Figure 30. Log-likelihood histories for the SPY IS fit (Pipeline A) at $K = 18$ across the three emission families (CHMM-N Gaussian classical EM; CHMM-t Student-t ECM with per-state ν_k golden-section sub-step; CHMM-L Laplace weighted-EM). All three curves are monotonically non-decreasing and plateau well before the `max_iter` $= 60$ budget.

C.12 Rydén K = 2 Replication

Table 18 reports the direct replication of Rydén et al.'s 1998 $K = 2$ setting on SPY IS under both initialization strategies, as discussed in Section 7. The quantile-based initialization used in the main body is row 1; five independent random-seed initializations are rows 2–6. Random init here draws per-state (μ_k, σ_k) from a moderate perturbation of the sample moments, which is representative of the random-init practice common in the 1990s HMM literature.

Table 18. Rydén replication: $K = 2$ CHMM-N on SPY IS under quantile vs random initialization (seed sub-index varies the random init).

Init	KS IS (%)	AD IS (%)	ACF-MAE	Kurt IS	KS OoS (%)
Quantile (main body)	79.0	71.5	0.0501	2.57	72.4
Random seed = 1	78.7	70.7	0.0502	2.57	73.2
Random seed = 7	75.3	69.6	0.0498	2.56	74.0
Random seed = 42	77.6	70.0	0.0498	2.59	71.8
Random seed = 100	76.2	69.4	0.0500	2.59	72.3
Random seed = 2024	76.4	69.8	0.0499	2.54	72.9

The key observation is that ACF-MAE is essentially constant at ≈ 0.050 across all six rows, matching the $K = 18$ CHMM-N of the main body. IS KS pass rate remains below the $K = 18$ operating point (75.3–79.0%) because two Gaussian components cannot tile the full multi-regime return distribution. The combination confirms that the Rydén low-K limitation is a distributional (not temporal) artefact: the ACF is already well reproduced at $K = 2$, and it is distributional fidelity that drives the preference for $K = 18$.

D Cross-Asset Dependence Supplementary Material

D.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 5.8 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 5.7) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [20, 29] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [30].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi\tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [22, 42]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \ldots, x_d)^\top$, is

$$\begin{aligned}\log c_t(\mathbf{u}; \Sigma, \nu) \;=\; & \log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1)\, \log \Gamma\!\left(\tfrac{\nu}{2}\right) - d\, \log \Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2} \log |\Sigma| \\ & - \tfrac{\nu+d}{2}\, \log\!\left(1 + \tfrac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2}\, \log\!\left(1 + \tfrac{x_j^2}{\nu}\right). \end{aligned} \tag{27}$$

Summing over $t = 1, \ldots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (27) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 5.8 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 19 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 5.8. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation in equation (5) re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 6.

Table 19. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$). Coefficients $\hat{\alpha}, \hat{\beta}$ and R^2 are reported for the factor equation used in equation (5).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

D.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 31 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 5.8. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature.

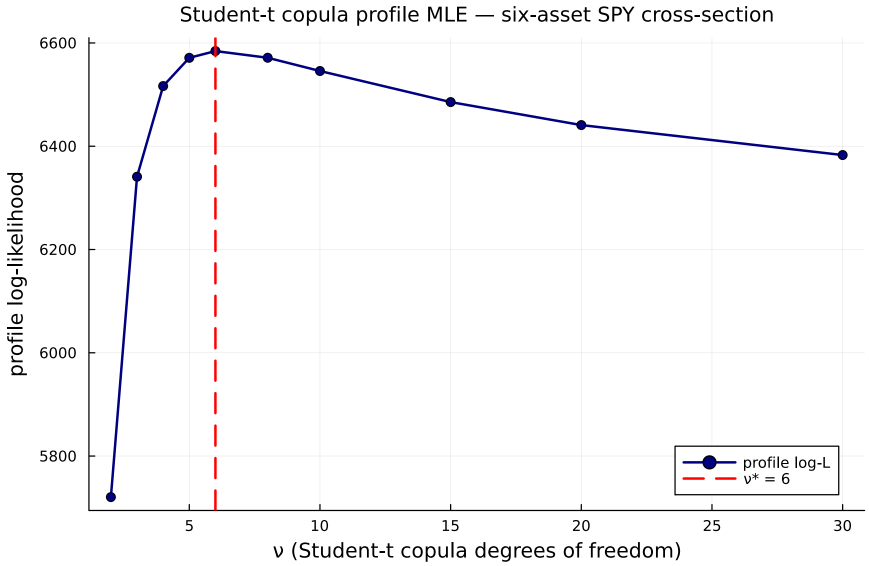


Figure 31. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

D.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 6)

Figure 32 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 5.8. The panel ordering is SIM, Gaussian copula, Student-t copula (each reported quantitatively in Table 6), plus a truncated C-vine variant with SPY as root for visual comparison. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.

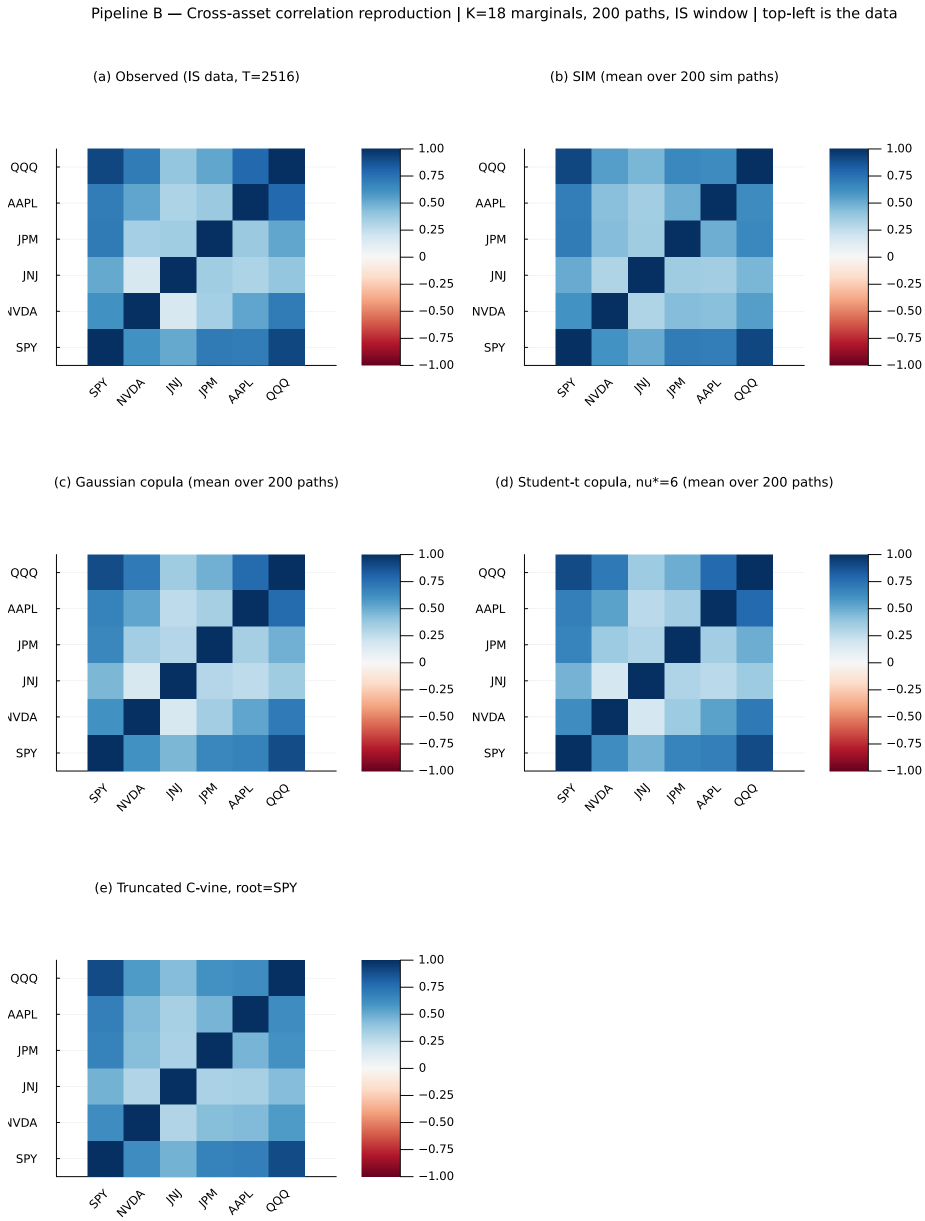


Figure 32. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 6). Panel (a), top-left: *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\mathrm{IS}} = 2{,}516$). Panel (b), top-right: path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c), middle-left: path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d), middle-right: path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e), bottom-left: path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 5; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

D.4 Per-Asset KS Bar Chart (Pipeline B, Sanity Check for Table 6)

Visualisation of the per-asset KS pass rates from Table 6; the SIM-vs-copula gap on JPM and AAPL is the visual headline.

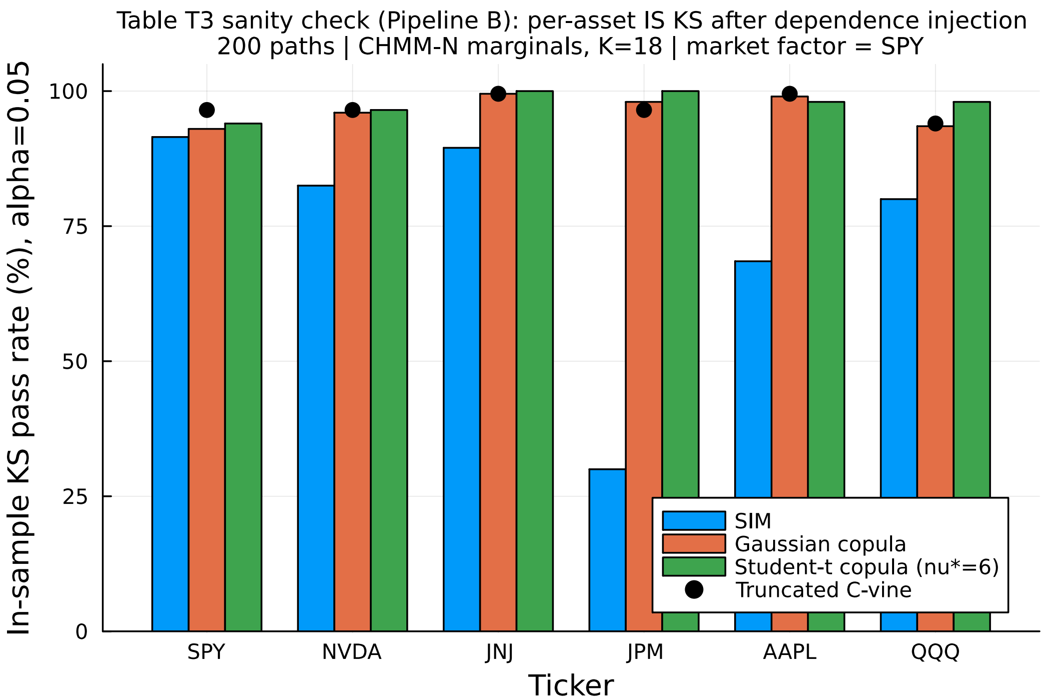


Figure 33. Per-asset IS KS pass rates after dependence injection (Pipeline B sanity check for Table 6). For each of the six tickers and each of the three dependence constructions (SIM, Gaussian copula, Student-t copula with $\nu^* = 6$), the bar reports the percentage of 200 simulated paths whose marginal distribution passes the two-sample KS test against the observed IS series at $\alpha = 0.05$. The market-factor asset SPY is approximately tied across constructions (it is the SIM engine). For non-market assets the copulas dominate because rank reordering preserves each CHMM marginal exactly; the SIM's affine propagation visibly degrades JPM and AAPL.

E Baseline and Per-Ticker Supplementary Material

E.1 OoS KS Test-Power Calibration

Table 20 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 5.5. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\text{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 20. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\text{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\text{OoS}} = 572$ (nearly correct)	90.0
Gaussian$(\hat{\mu}_{\text{IS}}, \hat{\sigma}_{\text{IS}})$ at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 5.5 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

E.2 Walk-Forward Rolling-Window Re-Estimation (Pipeline A)

This robustness check lives inside Pipeline A: the CHMM-N is still fit per ticker independently, only the fit window rolls forward. It does not introduce cross-asset coupling, and therefore should not be interpreted alongside the Pipeline B results in Section 5.8. Table 21 reports the quarterly walk-forward re-estimation results referenced in Section 5.7. The walk-forward protocol refits the CHMM-N every 63 trading days (one quarter) across the 572-day OoS window, feeding the realized returns of the prior quarter into the training set before each refit. The rolling protocol recovers a modest ~ 1 percentage point on JNJ and a substantial ~ 15 percentage points on JPM relative to the fixed-IS baseline, confirming that the stationarity assumption is the principal contributor to the JPM OoS gap while JNJ's OoS gap is already small under the fixed fit.

Table 21. Walk-forward (quarterly CHMM-N refit) vs fixed-IS CHMM-N on JNJ and JPM (Pipeline A, per-ticker independent fits; walk-forward only rolls the fit window, it does not introduce cross-asset dependence). $K = 18$, 1,000 simulated paths per (ticker, mode), seed = 20260420. IS window 2014-01-03 through 2024-01-03; OoS window 2024-01-04 through 2026-04-17 ($T_{\mathrm{OoS}} = 572$). The walk-forward mode refits the CHMM-N every 63 trading days across the OoS window, feeding each completed quarter's realized returns back into the training set before the next refit. Columns report percent KS / AD pass rate at $\alpha = 0.05$, excess kurtosis, ACF-MAE of $|G_t|$, Wasserstein-1, and Hellinger distance.

Mode	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
JNJ fixed-IS	99.3	98.2	94.0	92.1	5.46	4.98	0.0322	0.095	0.0786
JNJ walk-forward	99.3	98.2	94.9	91.3	5.46	5.14	0.0322	0.095	0.0786
JPM fixed-IS	96.6	94.4	49.0	48.6	7.33	6.31	0.0449	0.165	0.0875
JPM walk-forward	96.6	94.4	64.3	63.6	7.33	5.96	0.0449	0.165	0.0875

E.3 Block-Bootstrap Baseline

Table 22 reports the full seven-metric panel for the stationary block bootstrap at block length $b = 10$ trading days, the temporally-aware non-parametric baseline added to Table 3.

Table 22. Stationary block bootstrap (block length $b = 10$ trading days) on SPY (1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	99.2	96.4	7.07	0.0568	0.092	0.0534	100.0
OoS	87.5	86.4	5.34	0.0512	0.225	0.1570	86.9

E.4 Discrete HMM with Bin-Conditional Student-t Emissions

Table 23 reports the full seven-metric panel for the Bin-T NJ variant referenced in Section 5.3: the discrete NJ transition matrix with $K_{\mathrm{disc}} = 13$ quantile bins, but with each bin emitting from a bin-conditional Student-t density rather than from the bin centroid.

The bin-conditional Student-t fit picks $\nu \approx 2.71$ for the two extreme bins (bins 1 and 13, corresponding to the lower and upper quantile tails of the IS distribution) and $\nu = 50$ (upper bracket)

Table 23. Discrete HMM with bin-conditional Student-t emissions on SPY (no jumps, $K_{\text{disc}} = 13$, 1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	95.4	92.1	26.18	0.0619	0.116	0.0929	89.9
OoS	86.7	93.2	10.76	0.0538	0.177	0.1614	93.9

for the eleven central bins, which explains both the large IS simulated kurtosis (26.18) and the very thin-scale central bins ($\sigma \sim 0.06$–0.28).

E.5 GRU Deep-Generative Baseline

Architecture and training details for the GRU baseline used in Section 5.3 are summarised here.

- **Architecture.** A single-layer Gated Recurrent Unit (GRU) with hidden dimension $H = 32$ followed by a Dense head with two outputs ($\mu_t, \log \sigma_t$). Input is a single feature (the standardised return) over a context window of $w = 20$ trading days.
- **Loss.** Per-window negative log-likelihood of a Gaussian next-step density, equation (3).
- **Optimiser.** Adam, learning rate 1.0×10^{-3}.
- **Training schedule.** 20 epochs, full pass per epoch over $T_{\text{IS}} - w$ training windows, randomly shuffled at each epoch.
- **Numerical stability.** $\log \sigma_t$ is clamped to $[-6, 4]$ inside the loss to prevent gradient explosion when the network attempts to assign vanishingly-small variance to extreme observations.
- **Pre-processing.** The input return series is standardised by its IS mean and standard deviation; predictions are re-scaled back at sampling time.
- **Synthesis.** The trailing w IS returns seed the recurrence; for each generation step the predicted Gaussian is sampled, the sample is appended to the context window, and the procedure rolls forward. The first 64 steps are discarded as burn-in.
- **Implementation.** Julia using `Flux.jl` [36] with the `Optimisers.Adam` state. Random seed for weight initialisation, training shuffling, and per-path sampling: 20260420, with deterministic per-path sub-seeds.

The trained model is callable through the same `simulate_gru(model, n_steps; seed_window)` interface as the CHMM and discrete-HMM models, so the seven-metric panel evaluates the GRU on byte-identical windows to the other generators.

Table 24. GRU deep-generative baseline on SPY (single-layer, hidden $H = 32$, window $w = 20$, 20 epochs, Adam $\eta = 10^{-3}$, 1,000 paths, seed = 20260420). Final training NLL: 0.2005.

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	18.1	13.7	2.85	0.0518	0.196	0.0966	37.4
OoS	52.5	55.0	2.21	0.0500	0.293	0.1603	69.7

E.6 Per-Ticker OoS Price-Simulation Figures and Full Path-Level Metrics (Unconditional CHMM)

This appendix collects (i) the full per-family path-level probabilistic-forecast metrics deferred from Table 7 in Section 5.9, and (ii) the per-ticker price-fan and terminal-distribution figures for the five non-body tickers NVDA, JNJ, JPM, AAPL, and QQQ. All figures use the same plotting conventions as Figure 6 in the main body: blue shaded bands are the 5–95% and 25–75% quantile envelopes of the 500 simulated paths, blue solid line is the simulated median, red solid line is the observed OoS price track. Terminal-histogram panels show the cross-path distribution of $\widetilde{P}_T$ at T_{OoS} in gray, with the observed terminal P_T^{obs} as a red vertical line and the simulated median as a blue dashed line. For brevity, the figures show CHMM-N only (the representative family; differences across the three emission families in the price-level metrics are small, see Table 25).

Table 25. Full path-level probabilistic-forecast metrics for OoS equity-price simulation, all (ticker, family) cells ($K = 18$, 500 paths, $T_{\mathrm{OoS}} = 572$; global seed policy in Section 3.5). $\overline{C}_{0.90}$: fraction of OoS steps at which the observed price falls inside the simulated 5–95% band (calibration); $\mathrm{MAPE}_{0.50}$: mean absolute percentage error between observed path and simulated median (sharpness); $\overline{\mathrm{CRPS}}$: horizon-average bias-corrected sample CRPS of Gneiting and Raftery [51]; $\overline{\mathrm{CRPS}}/S_0$: CRPS scaled by S_0 to remove price-level effects. Lower MAPE and CRPS, higher $\overline{C}_{0.90}$, are better. **Headline:** no CHMM emission family dominates on all four metrics across all six tickers; differences between CHMM-N, CHMM-t and CHMM-L live mostly in the return tails and propagate only weakly into aggregated price-level scores.

Ticker	Family	$\overline{C}_{0.90}$	$\mathrm{MAPE}_{0.50}$	$\overline{\mathrm{CRPS}}$	$\overline{\mathrm{CRPS}}/S_0$
SPY	CHMM-N	1.000	0.100	39.25	0.0835
	CHMM-t	1.000	0.112	42.66	0.0908
	CHMM-L	1.000	0.102	38.68	0.0823
NVDA	CHMM-N	0.698	0.339	30.62	0.6424
	CHMM-t	0.636	0.383	34.77	0.7293
	CHMM-L	0.666	0.411	36.58	0.7674
JNJ	CHMM-N	1.000	0.101	12.55	0.0780
	CHMM-t	1.000	0.098	12.62	0.0784
	CHMM-L	1.000	0.101	12.65	0.0786
JPM	CHMM-N	1.000	0.195	32.23	0.1881
	CHMM-t	1.000	0.241	40.38	0.2357
	CHMM-L	1.000	0.211	35.25	0.2057
AAPL	CHMM-N	1.000	0.108	21.30	0.1156
	CHMM-t	1.000	0.075	18.90	0.1025
	CHMM-L	1.000	0.077	18.82	0.1021
QQQ	CHMM-N	0.998	0.073	30.48	0.0763
	CHMM-t	0.995	0.079	31.63	0.0792
	CHMM-L	1.000	0.118	41.12	0.1029

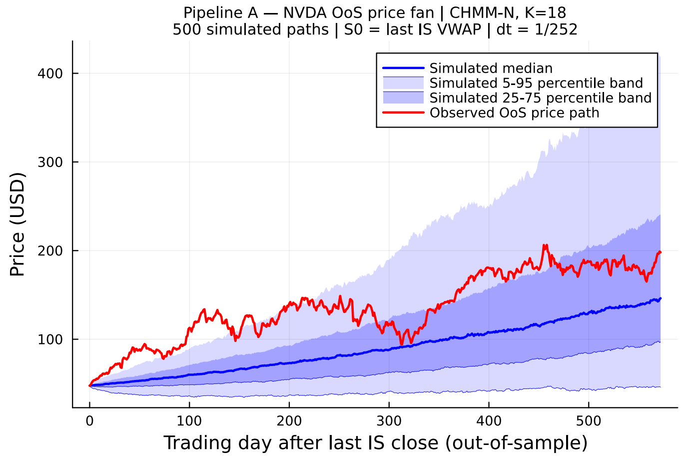
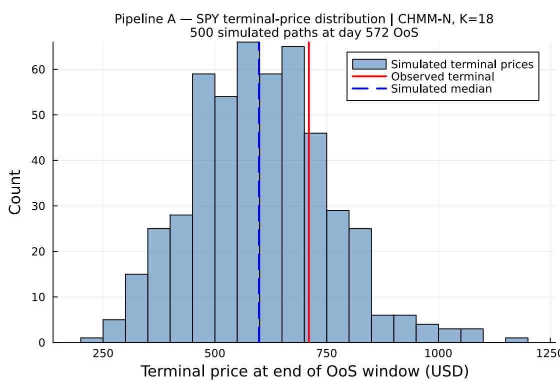
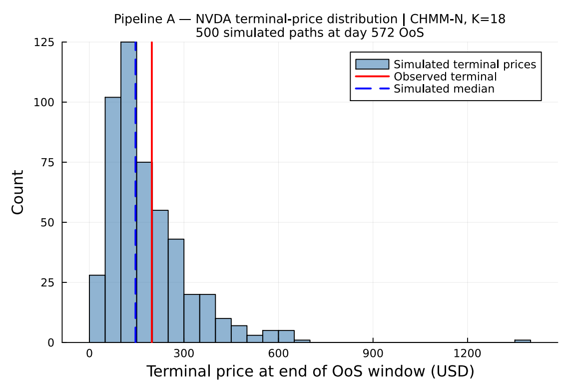


Figure 34. NVDA OoS price fan, CHMM-N ($S_0 = 47.67 \to P_T^{\text{obs}} = 198.07$, a 4.15× rally). The observed path exits the 25–75% interquartile band by mid-horizon but remains inside the 5–95% envelope; horizon-wide coverage under CHMM-N is 0.70 (CHMM-t: 0.64, CHMM-L: 0.67; Table 25), the lone calibration outlier in the panel.

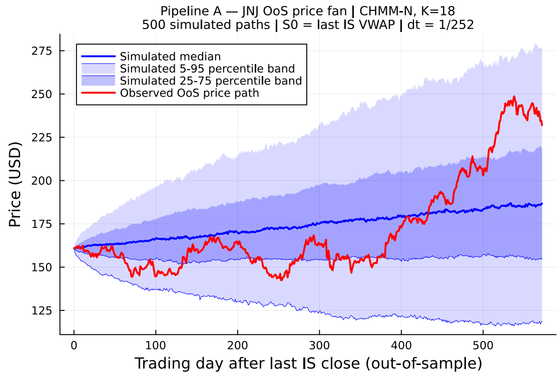


(a) SPY terminal.

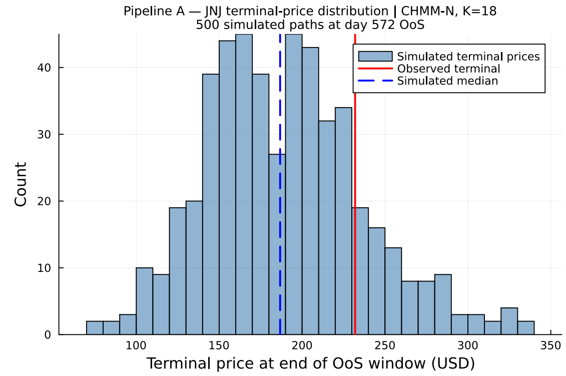


(b) NVDA terminal.

Figure 35. OoS terminal-price distributions under CHMM-N for SPY and NVDA. The right-skew is characteristic of log-returns aggregated to prices; the SPY terminal sits close to the simulated median, while the NVDA terminal sits in the upper-right of the simulated distribution (the realised rally is in the upper tail of the CHMM-N terminal density).

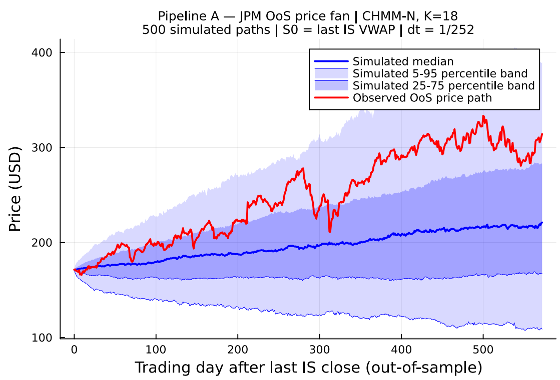


(a) Price fan.

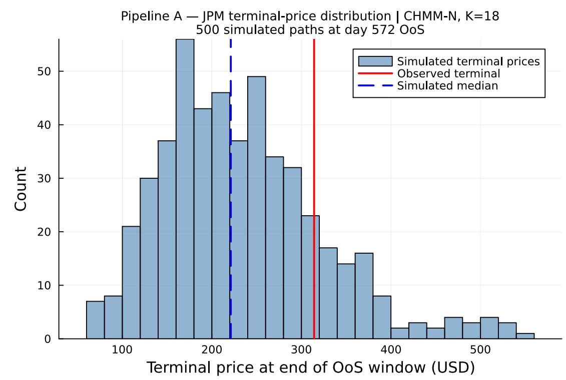


(b) Terminal distribution.

Figure 36. JNJ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as in Figure 6).

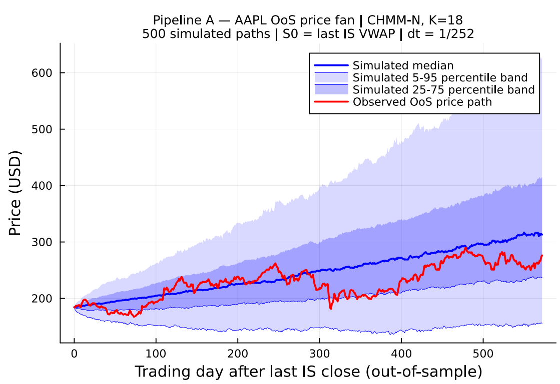


(a) Price fan.

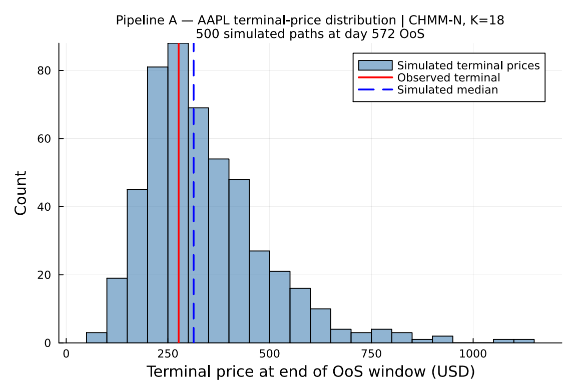


(b) Terminal distribution.

Figure 37. JPM OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as in Figure 6).

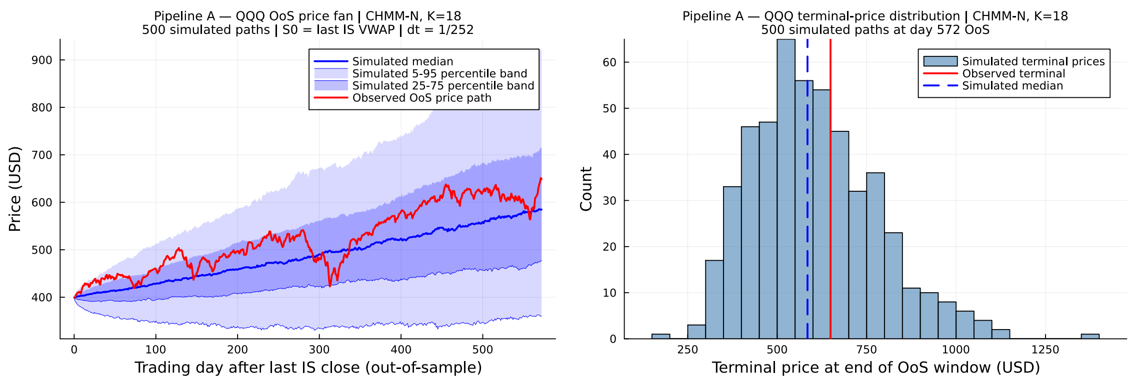


(a) Price fan.



(b) Terminal distribution.

Figure 38. AAPL OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as in Figure 6).



(a) Price fan.

(b) Terminal distribution.

Figure 39. QQQ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as in Figure 6).